

TECHNICAL MANUAL

O342e

UV PHOTOMETRIC
OZONE ANALYZER

- JANUARY 2020 -



WARNING

The information in this documentation is subject to change without notice.

ENVEA all rights reserved.

This document does not represent a commitment under part of ENVEA



HAZARD INFORMATIONS:

The presence of high intensity ultraviolet light could represent a hazard for eyes and skin: avoid eyes and skin exposure. Do not directly watch the light: use protection glasses.

SAFETY INSTRUCTIONS :

This device uses a UV LED confined under a protection cover inside the measurement chamber.

DO NOT REMOVE, IN ANY CASE, THE PROTECTION COVER OF THE UV LED LOCATED INSIDE THE MEASUREMENT CHAMBER, WHEN THE ANALYZER IS UNDER VOLTAGE.

Adjustment or carrying out performance of procedures other than those specified in this technical manual may present a risk of exposure to dangerous light rays when the unit is turned on.

Maintenance of the unit must be carried out by qualified people.

The O342e is of no hazard for the user.

In order to properly use the O342e analyzer, the user must thoroughly read this technical manual and keep it.

Warranty

Defects that fall under warranty

The seller shall undertake to remedy any operational malfunction resulting from a manufactured or material defect within the limits of the provision below.

The seller shall not be liable in the case of a defect caused either by materials supplied by the buyer or by a design imposed by the buyer.

Any warranty is also invalid in the case of damage resulting from normal wear and tear, accident, disaster, misuse, fault or negligence of or by Buyer, causes external to the Products such as, but not limited to, unauthorized repairs or part replacement, electrical power surges, improper storage of the Product, use of the Product in a manner for which it was not designed.

Duration and starting point of the warranty

Unless otherwise stipulated, the warranty period shall have a duration of twelve months from the date of delivery within the meaning of article 6 paragraph 2 of the «ENVEA: 2018 INTERNATIONAL GENERAL TERMS AND CONDITIONS OF SALES», even if the shipment or assembly is postponed for any reason outside the seller's control.

Unless agreed upon by both parties, the repair, modification or replacement of parts during the warranty period will not extend or renew the original equipment warranty period.

Buyer's obligation

In order to file a claim under warranty, the buyer must notify the seller immediately in writing of any defect in the equipment and supply evidence in proof thereof. The buyer must provide the seller with the opportunity to observe and remedy the said defects. In addition, the buyer must not carry out any repairs or have repairs made by a third party without the written agreement of the seller.

The buyer is required to check the equipment as soon as possible upon receipt and acceptance and no later than eight days following receipt. Failing to do so might invalidate any claims made later regarding a declared defect.

Any installation, maintenance, repair, service of the product performed by any person or entity other than seller without seller's prior written approval, or any use of replacement parts not supplied by seller, shall immediately void and cancel all warranties with respect to the affected product.

Exercising the warranty

Once notified of a defect, the seller shall be responsible for remedying the defect at its own expense. The seller, however, reserves the right to modify the mechanisms of the equipment as needed to comply with its obligations.

The work to satisfy the warranty obligation shall be carried out, principally, in the seller's workshop after the buyer has returned the equipment or the defective parts to the seller for the purposes of repair or replacement, whichever the seller deems best.

However, if the nature of the equipment is such that the repair has to be carried out at the location where it was installed, the seller shall only be responsible for the on-site labor costs involved in direct service of the analyzer itself. The buyer is responsible for the cost of any additional measures needed to provide unrestricted access.

The cost of transport of the equipment or the defective parts, as well as the return of the repaired or replaced equipment or part, shall be borne by the buyer. In the case of on-site repair, the buyer shall be responsible for any travelling and accommodation expenses of the seller's representative.

Defective parts replaced free of charge, under warranty, must be returned to the seller and shall become its property once again.

SUMMARY

CHAPTER 1. GENERAL – CHARACTERISTICS

1.1.	GENERAL	1–3
1.2.	CHARACTERISTICS	1–9

CHAPTER 2. PRINCIPLE OF OPERATION

2.1.	MEASUREMENT PRINCIPLE	2–3
2.2.	ANALYSIS	2–6
2.3.	AUTOMATIC RESPONSE TIME	2–9
2.4.	MEASUREMENT CONDITIONS	2–10
2.5.	FLUID CIRCUIT DIAGRAM	2–11
2.6.	SIMPLIFIED FLOWCHART OF MAIN PROGRAM	2–12
2.7.	ELECTRONIC ARCHITECTURE	2–13
2.8.	NETWORK LINK AND USB PORTS	2–14

CHAPTER 3. OPERATING INSTRUCTIONS

3.1.	INITIAL STARTUP	3–3
3.2.	PROGRAMMING THE O342e	3–12
3.3.	BROWSING ERGONOMY	3–16
3.4.	ANALYZER FUNCTION DESCRIPTION	3–21
3.5.	CALIBRATION	3–41

CHAPTER 4. PREVENTIVE MAINTENANCE

4.1.	SAFETY INSTRUCTIONS	4-3
4.2.	MAINTENANCE CALENDAR	4-4
4.3.	MAINTENANCE OPERATION SHEETS	4-5
4.4.	KITS AND SPARE PARTS SET FOR O342e MAINTENANCE	4-16

CHAPTER 5. CORRECTIVE MAINTENANCE

List of Figures

Figure 1-1 – ENVEA O342e (with screen).	1-2
Figure 1-2 – ENVEA O342e * (without screen).	1-2
Figure 1-3 – Color screen fitted with touch-screen	1-3
Figure 1-4 – Rear panel	1-4
Figure 1-5 – Internal view of O342e	1-6
Figure 1-6 – Measurement module diagram	1-7
Figure 1-7 – Links between units	1-10
Figure 1-8 – O342e dimensions (in mm)	1-11
Figure 2-1 – Spectrum absorption diagram	2-4
Figure 2-2 – General principle diagram	2-5
Figure 2-3 – I_0 , I and UV_0 signals acquisition	2-8
Figure 2-4 – Acquisition chronologic diagram for one measurement cycle	2-8
Figure 2-5 – O342e fluid circuit diagram	2-11
Figure 2-6 – Simplified flowchart of the main program	2-12
Figure 2-7 – Electronic architecture	2-14
Figure 3-1 – Fluid and electrical connections	3-3
Figure 3-2 – Installation of internal ozone generator (if option available)	3-4
Figure 3-3 – Installation of the sample gas inlet	3-5
Figure 3-4 – Filtering columns	3-42
Figure 3-5 – Schematic diagram of a typical UV photometric calibration system (1)	3-45
Figure 3-6 – Schematic diagram of a typical UV photometric calibration system (2)	3-45
Figure 3-7 – Gas flow system with zero span solenoid valve	3-55
Figure 3-8 – Gas flow system with Ozone generator	3-59
Figure 4-1 – Replacing of sample inlet dust filter	4-5
Figure 4-2 – KNF head pump dismantling	4-9
Figure 4-3 – Diaphragm replacing	4-10
Figure 4-4 – Replacing of valves and O-rings	4-11
Figure 4-5 – Pump reassembly	4-12
Figure 4-6 – Cleaning of flow restrictor	4-13
Figure 4-7 – Replacing of Teflon solenoid valve	4-14
Figure 4-8 – Replacement of selective filter grids	4-15
Figure 5-1 – Interconnection board diagram	5-13
Figure 5-2 – Automaton board diagram (above view)	5-15
Figure 5-3 – Photodiode board diagram (above view)	5-16
Figure 5-4 – LED UV board diagram (above view)	5-17
Figure 5-5 – Measurement board diagram	5-19
Figure 5-6 – Supply board diagram	5-20
Figure 5-7 – ARM20 board	5-21

List of Tables

Table 5-1 – List of faults and corrective actions	5-4
Table 5-2 – Interconnection board configuration	5-12
Table 5-3 – Automaton board configuration	5-14
Table 5-4 – Photodiode board configuration	5-16
Table 5-5 – LED UV board configuration	5-17
Table 5-6 – Measurement board configuration	5-18
Table 5-7 – Supply board configuration	5-20
Table 5-8 – ARM20 board description	5-21

INDEX OF PAGES

Page	Date	Page	Date	Page	Date
0-1	01.2020	3-15	04.2019	4-5	04.2019
0-2	04.2019	3-16	04.2019	4-6	04.2019
0-3	04.2019	3-17	04.2019	4-7	04.2019
0-4	04.2019	3-18	04.2019	4-8	04.2019
0-5	04.2019	3-19	04.2019	4-9	04.2019
0-6	04.2019	3-20	04.2019	4-10	04.2019
0-7	04.2019	3-21	04.2019	4-11	04.2019
0-8	01.2020	3-22	04.2019	4-12	04.2019
		3-23	04.2019	4-13	04.2019
		3-24	04.2019	4-14	04.2019
1-1	04.2019	3-25	04.2019	4-15	04.2019
1-2	04.2019	3-26	04.2019	4-16	04.2019
1-3	04.2019	3-27	04.2019		
1-4	04.2019	3-28	04.2019		
1-5	04.2019	3-29	04.2019		
1-6	04.2019	3-30	04.2019		
1-7	04.2019	3-31	04.2019	5-1	04.2019
1-8	04.2019	3-32	04.2019	5-2	04.2019
1-9	04.2019	3-33	04.2019	5-3	04.2019
1-10	09.2019	3-34	04.2019	5-4	04.2019
1-11	04.2019	3-35	04.2019	5-5	04.2019
1-12	04.2019	3-36	04.2019	5-6	04.2019
		3-37	04.2019	5-7	04.2019
		3-38	04.2019	5-8	04.2019
2-1	04.2019	3-39	04.2019	5-9	04.2019
2-2	04.2019	3-40	04.2019	5-10	04.2019
2-3	04.2019	3-41	04.2019	5-11	04.2019
2-4	04.2019	3-42	04.2019	5-12	04.2019
2-5	04.2019	3-43	04.2019	5-13	04.2019
2-6	04.2019	3-44	04.2019	5-14	04.2019
2-7	04.2019	3-45	04.2019	5-15	04.2019
2-8	04.2019	3-46	04.2019	5-16	04.2019
2-9	04.2019	3-47	04.2019	5-17	04.2019
2-10	04.2019	3-48	04.2019	5-18	04.2019
2-11	04.2019	3-49	04.2019	5-19	04.2019
2-12	04.2019	3-50	04.2019	5-20	04.2019
2-13	04.2019	3-51	04.2019	5-21	04.2019
2-14	04.2019	3-52	04.2019	5-22	04.2019
		3-53	04.2019		
		3-54	04.2019		
3-1	04.2019	3-55	01.2020		
3-2	04.2019	3-56	04.2019		
3-3	04.2019	3-57	04.2019		
3-4	04.2019	3-58	04.2019		
3-5	04.2019	3-59	04.2019		
3-6	04.2019	3-60	04.2019		
3-7	04.2019				
3-8	04.2019				
3-9	04.2019				
3-10	04.2019				
3-11	04.2019	4-1	04.2019		
3-12	04.2019	4-2	04.2019		
3-13	04.2019	4-3	04.2019		
3-14	04.2019	4-4	04.2019		

CHAPTER 1

GENERAL – CHARACTERISTICS

1.1	GENERAL	1-3
1.1.1	INTRODUCTION	1-3
1.1.2	DESCRIPTION	1-3
1.1.2.1	Front panel	1-3
1.1.2.2	Rear panel	1-4
1.1.2.3	Internal view	1-5
1.1.3	VARIOUS OPERATING MODES	1-8
1.1.3.1	Standard	1-8
1.1.3.2	Options	1-8
1.1.4	ASSOCIATED EQUIPMENT (OPTIONAL)	1-8
1.2	CHARACTERISTICS	1-9
1.2.1	TECHNICAL CHARACTERISTICS	1-9
1.2.2	STORAGE CHARACTERISTICS	1-10
1.2.3	INSTALLATION CHARACTERISTICS	1-10
1.2.3.1	Links between units	1-10
1.2.3.2	Dimensions and weight	1-10
1.2.3.3	Handling and storage	1-10
1.2.4	STORAGE	1-10

Figure 1-1 – ENVEA O342e (with screen).	1-2
Figure 1-2 – ENVEA S.A O342e * (without screen).	1-2
Figure 1-3 – Color screen fitted with touch-screen	1-3
Figure 1-4 – Rear panel	1-4
Figure 1-5 – Internal view of O342e	1-6
Figure 1-6 – Measurement module diagram	1-7
Figure 1-7 – Links between units	1-10
Figure 1-8 – O342e dimensions (in mm)	1-11

1 GENERAL – CHARACTERISTICS



Figure 1-1 – ENVEA O342e (with screen).

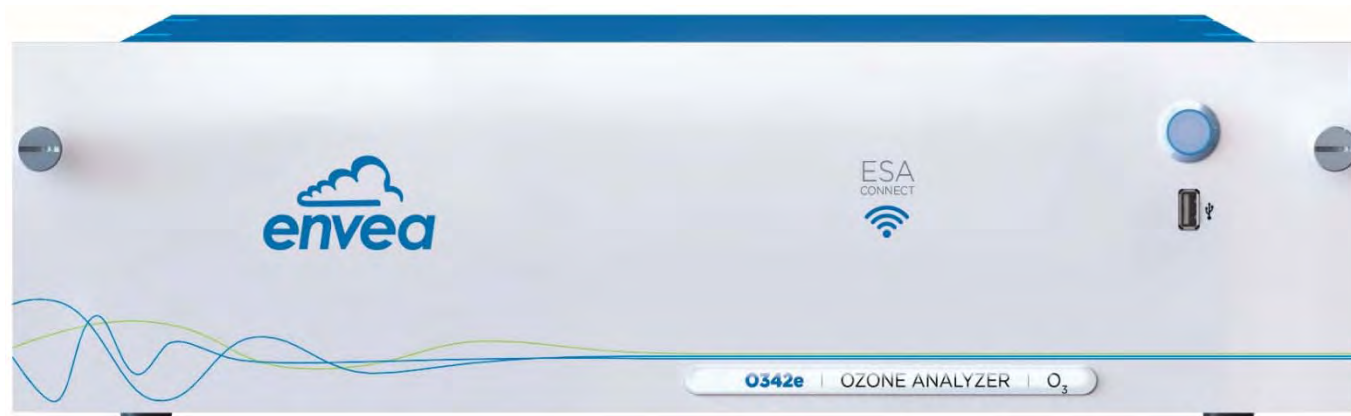


Figure 1-2 – ENVEA O342e * (without screen).

1.1 GENERAL

1.1.1 INTRODUCTION

The O342e is a continuous ozone analyzer (with a detection limit of 0.2 ppb).

Its measurement principle is based on ozone detection by direct absorption in ultraviolet light.

With the incorporation of recent optical and electronic technologies, the analyzer offers many advantages while requiring only limited maintenance.

The sample is taken using a Teflon tube (6 mm external diameter) connected to the rear panel of the analyzer. The sample is aspirated by an internal pump.

The measurement is displayed on a color display equipped with a touch screen.

This analyzer is also available in O342e * (without screen) version.

1.1.2 DESCRIPTION

1.1.2.1 Front panel

The front panel includes the following:

- General switch.
- Backlit color TFT LCD (Thin Film Transistor Liquid Crystal Display):
 - Resolution of 800 (RGB) x 480, size equal to 7" size.
 - The display provides the measurement values and other system parameters, as desired.
- Capacitive touch-screen projected on glass.
- Two-color LED as an operation indicator, placed below the switch.



Figure 1-3 – Color screen fitted with touch-screen

1.1.2.2 Rear panel

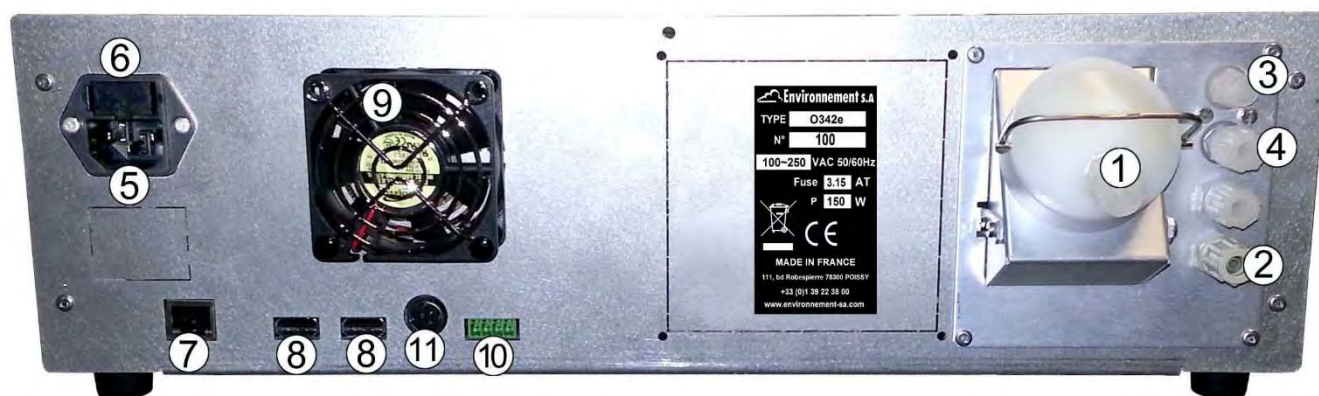
All of the O342e electrical connectors and gas inlets/outlets are located on the rear panel. Refer to Figure 1–4.

Gas inlets / outlets (right-hand side):

- The inlet for the sample to be analyzed (1) is composed of 4 mm (I.D.) and 6 mm (O.D.) Teflon tube fittings, attached to the dust filter holder. The dust filter holder is equipped with a Teflon filtering diaphragm.
- The pump outlet (2) to exhaust the analyzed sample, consists of 4 mm (I.D.) and 6 mm (O.D.) Teflon fitting.
- The fan (9).
- The "span / zero" inlets (3) and (4) are composed of 4/6 mm pneumatic tube fitting. They are used either to connect a "ZERO AIR" external supply free of ozone, or to connect a device enabling ozone generation (both gases are always at atmospheric pressure).

Electrical equipment and connections (left-hand side):

- One main power supply block consisting of a three-contact socket (5) to connect a standard power cable and the general fuse (6): 3.15 A/250 V.
- One Ethernet output (7) and two USB ports (8).
- One 4-points BL connection to connect the optional solenoid valve for external calibration (10),
- One connection for the 24 V power supply for ESTEL board option (11).



(1) sample inlet, (2) pump outlet, (3) span inlet and (4) zero inlet, (5) three-contact socket for main power supply, (6) general fuse, (7) Ethernet output, (8) two USB ports, (9) fan, (10) duplicate zero and span solenoid valves for the optional solenoid valve of external calibration, (11) 24 V power supply for ESTEL board option.

Figure 1–4 – Rear panel

1.1.2.3 Internal view

The internal components of the analyzer are accessed by simply unscrewing the single knurled screw at the rear panel and sliding the upper cover.

The internal elements include (refer to Figure 1–5):

- Dust filter of the sample inlet mounted on the rear panel (1),
- An optional solenoid valve-filter assembly (not shown),
- Measurement module (2),
- Pump (3),
- HMI board (4),
- Interconnection board (5),
- 24 VDC power supply unit (6), with automatic universal input voltage 100–250 Vac 50/60Hz.
- Measurement board (7) that uses the + 24 V, + 12V, + 5V and + 3.3 V power supplies. That supports analog signal acquisitions and digital signal processing.
- Pressure sensor board (8).
- Optional Zero-Span SV block (9).

SAFETY INSTRUCTIONS :

This device uses a UV LED confined under a protection cover inside the measurement chamber.

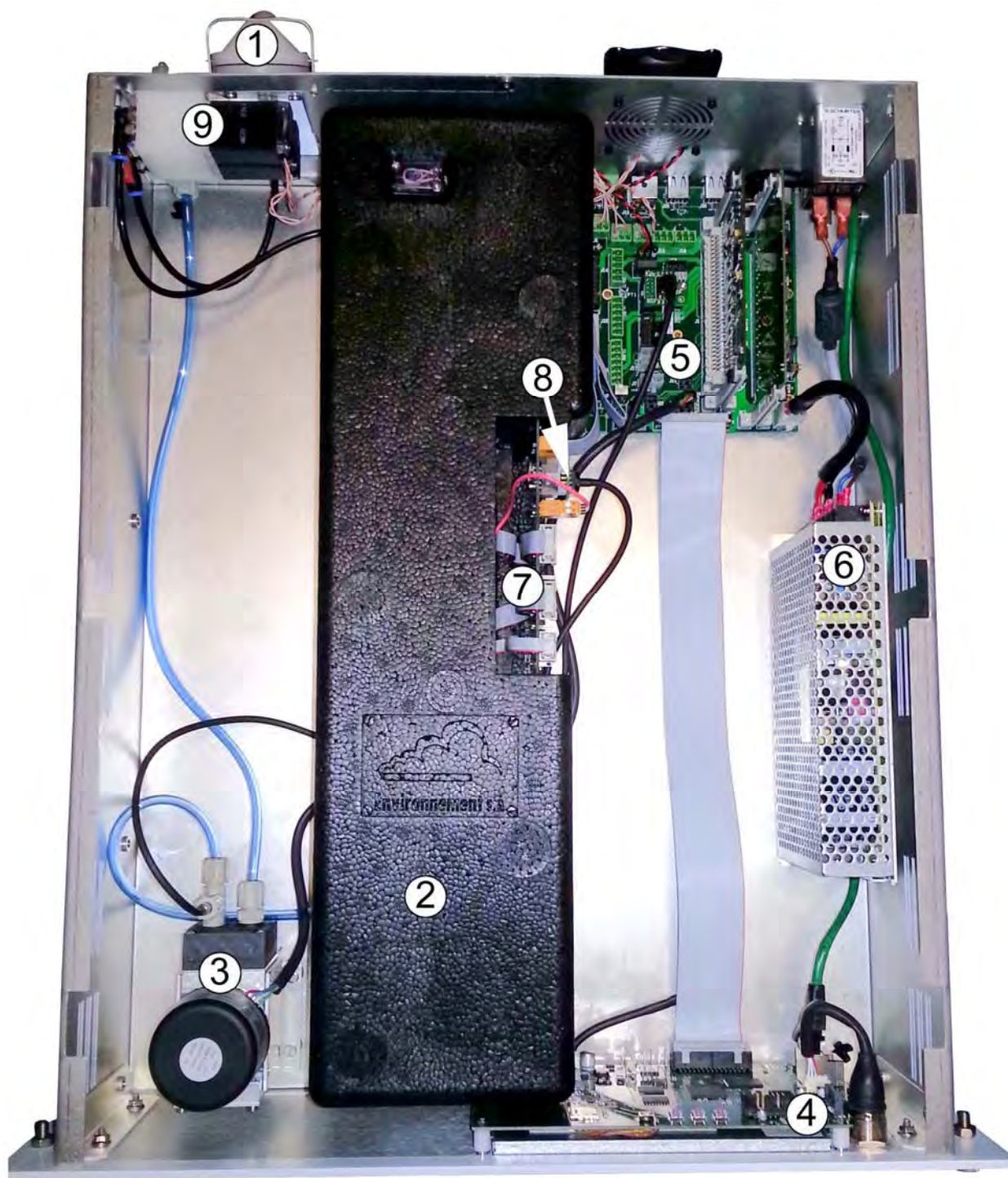
DO NOT REMOVE, IN ANY CASE, THE PROTECTION COVER OF THE UV LED LOCATED INSIDE THE MEASUREMENT CHAMBER, WHEN THE ANALYZER IS UNDER VOLTAGE.

Adjustment or carrying out performance of procedures other than those specified in this technical manual may present a risk of exposure to dangerous light rays when the unit is turned on.

Maintenance of the unit must be carried out by qualified people.

The O342e is of no hazard for the user.





(1) sample inlet dust filter, (2) measurement module, (3) pump, (4) HMI board, (5) interconnection board, (6) 24 VDC power supply unit, (7) measurement board, (8) pressure sensor board, (9) zero-span solenoid valve block.

Figure 1-5 – Internal view of O342e

After passing through the dust filter, the sample to be analyzed moves towards the measurement module which includes the main following elements (refer to Figure 1–6):

- Monochromatic emission LED at 255 nm in UV, mounted on LED board (1), is located under a protective cover fixed with three screws. This board is directly connected to the Measurement board.
- Two photodiode boards: reference photodiode (2a) board dedicated to measurement of incident energy emitted by LED (UV0), and measurement photodiode (2b) board dedicated to measurement of UV absorption which allows I and I_0 signal acquisition (refer to chapter 2.2). Each of these boards is mounted under a protective cover which isolates them from visible light.
- Optical block (3) consisting of one beam splitter and one flat convex lens to focus light on Reference (2a) photodiode. Optical block is used to separate light emitted by LED (1) to Reference (2a) photodiode and absorption medium (4) by means of a beam splitter.
- Absorption medium (4), consisting of one glass tube and two mechanical parts at inlet and outlet, inside which light absorption emitted by LED is performed. Optical path length traveled by gas to be measured is 400 mm.
- Cycle solenoid valve (5) allowing cyclically or alternatively switching gas to be analyzed to i cycle and i_0 cycle channels (refer to chapter 2.2 Analysis).
- Flow restrictor (6) controlling gas flow to $55 \text{ NI/h} \pm 5 \text{ NI/h}$. Restrictor is directly screwed on absorption medium fluid outlet.
- Ozone purifier filter (7) removes any ozone traces in gas to be analyzed.
- Pressure sensor connector (8).
- Temperature sensor (9), PT1000 component, is placed inside a thin metallic finger shape object. This temperature sensor is in direct contact with the gas flow.
- Sample gas inlet (10).

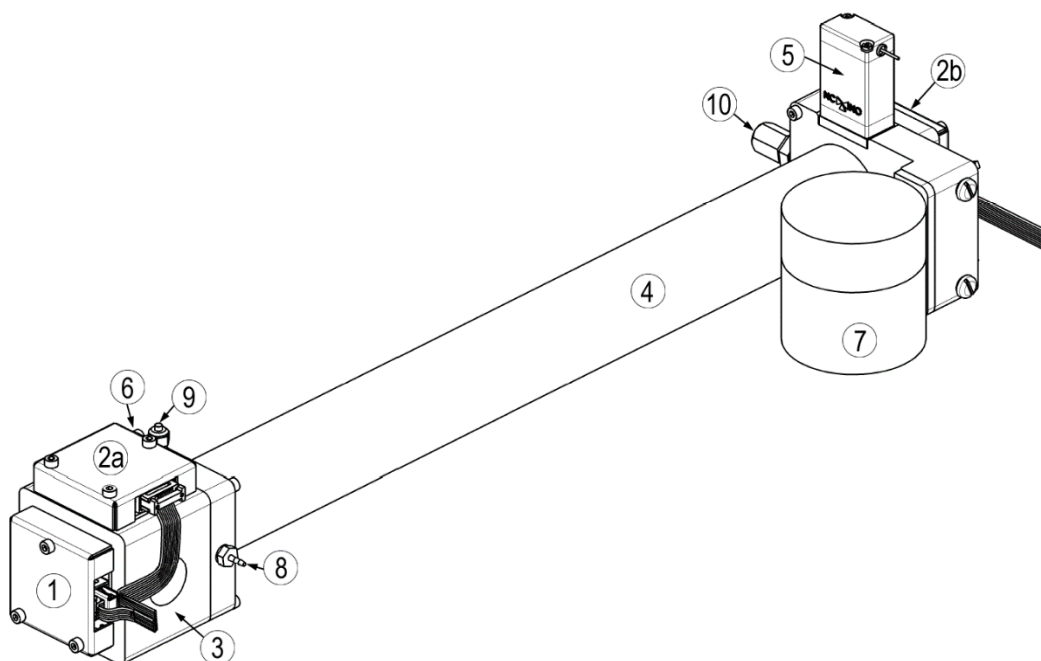


Figure 1-6 – Measurement module diagram

1.1.3 VARIOUS OPERATING MODES**1.1.3.1 Standard**

- Programmable measurement range from 0 to 10 ppm, with a detection limit of 0.2 ppb for response time 50 seconds.
- Automated monitoring of parameters influencing metrology (UV emitted energy, gas flow, temperature and pressure) and correct operation tests.
- Measurement values in ppb or $\mu\text{g}/\text{m}^3$ (integrated conversion coefficient).
- Memory storage of average measurements with programmable periods.

1.1.3.2 Options

The following analyzer options are available:

- One ESTEL board case, as accessory, and to be directly connected to USB ports on rear panel,
- One RS232 USB converter delivered as an accessory and to be directly connected to USB ports on rear panel.

1.1.4 ASSOCIATED EQUIPMENT (OPTIONAL)

- Analog recorders and data loggers,
- Digital data acquisition system.

1.2 CHARACTERISTICS

1.2.1 TECHNICAL CHARACTERISTICS

Measurement range :	– 0 to 10 ppm
Units :	– ppb or $\mu\text{g}/\text{m}^3$ (programmable)
Noise (σ) :	– 0.1 ppb. (response time : 50 sec)
Minimum detectable limit (2σ) :	– 0.2 ppb. (response time : 50 sec)
Response time (0-90 %) :	– 20-90 s (programmable)
Zero drift :	– < 1 ppb / 7 days
Span drift :	– < 1 % / 7 days
Linearity :	– ± 1 %
Sample flow-rate :	– approximately 1 liter/min.
Temperature and pressure influence :	– Automatic temperature/pressure compensation
Display :	– TFT LCD color screen, resolution : 800 (RGB) x 480, size : 7 inches
Control :	– Touch screen
Output signals (option) :	– 4 analog outputs (0-1 V, 0-10 V, 0-20 mA, 4-20 mA)
Power supply :	– 100~250Vac, 50/60Hz + ground
Consumption :	– 50 W
Working temperature :	– + 0 °C to 35 °C
Memory storage of measurement values :	– Capacity : 1 year based on 15-min intervals – Flash EPROM storage
Alarms checks :	– Detection and identification of anomalies in continuous: temperature, flow rate, electric parameters, out of O ₃ programmable measurement threshold.
Tests and diagnostics of maintenance :	– Direct selection with the touch screen and/or remote selection on the embedded Web server.
Back-up saving time for the real-time clock and RAM-stored data:	– 1 year maximum.
Ethernet output:	– RJ45 socket, UDP protocol.
USB port :	– Type A USB socket: 2.0 (3.0 compliant)
Zero/span external SV control	– Contact connector with screw terminals

SV : Solenoid valve

1.2.2 STORAGE CHARACTERISTICS

Temperature: – 10°C to +60°C.

1.2.3 INSTALLATION CHARACTERISTICS

1.2.3.1 Links between units

The O342e analyzer uses the external links and power supplies illustrated in Figure 1–7 :

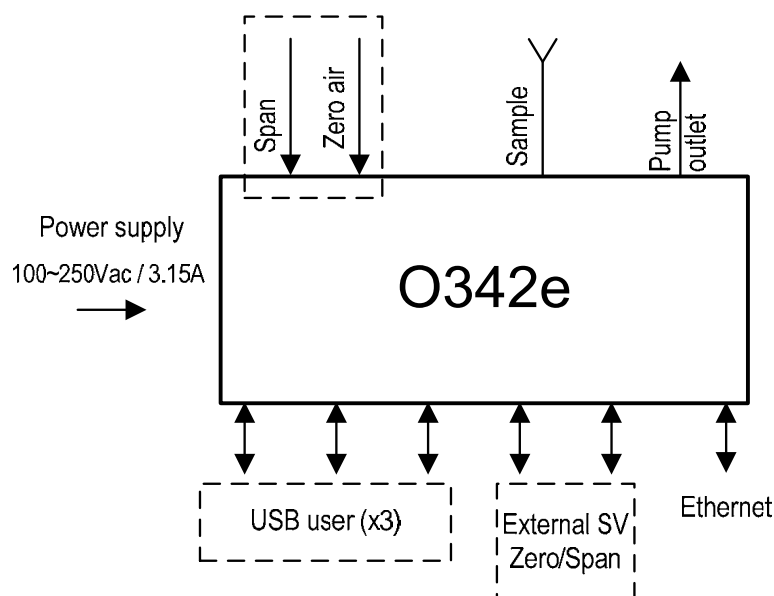


Figure 1–7 – Links between units

1.2.3.2 Dimensions and weight

The analyzer is a standard 19-inch, three-unit high rack.

Length	:	606 mm
Width	:	483 mm
Height	:	133 mm
Weight	:	9.5 Kg

1.2.3.3 Handling and storage

The O342e analyzer must be handled with care to avoid damage to the various connectors and fittings on the rear panel.

1.2.4 STORAGE

The unit should be stored in the foam-packed case provided for this purpose.

Make sure that the fluid inlets and outlets of the analyzer are protected with caps during storage.

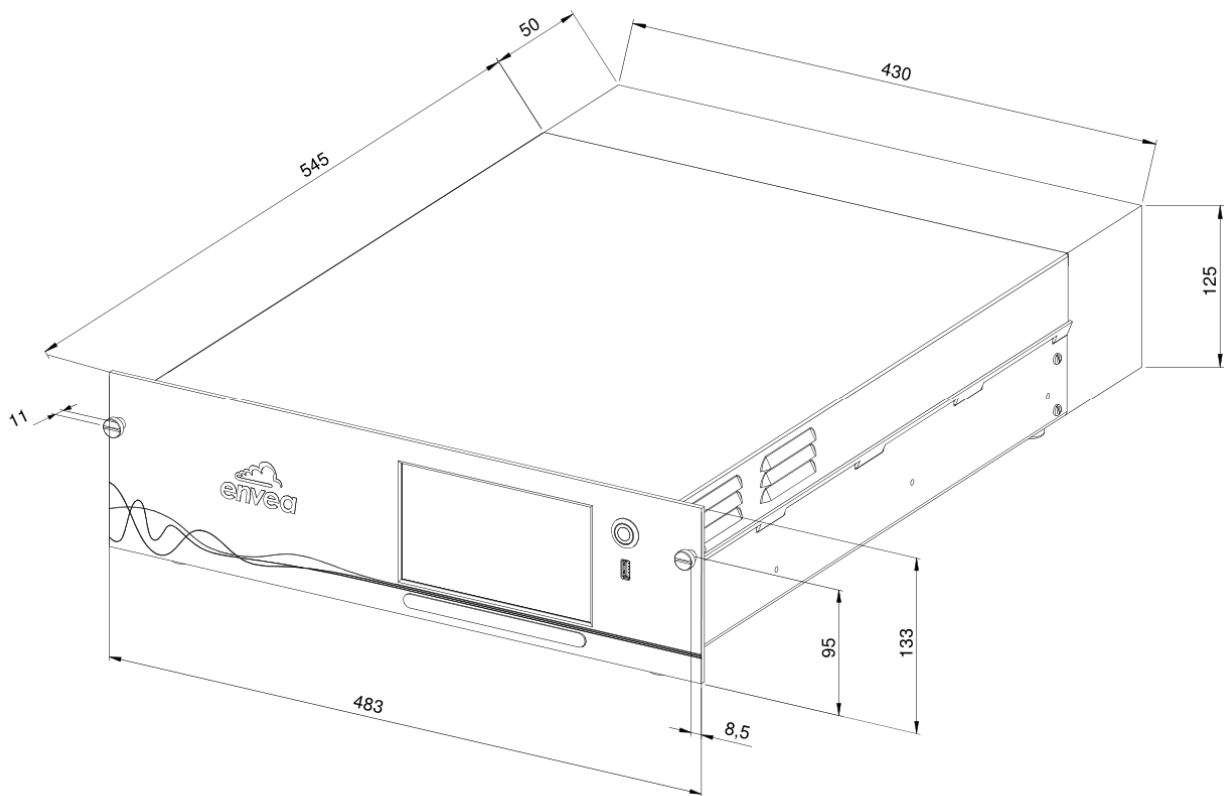


Figure 1-8 – O342e dimensions (in mm)

Page intentionally left blank

CHAPTER 2.

PRINCIPLE OF OPERATION

2.1.	MEASUREMENT PRINCIPLE	2-3
2.2.	ANALYSIS	2-6
2.3.	AUTOMATIC RESPONSE TIME	2-9
2.3.1.	PRINCIPLE	2-9
2.3.2.	PROGRAMMING THE RESPONSE TIME	2-9
2.4.	MEASUREMENT CONDITIONS	2-10
2.5.	FLUID CIRCUIT DIAGRAM	2-11
2.6.	SIMPLIFIED FLOWCHART OF MAIN PROGRAM	2-12
2.7.	ELECTRONIC ARCHITECTURE	2-13
2.8.	NETWORK LINK AND USB PORTS	2-14

Figure 2-1 – Spectrum absorption diagram	2-4
Figure 2-2 – General principle diagram	2-5
Figure 2-3 – I_0 , I and UV_0 signals acquisition	2-8
Figure 2-4 – Acquisition chronologic diagram for one measurement cycle	2-8
Figure 2-5 – O342e fluid circuit diagram	2-11
Figure 2-6 – Simplified flowchart of the main program	2-12
Figure 2-7 – Electronic architecture	2-14

Page intentionally left blank

2. PRINCIPLE OF OPERATION

2.1. MEASUREMENT PRINCIPLE

Refer to Figure 2–1 and Figure 2–2.

O342e measurement principle is based on the Beer-Lambert law related to the direct UV photometry technique. Ozone absorption spectrum is intense in the 250 and 270 nanometers (nm) wavelength range. UV LED monochromatic source used in O342e for ozone measurement is centered around 255 nm. Thus, it corresponds to the maximum range of ozone absorption.

Ozone absorption coefficient, α , was defined experimentally at 255 nm, for normal P_0 pressure and to temperature conditions, to be:

$$\alpha = 309 \text{ atm}^{-1} \cdot \text{cm}^{-1}$$

With: $t_0 = 273 \text{ K (0 °C)}$
 $P_0 = 101.3 \text{ kPa}$

Ozone concentration calculation, after application of the Beer Lambert law is:

$$[O_3]_{(Theoric)_{ppm}} = \frac{10^6}{\alpha \ell} \times \ln \left[\frac{I_0}{I} \right]$$

Correcting the O_3 concentration theoretically calculated as function of temperature and pressure measurement conditions, the corrected ozone concentration can be written as:

$$[O_3]_{(Corrected)_{ppm}} = \frac{10^6}{\alpha \ell} \times \ln \left[\frac{I_0}{I} \right] \times \frac{P_0}{P} \times \frac{t}{t_0}$$

With:

- α : ozone absorption coefficient at 255 nm.
- ℓ : absorption optical path length in cm.
- I_0 : UV energy measured through the absorption medium when the sample is ozone free (by passing through the ozone scrubber, i_0 cycle).
- I : UV energy measured when the sample contains ozone, i.e. direct absorption (i cycle).
- P_0 : atmospheric pressure at normal conditions, i.e. equal to 101.3 kPa.
- P : atmospheric pressure at measurement conditions, in kPa.
- t_0 : ambient temperature at normal conditions, i.e. equal to 273 K (0 °C).
- t : ambient temperature at measurement conditions, in K.

O342e span coefficient, K , is defined by the following formula:

$$K = \alpha \times \ell$$

Because of its mechanical design, the path-length is equal to 40 cm. Accordingly, the calibration factor is:

$$K = 309 \text{ cm}^{-1} \times 40 \text{ cm} = 12360$$

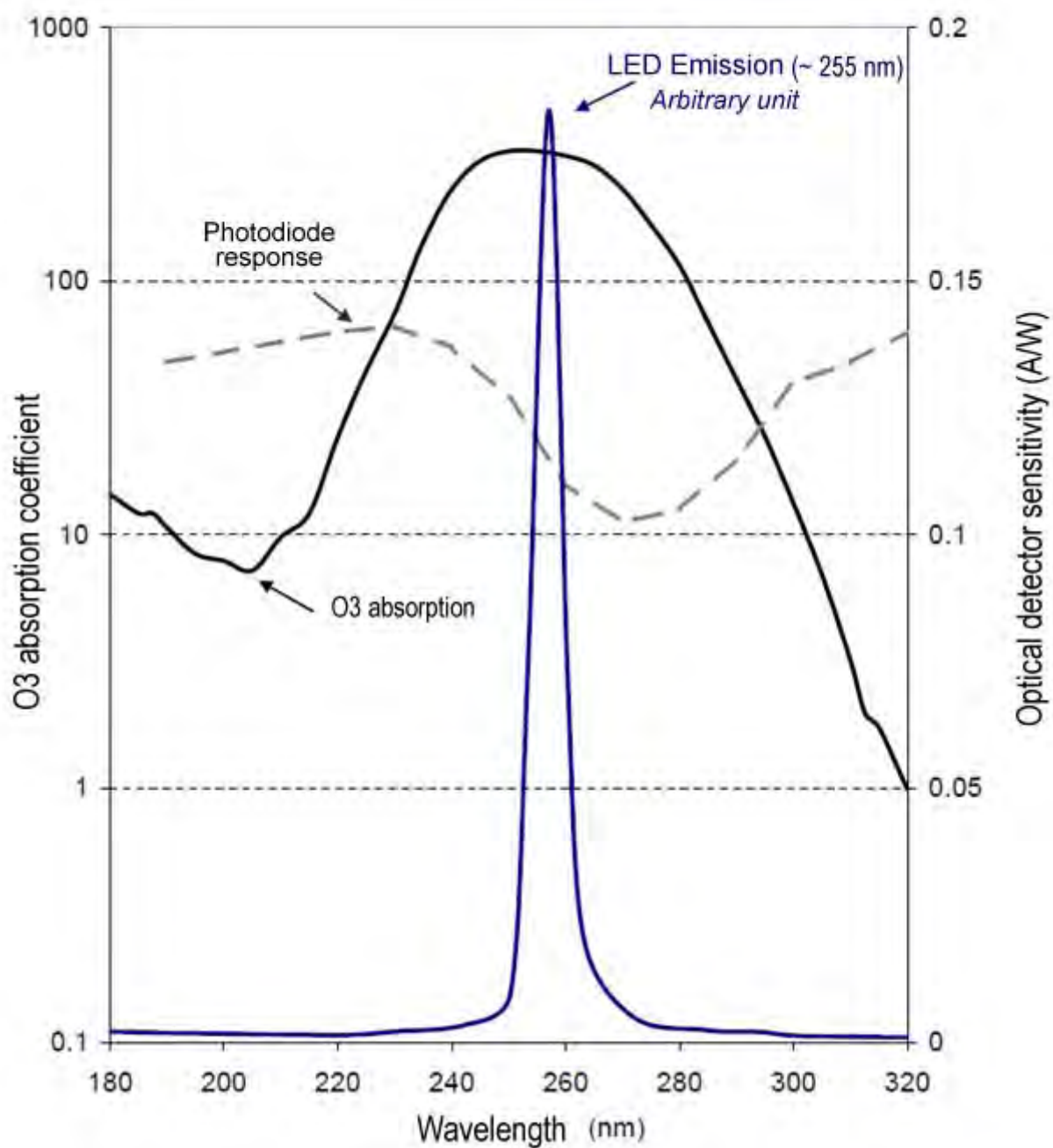
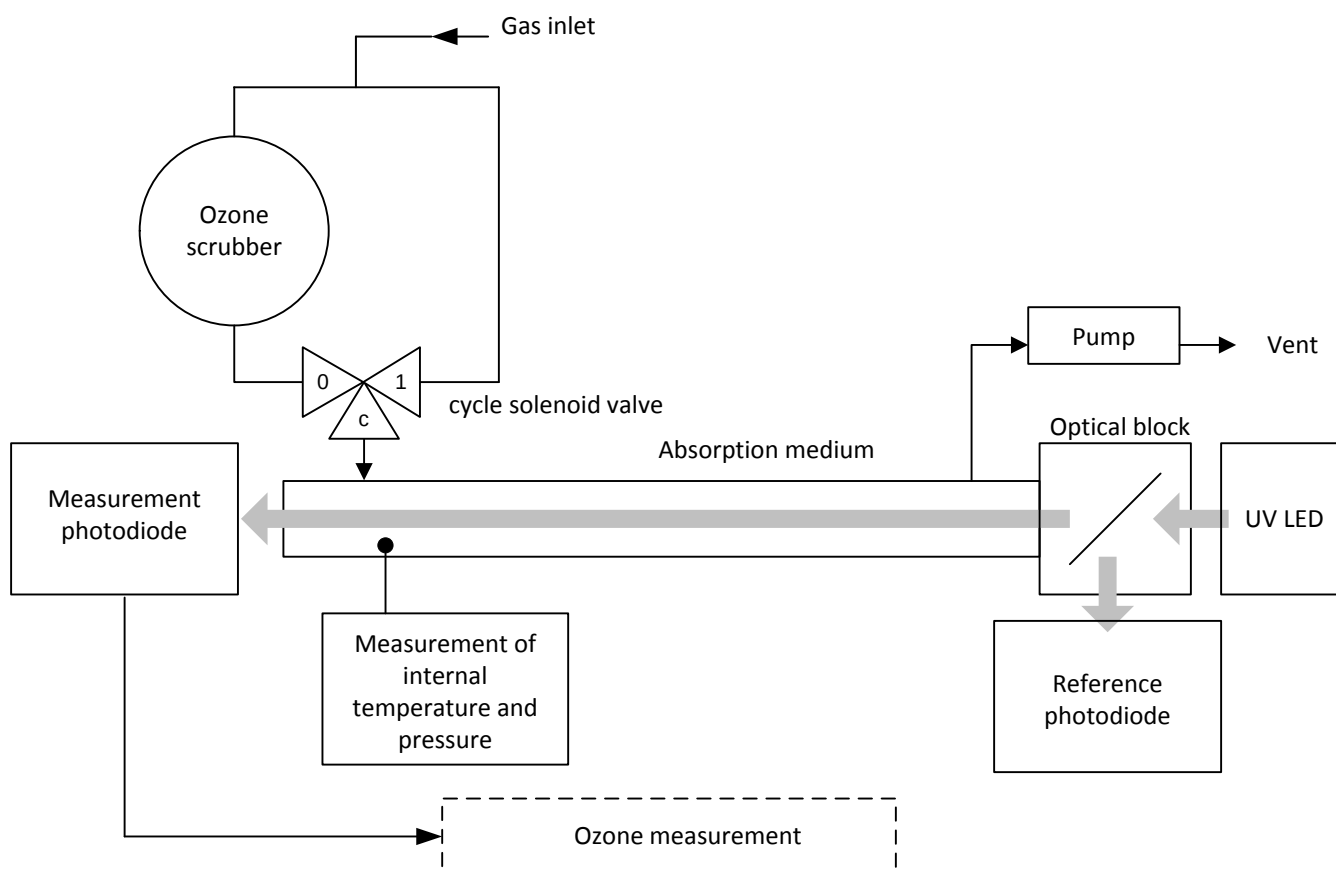


Figure 2-1 – Spectrum absorption diagram



«UV» means «ultraviolet»

«LED» means «light-emitting diode»

Figure 2-2 – General principle diagram

2.2. ANALYSIS

Whatever the gas to be analyzed (sample, zero air, span), the incident energy emitted by the LED is constant.

Indeed, O342e implements a LED current control system based on its emitted energy. A « reference » photodiode measures in continuous the emitted LED energy and it offsets the LED control current during I and I_0 measurements, which ensures a constant level of UV energy.

A measurement cycle lasts 10 seconds and corresponds to two half-cycle sequences of i_0 and i measurement, each lasting 5 seconds (cycle duration programmed by default).

- i_0 half-cycle: solenoid valve cycle status at 0 :
 - Gas to be analyzed passes through the O₃ scrubber, acquisition is timed for 3 seconds to purge the absorption medium.
 - I_0 measurement acquisition is realized by the measurement photodiode.
 - LED current is adjusted to allow UV reference measurement by the reference photodiode.
- i half-cycle: solenoid valve cycle status at 1 :
 - Cycle solenoid valve is switched on.
 - Gas to be analyzed passes directly through the absorption medium (i. e. without passing through O₃ scrubber), where acquisition is timed for 3 seconds in order to purge.
 - I measurement acquisition is carried out by measurement photodiode.
 - LED current is adjusted to allow UV reference measurement by the reference photodiode.
 - Cycle metrology cycle is calculated.

I_0 and I energies acquisition (refer to Figure 2–3 and Figure 2–4):

The analog/digital (A/D) linear converters convert measurement and reference signals (UV_0 reference and I_0 / I measurement) from the UV photodiodes.

The average of 2000 measured values (for example I_0) is first calculated. Then, this average is offset by the average of 2000 measurement values of UV_0 reference. This is the calculation procedure to obtain the I and I_0 metrology offset in real time by the LED energy measurement.

The next half-cycle during which the gas sample directly passes (i.e. without passing through ozone scrubber) through the absorption medium (cycle solenoid valve status to 1), gives I value.

The raw ozone concentration is calculated according to the Beer Lambert law:

$$[O_3]_{(Raw) ppm} = \frac{10^6}{K} \times \ln \left[\frac{I_0}{I} \right]$$

A barometric sensor measuring the absorption medium pressure ($P_{(Chamber)}$) allows the pressure offset.

A sensor measuring the gas temperature ($T_{(Gas)}$) is used to perform the temperature offset.

The ozone corrected concentration is given by:

$$[O_3]_{(Corrected) ppm} = [O_3]_{(Raw) ppm} \times \frac{273 + T_{(Gas)}}{273} \times \frac{1013}{P_{(Abs.med.)}}$$

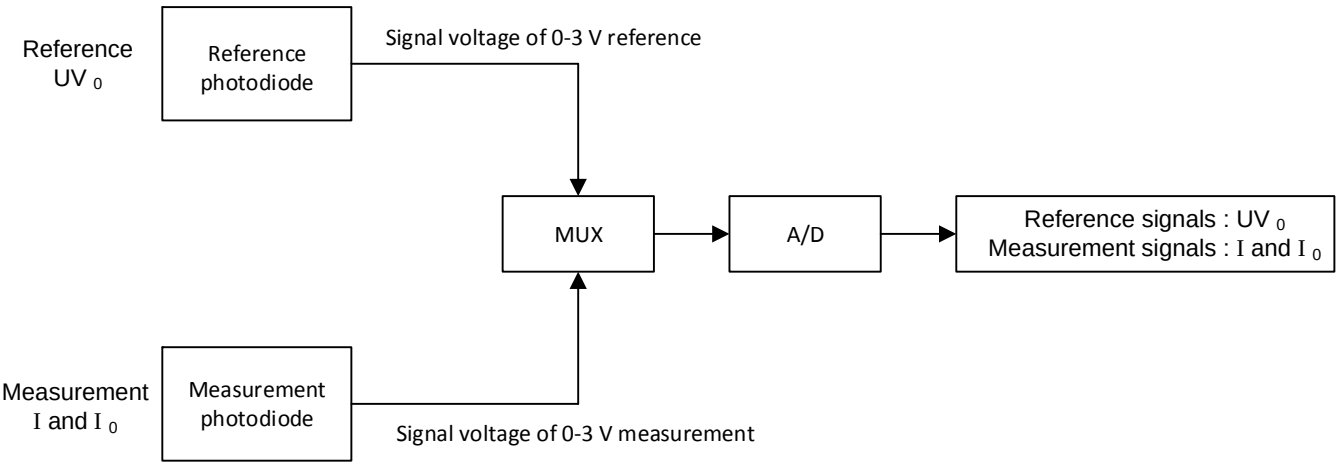


Figure 2-3 – I_0 , I and UV_0 signals acquisition

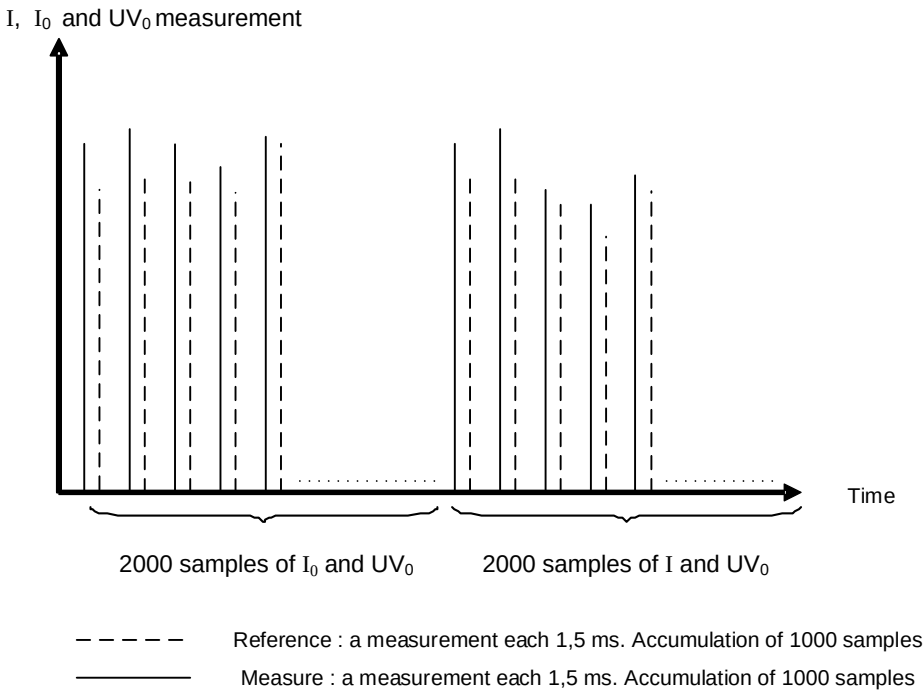


Figure 2-4 – Chronologic acquisition diagram for one measurement cycle

2.3. AUTOMATIC RESPONSE TIME

In order to optimize its metrology, the O342e analyzer is equipped with a software function called *Automatic Response Time*, which enables the filtration of measurements.

2.3.1. PRINCIPLE

First, an average of the instantaneous readings is carried out corresponding to the minimum response time.

$$[MEAS]_{AVERAGE} = \frac{1}{n} \sum_1^n [MEAS]_{INSTANTANEOUS}$$

n = number of instantaneous measurements which is determined by the programmed response time $[RT]_{MIN}$.

Then, a weighted average between the filtered measurement values ($[MEAS]_{FILTERED}$) and the average measurements ($[MEAS]_{AVERAGE}$) is recursively calculated according to the following equation:

$$[MEAS]_{DISPLAYED} = [MEAS]_{FILTERED(new)} = X[MEAS]_{FILTERED(previous)} + Y[MEAS]_{AVERAGE}$$

$$X + Y = 1$$

When the difference ($[MEAS]_{FILTERED(previous)} - [MEAS]_{AVERAGE}$) exceeds a determined threshold, the Y value is increased up to a maximum value of 0.99 which corresponds to a fixed response time, RT_{MIN} .

When $[MEAS]_{FILTERED(previous)} - [MEAS]_{AVERAGE}$ is below the threshold, the Y value is progressively decreased.

2.3.2. PROGRAMMING THE RESPONSE TIME

The automatic response time function may be activated or de-activated in the « *Response time* » field of the « *Advanced analyzer settings* » screen.

The minimum response time may also be modified in the mentioned menu.

2.4. MEASUREMENT CONDITIONS

In order to avoid wrong ozone metrology, be careful to take into account the following conditions:

- Ozone is highly reactive, therefore there may be concentration losses by contact with active surfaces.
- Teflon inlet dust filter may trap a few ppb levels of ozone due to dust deposit. If the dust amount is very low, it is possible not to install filter membrane. If the dust amount is important, for low O₃ concentration measurements (less than 20 ppb), it will be necessary to saturate the areas likely to absorb by exposing them to a high ozone concentration (400 to 800 ppb) during 1 or 2 hours. Long external tubes are also to be avoided. Regularly check the dust amount of inlet filter and change the membrane more frequently.
- The recommended replacement frequency of the Teflon inlet filter is 2 to 4 weeks depending on the site conditions.
- Ozone measurement is carried out during a 10-second complete cycle. Differences in pressure or variations in air turbulence in the absorption medium can cause measurement noise. In order to avoid this problem, it is imperative to supply the gas sample at atmospheric pressure on the analyzer inlet.
- By its principle, the analyzer has no long-time calibration or zero drift. Only sudden changes in pressure or sudden and abnormal changes in concentration of the other gases absorbing in this spectral area and punctually interfering in a cycle, can cause noise on the measurement.
- Fluid circuit is designed to be as symmetrical as possible.
- Tightness of the 3-way solenoid valve and selective filter should be checked periodically, as well as the pressure loss balance in I and I_0 measurement mode.
- It is important to replace the Ozone scrubber routinely once a year.

Current maintenance details of O342e analyzer are described in Chapter 4 of this manual.

SUMMARY:

- 1 – Ozone is reactive and very easily trapped. Consequently, any part change requires O₃ passivation time.
- 2 – Fluid circuit, solenoid valve and absorption medium must always be kept clean.
- 3 – Measurement is cyclical: pressure must be balanced on both channels, pressure loss check is necessary on both channels.
- 4 – Ozone scrubber must be replaced periodically by the user.

2.5. FLUID CIRCUIT DIAGRAM

Below is the O342e fluid circuit diagram.

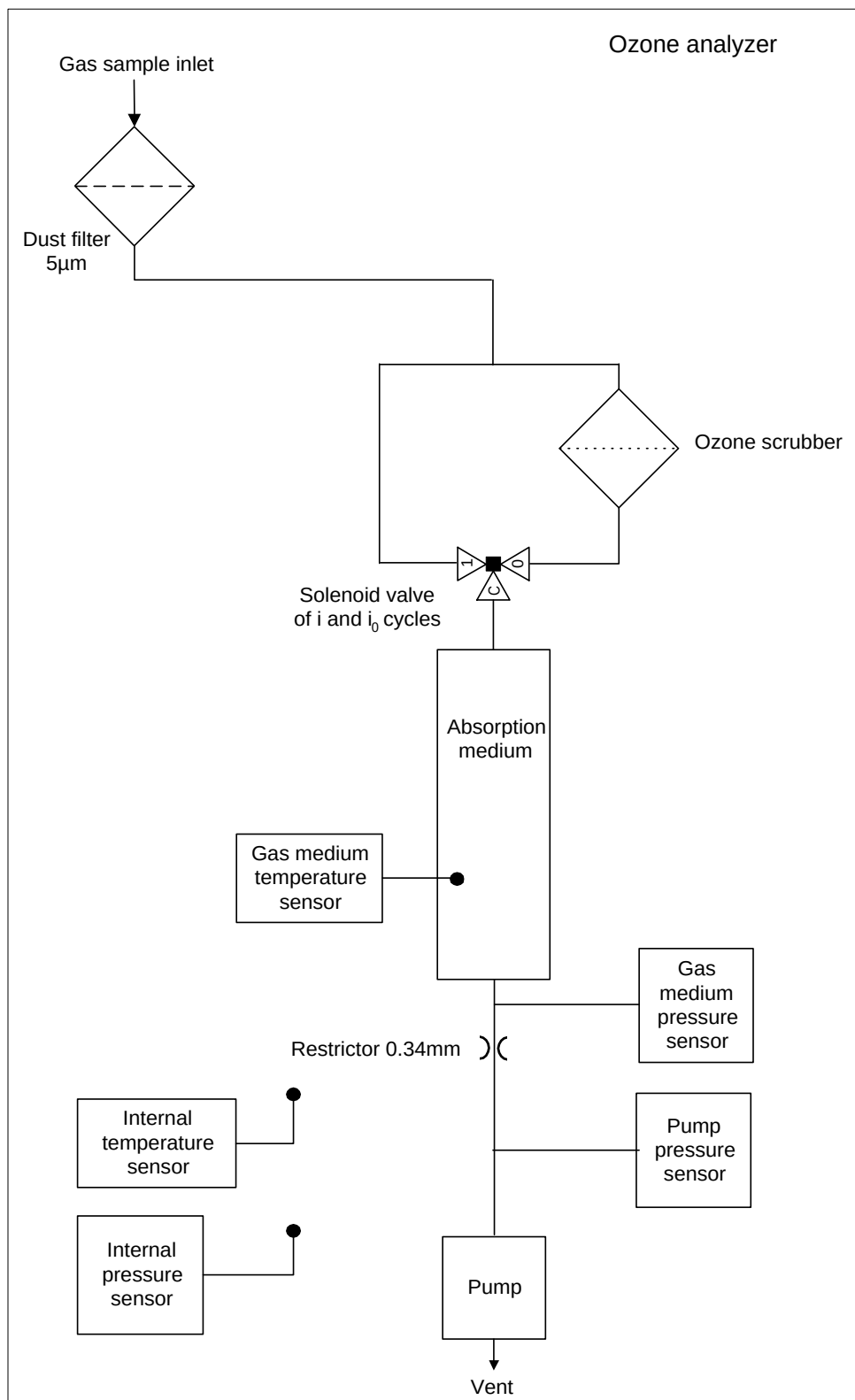


Figure 2-5 – O342e fluid circuit diagram

2.6. SIMPLIFIED FLOWCHART OF MAIN PROGRAM

The succession of the various sequences of the program is schematically represented below in Figure 2–6.

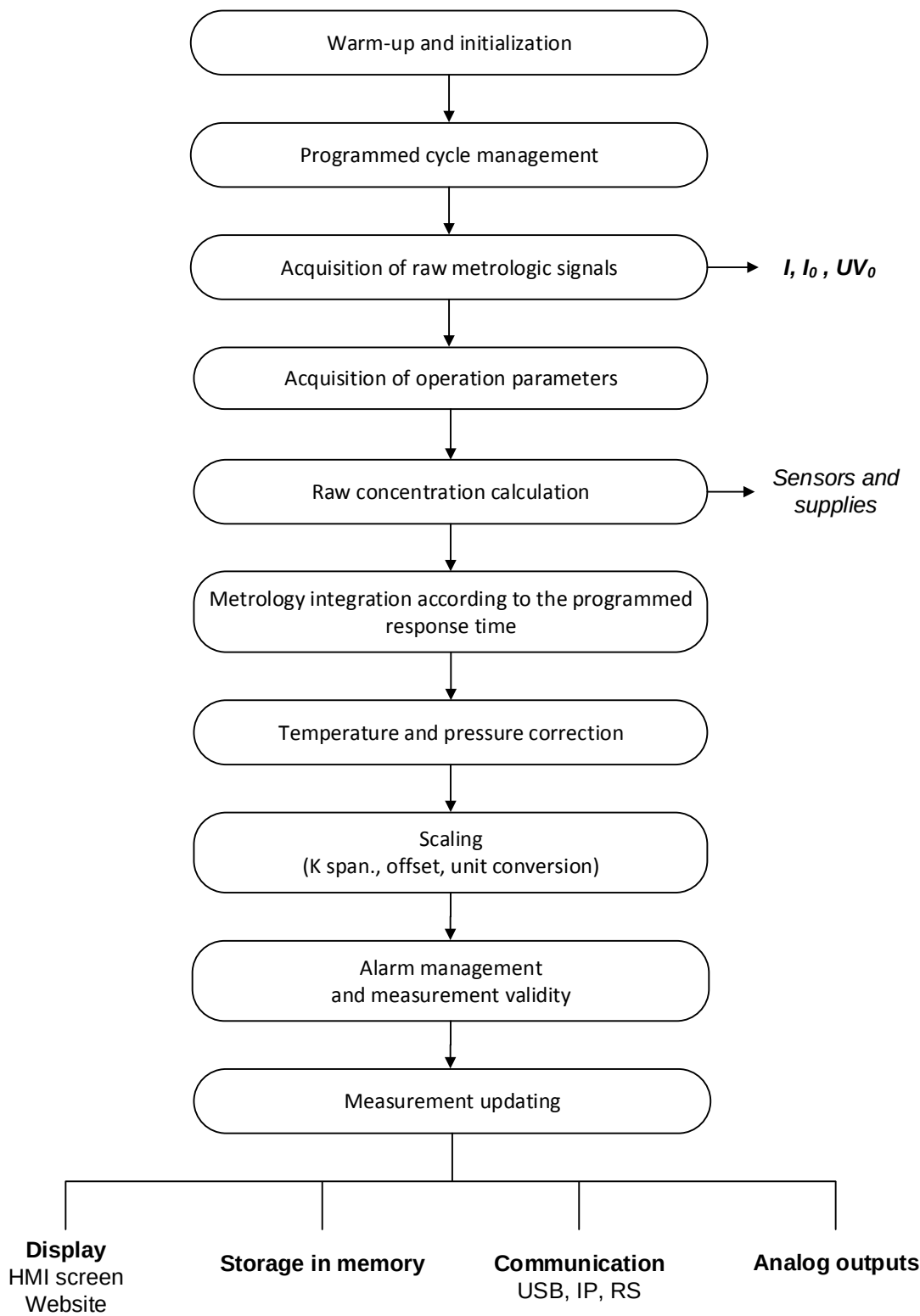


Figure 2–6 – Simplified flowchart of the main program

2.7. ELECTRONIC ARCHITECTURE

Communication between the ARM20 board and the other electronic boards (Measurement board, Interconnection board, Power supply board, Controller board, pressure sensor board....) follows the schematic presented in Figure 2–7.

The ARM20 board is a rapid calculation and interfacing (communication) board for the measurement modules of “e” series. It is provided in all analyzers and offers one Ethernet output (RJ45 socket), and three USB ports (one on the front panel, and two on the rear panel through the Interconnection board). For these analyzers, it is the central element for electronic and metrological operation, and the outward communication.

The Measurement board carries out measurement and operation parameter acquisitions. All metrological parameters are sent to the ARM20 board through a USB communication protocol.

The power supply board provides the analyzer internal voltages from the 24 V cut-out power supply, as follows:

- +24 V filtered and secured,
- +12 V, for LED and ARM20 board supplying,
- +5 V,
- +3.3 V.

The Controller board ensures the main following functions:

- It provides (+ 24V) various power control for the internal solenoid valve switchings.
- It allows temperature acquisition and control management of the various internal heating elements.
- It allows pressure signal acquisition from the various pressure sensor boards.
- It integrates one USB hub to ensure communications between the various boards.

The Interconnection board allows connections of all the connectors and sensors together.

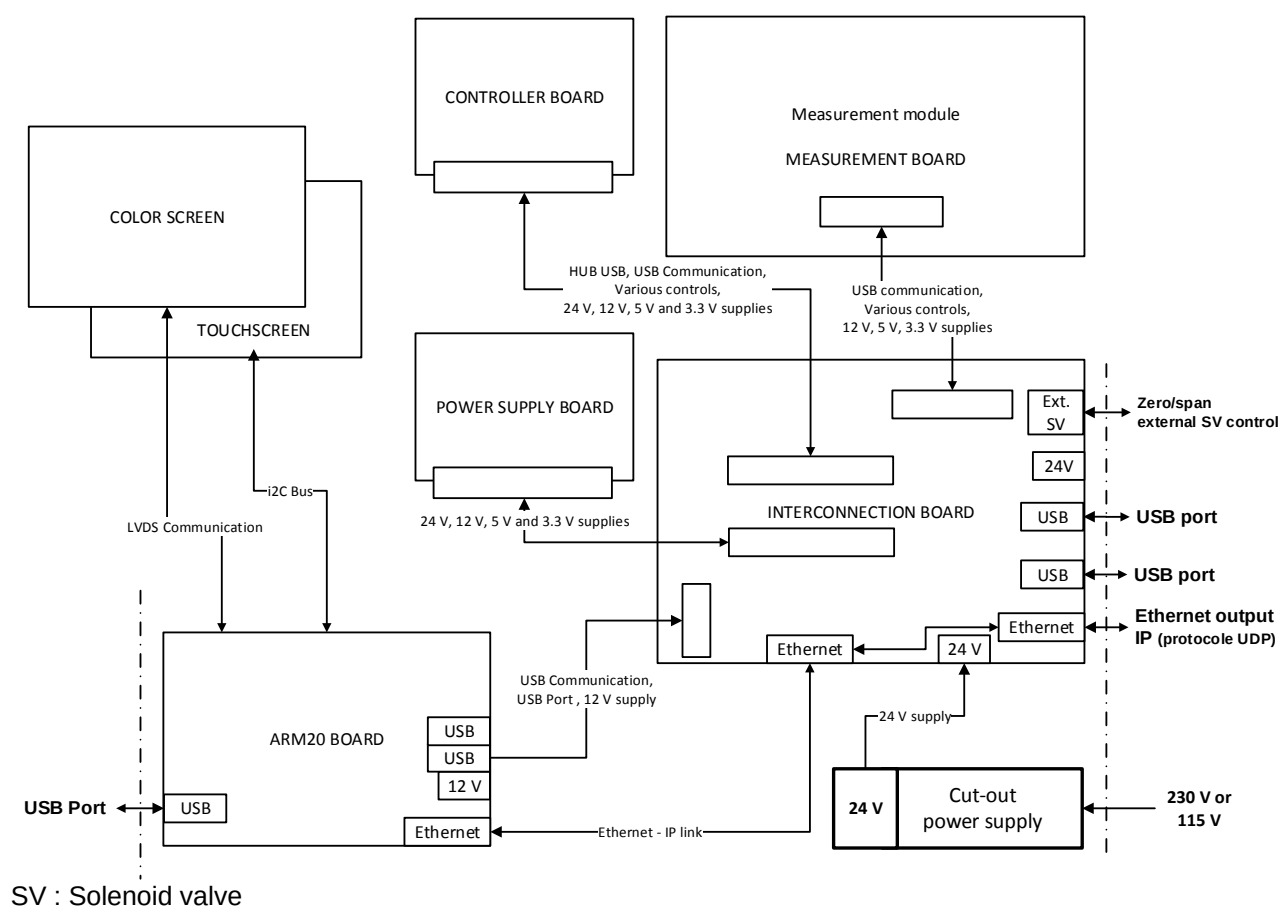


Figure 2-7 – Electronic architecture

2.8. NETWORK LINK AND USB PORTS

– Network link (Ethernet) :

One RJ45 connector is interfaced on the analyzer rear panel via the Interconnection board. Network communication (Ethernet) uses UDP protocol.

– USB ports:

Three USB ports are available in the “e” series electronic architecture.

CHAPTER 3

OPERATING INSTRUCTIONS

3.1.	INITIAL START-UP	3-3
3.1.1.	PRELIMINARY OPERATIONS	3-3
3.1.2.	STARTING UP THE UNIT	3-6
3.2.	PROGRAMMING THE O342E	3-13
3.2.1.	SCREEN AND KEYBORAD DESCRIPTION AND USE INSTRUCTIONS	3-13
3.2.1.1.	Definition of the control screen areas	3-13
3.2.1.2.	Contextual help functioning	3-14
3.2.1.3.	Virtual keyboard input	3-14
3.2.2.	OPERATION PARAMETER PROGRAMMING	3-16
3.2.2.1.	Field requiring character input	3-16
3.2.2.2.	Fields requiring parameter selection in a list	3-16
3.2.2.3.	OFF/ON status modification button	3-16
3.3.	BROWSING ERGONOMY	3-17
3.4.	ANALYZER FUNCTION DESCRIPTION	3-22
3.4.1.	STANDARD FUNCTIONS	3-22
3.4.1.1.	Analyzer controls	3-22
3.4.1.2.	Measurement channel display	3-24
3.4.1.3.	Measurement channel curve display	3-26
3.4.1.4.	Analyzer synoptic diagram	3-29
3.4.2.	ADVANCED FUNCTIONS	3-30
3.4.2.1.	Password management	3-30
3.4.2.2.	General configuration	3-31
3.4.2.3.	Automatic cycle configuration	3-32
3.4.2.4.	Language selection	3-33
3.4.2.5.	Find and display recorded averages	3-33
3.4.2.6.	Delete all recorded averages	3-35
3.4.2.7.	List of connected clients	3-35
3.4.2.8.	Diagnostic functions	3-35
3.4.2.9.	Set the analyzer in standby mode	3-38
3.4.2.10.	Set the analyzer in maintenance mode	3-38
3.4.2.11.	Advanced analyzer configuration	3-39

3.4.2.12. Automatic linearization of the flow	3–40
3.4.2.13. Airproof test	3–41
3.5. CALIBRATION	3–42
3.5.1. OVERVIEW OF CALIBRATION AND CONCEPTS	3–42
3.5.1.1. Zero air generation	3–43
3.5.1.2. Ozone standard generation	3–44
3.5.1.3. Description of a complete calibration apparatus	3–44
3.5.1.4. Preparation and verification of calibration apparatus	3–47
3.5.1.5. O ₃ concentrations assessment using a photometer	3–48
3.5.1.6. Internal valve for zero and span switching	3–49
3.5.2. CHECK OF ZERO AND SPAN POINT	3–49
3.5.2.1. Required Equipment	3–49
3.5.2.2. Procedure	3–50
3.5.2.3. Use of automatic cycles	3–50
3.5.3. TWO POINT CALIBRATION	3–51
3.5.3.1. Equipment required	3–51
3.5.3.3. Procedure	3–52
3.5.4. MULTIPOINT CALIBRATION	3–53
3.5.4.1. Equipment required	3–53
3.5.5. OVERVIEW	3–54
3.5.5.1. Procedure	3–54
3.5.6. INTERNAL OZONE GENERATOR	3–57
3.5.6.1. General operating principle	3–57
3.5.6.2. Description of the operating mode	3–57
3.5.6.3. Remarks on use of Ozone generator as calibration standard	3–57
3.5.6.4. Ozone generator configuration and starting-up	3–58
Figure 3–1 – Fluid and electrical connections	3–3
Figure 3–2 – Installation of internal ozone generator (if option available)	3–4
Figure 3–3 – Installation of the sample gas inlet	3–5
Figure 3–4 – Filtering columns	3–43
Figure 3–5 – Schematic diagram of a typical UV photometric calibration system (1)	3–46
Figure 3–6 – Schematic diagram of a typical UV photometric calibration system (2)	3–46
Figure 3–7 – Gas flow system with zero span solenoid valve	3–56
Figure 3–8 – Gas flow system with Ozone generator	3–60

3. OPERATING INSTRUCTIONS

3.1. INITIAL START-UP

The analyzer is checked before delivery. Calibration is verified at the factory.

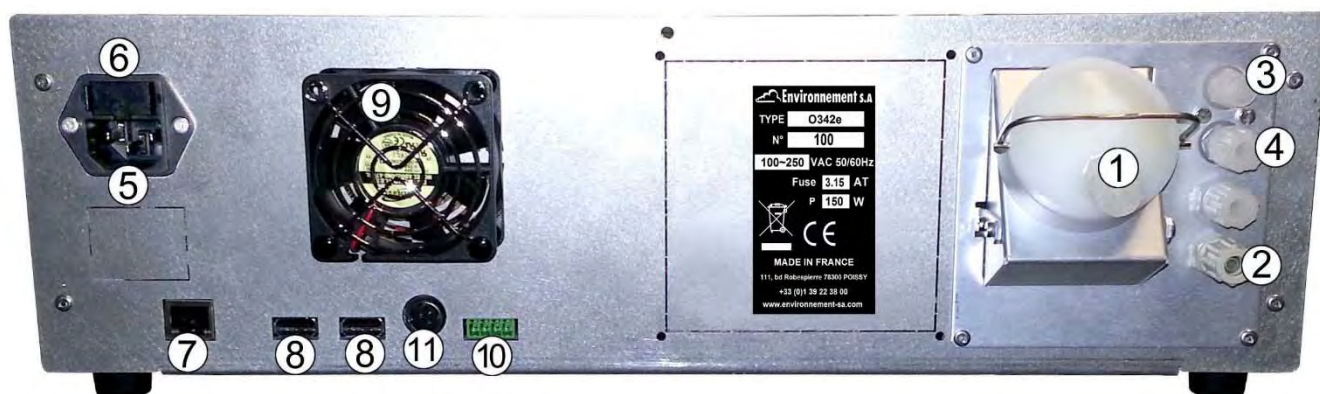
3.1.1. PRELIMINARY OPERATIONS

Refer to Figure 3–1.

It is necessary to leave a free space of at least 10 cm (4 in.) between the rear panel and the wall behind the analyzer to ensure correct ventilation of the analyzer.

Start-up involves the following preliminary operations:

- Visually examine the inside of the instrument to make sure no damage occurred during transport.
- Remove the caps from the gas inlet and outlet (1) (2) on the unit (set aside for future storage).
- Make sure a Teflon filter diaphragm is inside the dust filter (1), and then connect the 4 mm (I.D.) and 6 mm (O.D.) air sampling tube to the sample inlet (1).
- Connect the ETHERNET output (7).
- If necessary, connect accessories via USB ports (8),
- Connect the main power supply cable to a socket 230 V, 50 Hz + ground or 115 V, 60 Hz + ground.
- Connect the pump outlet (2) to the vent.



(1) sample inlet, (2) pump outlet, (4) zero inlet and (3) span inlet, (5) three-contact socket for main power supply, (6) general fuse, (7) Ethernet output, (8) two USB ports, (9) fan, (10) duplicate zero and span solenoid valves for the optional solenoid valve of external calibration, (11) 24 V power supply for ESTEL board option.

Figure 3–1 – Fluid and electrical connections

Analyzer equipped with the ozone generator option :

WARNING Always **switch off** the analyzer before installing the ozone generator.

The user must install the ozone generator as follow:

- Fix the internal generator in the analyzer frame, by using the screws (1),
- Connect the electrical flat cable to the interconnection board (J26),
- Disconnect the plug (Ø4) (2) from the SV block and connect the black Teflon tube,
- Connect the plug (Ø6) (3) to the span inlet,
- Connect the 4/6 mm tube (4) to the option inlet.

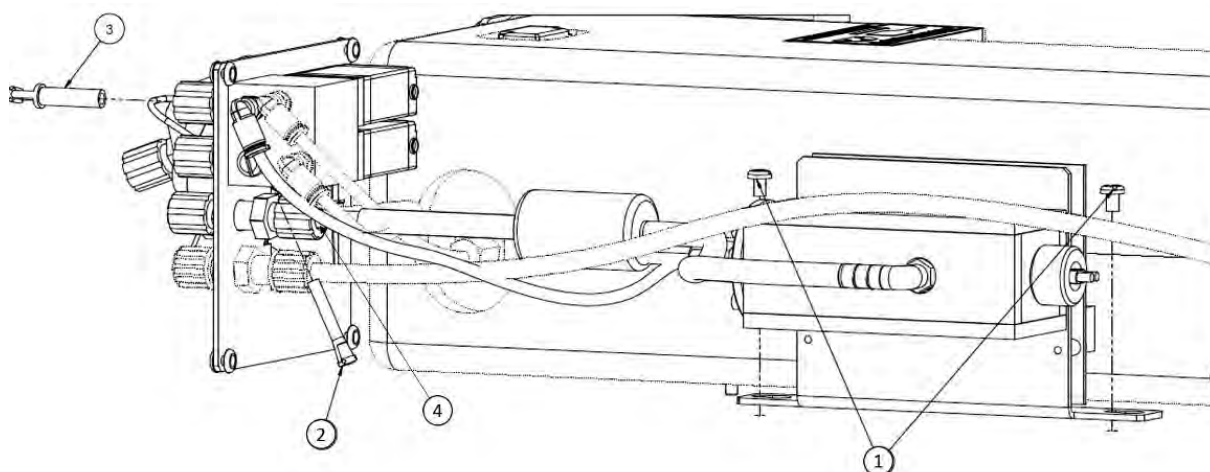


Figure 3-2 – Installation of internal ozone generator (if option available)

NOTE : In case of using an external ozone generator to calibrate the analyzer, it is necessary to connect it to the span inlet, and to seal the option inlet with the plug.



The internal ozone generator is fitted with a mercury lamp.
DO NOT THROW AWAY THIS MERCURY LAMP IN THE BIN.

Connect the activated carbon cartridge to the inlet option.

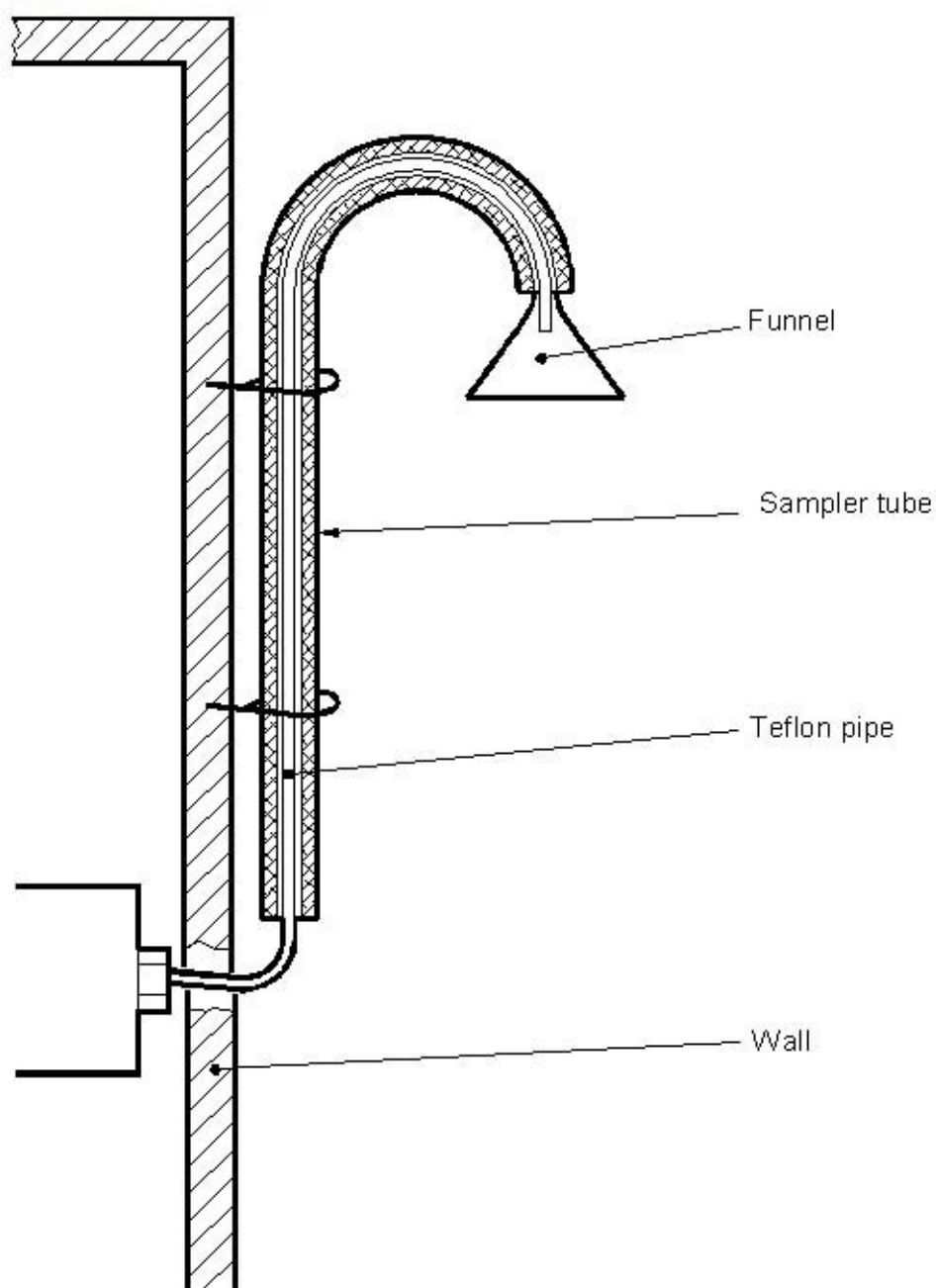


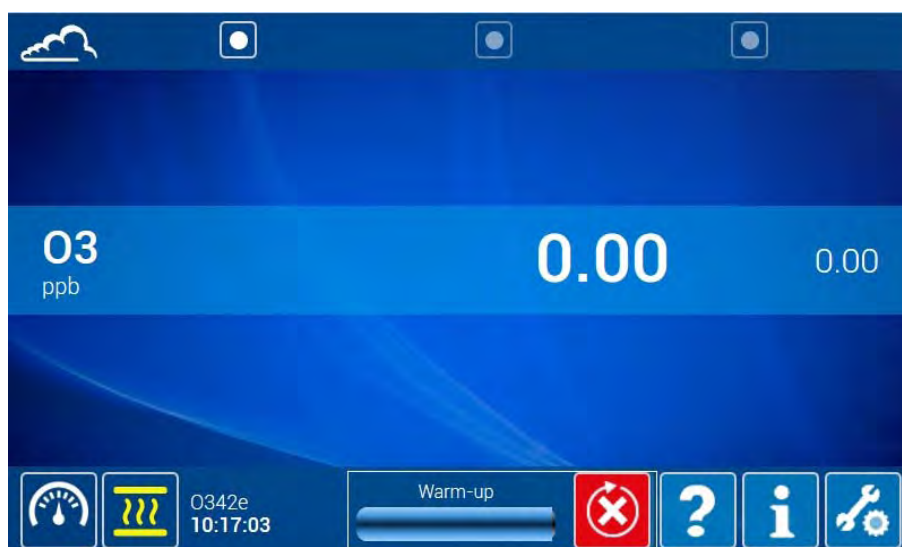
Figure 3-3 – Installation of the sample gas inlet

NOTE : Recommended height for sampling tap: 2.50 m
Maximum recommended length of Teflon sampling pipe: 6 m.

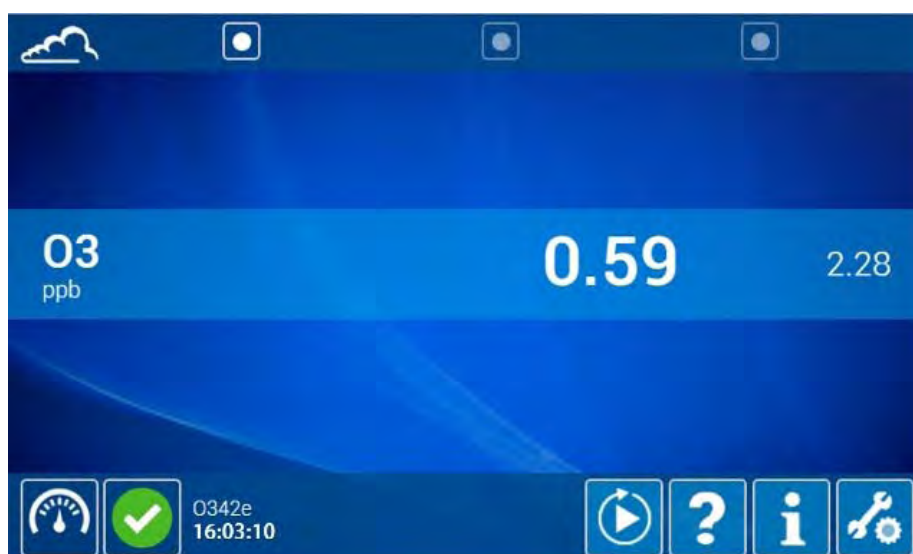
3.1.2. STARTING UP THE UNIT

Starting up the unit consists in carrying out, in the following order, the INSTALLATION / COMMISSIONING procedure described here-below.

1/ Press the ON/OFF push button located on the front panel. The analyzer starts running, it passes into warm-up cycle (duration of which is 1800 seconds maximum). The homepage is displayed: it allows to visualize the warm-up progress period.



The warm-up period is ended when all the metrology parameters are within the operational limits. The warm-up progress icon will disappear and the following homepage is displayed:

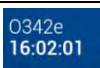


This screen is the analyzer standard function homepage. It contains additional pages that are displayed by activating the buttons  to , present at the top of the screen. These pages allow the measurement channel display, the « *Real-time graph* » of the instantaneous measurement channel values, and the « *Analyzer synoptic* » giving the main parameter values.

By using the touch screen mounted on the analyzer front panel: no password is required to activate the standard functions.

By using a remote PC: a password is required to be input in the pop-up that will appear. Enter the User password (see next page), and validate with .

Button and icon definitions specific to these screens

	Displays the selection buttons of the analyzer fluid inlets:  sample inlet,  zero inlet,  span inlet.
	Warm-up status icon: indicates that analyzer is warming-up.
	Normal operation status icon: indicates that the analyzer operates normally.
	Information area: indicates the analyzer model and time.
	Allows to visualize the warm-up progress status
	Stop button of the current cycle
	Displays the starting buttons of the analyzer cycles:  starts calibration cycle,  starts zero reference cycle.
	Displays the contextual help
	Displays analyzer information panel.
	Gives access to the advanced analyzer functionalities.
	Symbolizes the analyzer password during operation and displays the pop-up to change password if necessary.

Access to functions depends on the hierarchical level of the user's password. There are three hierarchical levels : User, Advanced, Expert. **The factory passwords delivered with the analyzer are:**

User level	12345	
Advanced level	78300	
Expert level	00007	

On the initial starting-on, it is advised to change the factory passwords. To do that, activate  to display the password management pop-up, enter the new passwords and activate  to validate and close the pop-up.

 closes the pop-up without validating.

2/ Access to the advanced analyzer functionalities

By using the touch screen mounted on the analyzer front panel: touch the button  in order to open the User password input pop-up (1). Touch the field (2) to display the QWERTY keyboard (3) for English, and AZERTY for French, in the half below part of the screen.

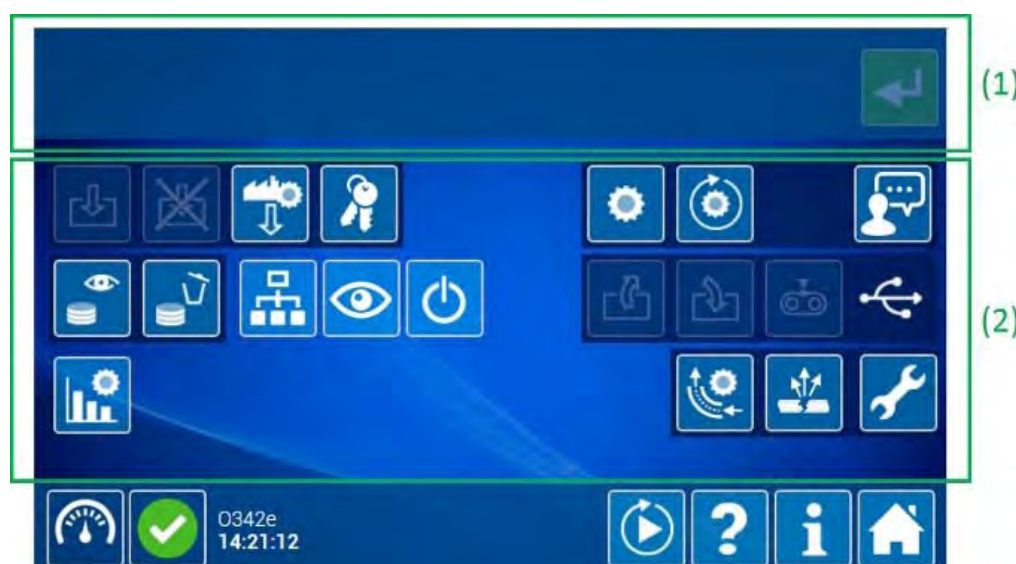
Touch  in order to switch alphanumeric keyboard to numeric keyboard. Enter the User password and validate with : the pop-up and the keyboard closes, and the advanced functionalities homepage opens.

By using a remote PC: do the same with the PC keyboard and mouse.



The button  allows to close the pop-up without input validation.

The home page of the advanced analyzer functionalities is presented below:



The accessible functions for the User appear in white with a blue background in zone (2). The function access to the functions depends of the password's level that was entered.

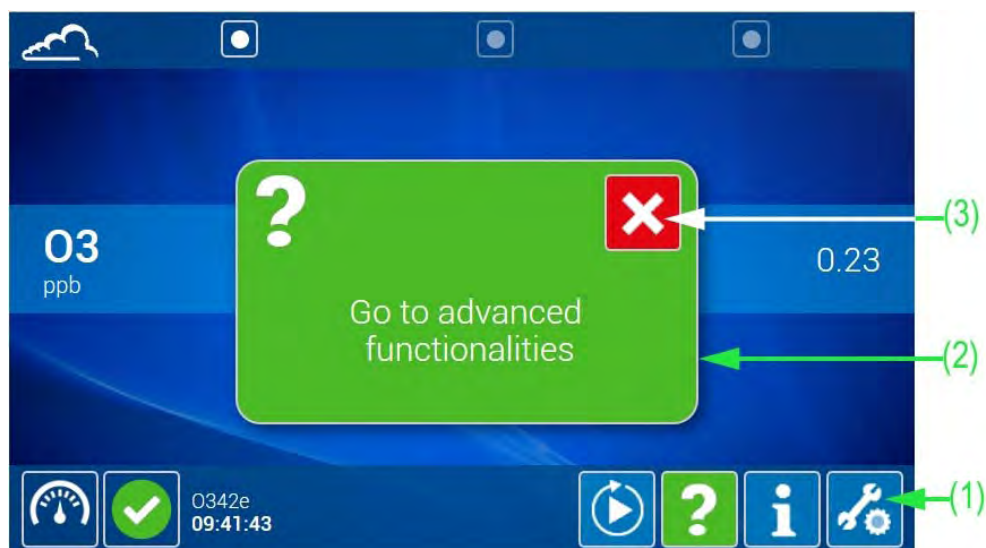
Each accessible function is activated by contact, it then changes color to green. The name of the selected function will appear in zone (1), and the button  is highlighted . By double clicking on the function's icon or by using the button , the user can access to the corresponding screen.

3/ Contextual help functioning

The contextual help is directly accessible through all the screens. The user activates it by touching / clicking on the button  that will become green . Then, the user touches / clicks on the buttons or on a parametric screen part in order to display a pop-up that describes the use of the selected function or button. The pop-up is closed by clicking on .

It is again necessary to touch / click on  to deactivate the contextual help.

See the example below:

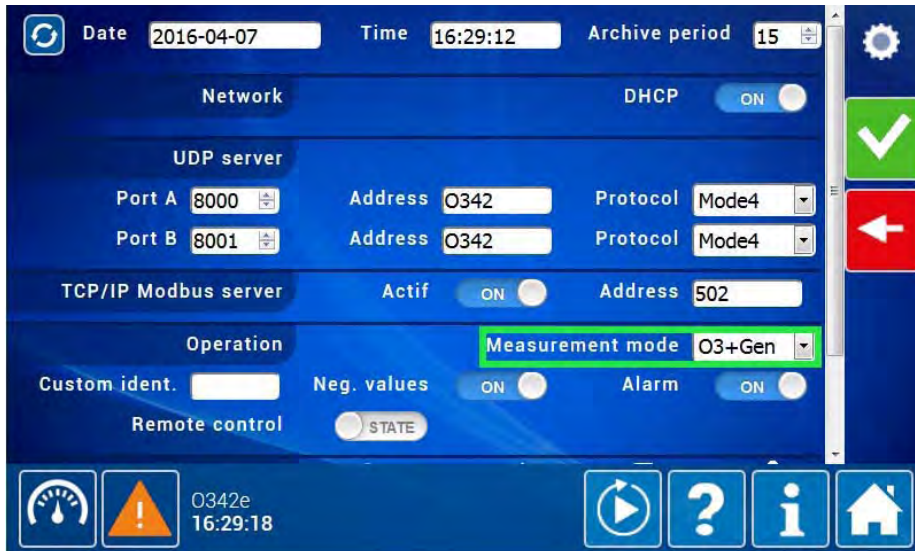


(1) Activated button (highlighted), (2) pop-up message explaining the function of the activated button, (3) icon to be used in order to close the pop-up.

4/ Ozone generator configuration and starting-up

After having installed the ozone generator in the unit, proceed in the following order :

- From the home page, touch/ click on  to access the advanced functions, then on  to display the « General configuration » screen and configure the « Measurement mode » field to « O3+Gen », as shown below (see green square).



- Valid the action by touching/clicking on the button .
- Close the span inlet on the analyzer rear panel to avoid dilution of the generated ozone concentration.
- Wait until the ozone generator temperature reaches 50°C. Check this temperature in the « Synoptic » screen (10) : this screen is displayed in touching / clicking on , then on the last button  in the top right part of the screen, which is then activated to  (see the green arrow below).



- When the temperature of 50° C is reached, touch/click on  to return to the advanced functions, then on  to display the "Advanced analyzer configuration" screen. The "Ozone Generation", "Concentration 1" and "Concentration 2" fields appear at screen bottom. The "Concentration" fields allow to enter the desired concentration between 30 and 200 ppb.



- Enter the desired concentration 1 and validate by clicking/touching . Then press  to start the ozone generator calibration. During calibration, the analyzer switches into maintenance mode for about 25 minutes. Do the same with concentration 2 if necessary.

When this calibration is finished, the user must update the "Ozone Generation" field: to do this, he click/ touch  to exit the screen then click/touch  to return again to the "Advanced analyzer configuration" screen. The concentrations 1 and 2 are refreshed in the "Ozone Generation" field (see green frame). The user will select them when a new calibration is needed.

NOTE : The orange-written values are obtained from the previous ozone generator calibration and are reminded here.



- Switch on the generator by clicking/touching  to select the span inlet: Ozone is generated 2 minutes later.

3.2. PROGRAMMING THE O342E

3.2.1. SCREEN AND KEYBORAD DESCRIPTION AND USE INSTRUCTIONS

The touch screen mounted on the front panel is alternately used to visualize the control screens and the virtual input screen allowing input and modification of the analyzer parameters. The control button activation is done by touching.

When the analyzer is connected to a remote computer, the parameters are modified using the computer keyboard, and the control buttons are activated by clicking with the mouse.

3.2.1.1. Definition of the control screen areas



The areas of control screens are defined as follow:

(1)	Browsing area proper to the current screen: when the screen has more than one page, activate the button to allow to display the corresponding page.
(2)	Measurement or configuration area. It displays the measurement parameters (gas, value, units...) or the configurable parameters associated with the selected menu.
(3)	Manual control area: inlet fluid selection and manual launching of calibration and zero reference cycles. Information area and advanced functionality access.

Control and information area (3) description

This area is present in all the screens.

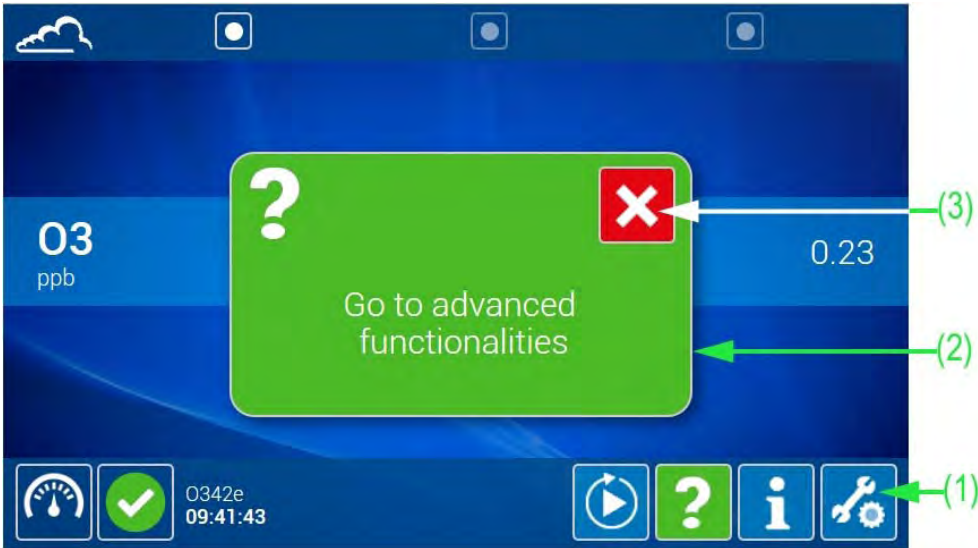
	Activating this button displays the buttons allowing to manually select the fluid inlets of the analyzer: sample inlet, zero inlet, span inlet.
	The icon displayed in this place indicates the currently operating status of the analyzer. The various status icons are: normal operation, warm-up, control , alarm , network disconnected , maintenance , stand-up .
	Information area indicating model and current time of the analyzer.
	Activating this button displays the buttons allowing to manually start the analyzer cycles: starts calibration cycle, starts zero reference cycle.

	Activating this button displays the contextual help.
	Activating this button displays the analyzer information panel.
	Activating this button gives access to the advanced analyzer functionality home page.

3.2.1.2. Contextual help functioning

The contextual help is directly accessible through all the screens. The user activates it by touching / clicking on the button  that will become green . Then, the user touches / clicks on the buttons or on a parametric screen part in order to display a pop-up that describes the use of the selected function or button. The pop-up is closed by clicking on .

See the example below:



(1) Activated button (highlighted), (2) pop-up message explaining the function of the activated button, (3) icon to be used in order to close the pop-up.

3.2.1.3. Virtual keyboard input

The virtual input screen is used to modify the control screen parameters displayed from the front panel touchscreen. This screen is displayed instantly in the lower half-part of the screen when the user touches an input field to be modified. Two touch-button keyboard types are available: numeric and alphanumeric QWERTY. They are displayed depending on the input field nature: numeric keyboard if the field is numeric type, alphanumeric keyboard if the field requires number and letter inputs.

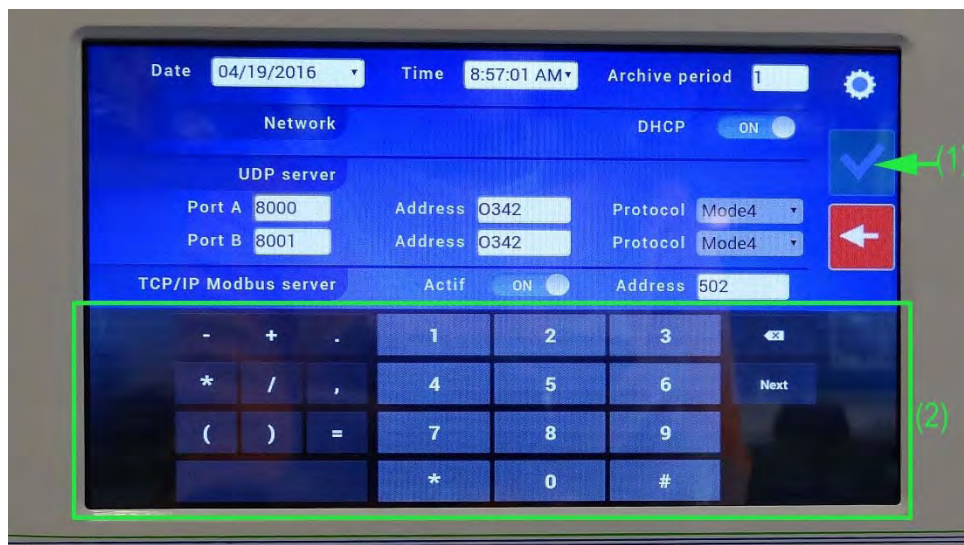
The examples below are showing the keyboard display in the « *General configuration* » screen. The keyboard operation is identical for all screens.

When the user presses down a key, the button (1)  at the top right corner is activated and highlighted to . It is used to validate the whole inputs done in the screen, and to close the screen after.

NOTE : Field inputs are indexed.
These keyboard keys work like Android system keys found on tablets and smart-phones.

The numeric keyboard:

It is displayed in the area (2) of the screen below:



(1) button for input validation, (2) numeric keyboard with touchscreens.

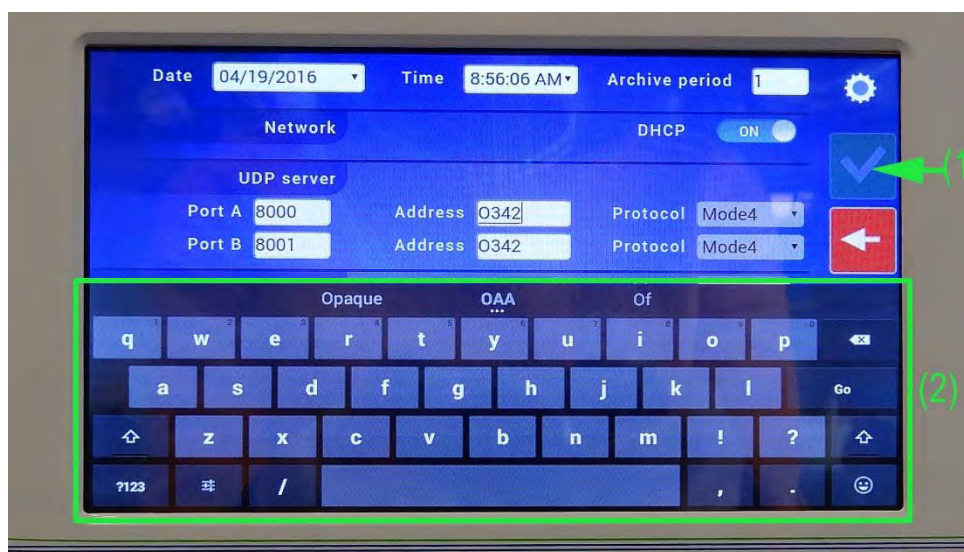


: clears digits during input.

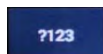


: used to move from an input field to another according to its indexation.

– The alphanumeric keyboard:



(1) button for input validation, (2) alphanumeric keyboard with touchscreens.



: switches from alphanumeric to numeric keyboard.



: switches from numeric to alphanumeric keyboard.



: switches from numeric to symbol keyboard.



: validates input of the considered field.

3.2.2. OPERATION PARAMETER PROGRAMMING

The standard functionalities cannot be modified. The user can only modify the advanced functionality parameters.

3.2.2.1. Field requiring character input

- Using the front panel touchscreen :

Touch the input field to be modified: if the field is alphanumeric, the QWERTY keyboard is displayed, if the field is numeric, the numeric keyboard is displayed.

Input with the touch keys, and validate with . The keyboard closes and the input field is displayed properly filled in the control screen.

- From a remote PC:

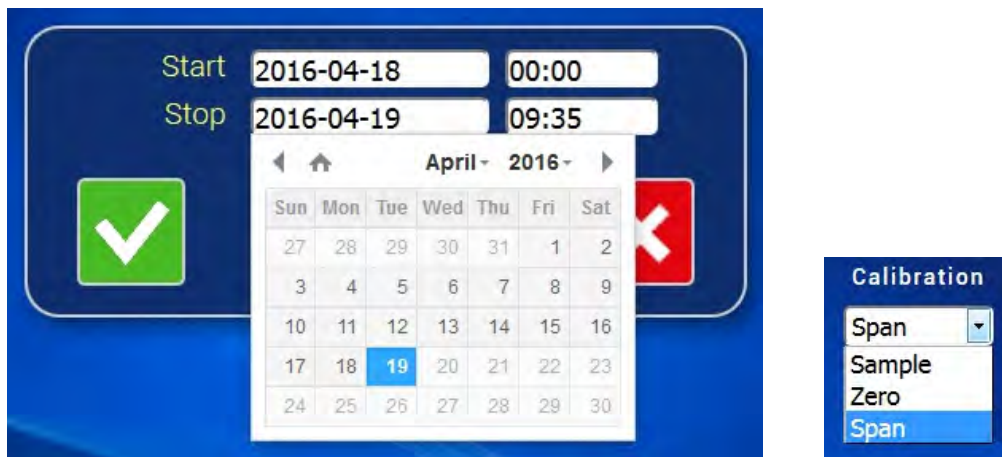
Use keyboard and mouse of remote PC to input, modify and validate the field characters.

3.2.2.2. Fields requiring parameter selection in a list

Many cases are possible, depending on the input type.

Touch the field to be modified. The parameter or value list to be selected is displayed: the current parameter or value is displayed white in blue background. Touch (or click on with the remote computer mouse) the new parameter or value to be selected. The field is displayed white in blue background.

Validate with . The list closes, and this new parameter or this new value replaces the previous.



3.2.2.3. OFF/ON status modification button

To switch OFF to ON, touch (or click on) the white button. The ON field becomes white in blue background. And vice versa.



NOTE : When many modifications are performed in the same screen, the user must ALWAYS validate with the key  to take into account the whole modifications.

3.3. BROWSING ERGONOMY

The O342e home page is the following:



By default, the home page displays the O₃ measurement channel.

From the front panel touchscreen, the homepage allows free (i.e. without password) and direct (i.e. the first browsing level) access to the standard functionalities of the analyzer.

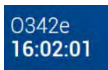
Activating the buttons of top screen  to  allows to drag display sideways to view the « *Real-time graph* » of the instantaneous values of the selected measurement channels, and the « *Analyzer synoptic* » indicating the main operation parameter values.

From a remote PC, the page below is displayed first:

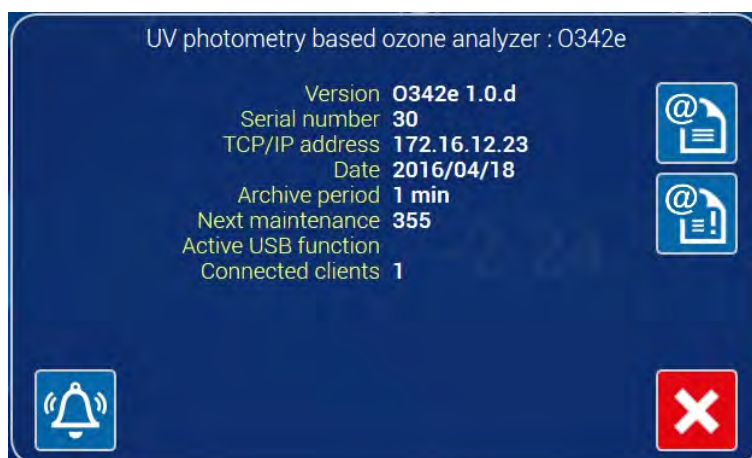


The user enters the password in the field (1) and validates with  to open the home page.

Standard functionalities are the followings:

	Display of fluid inlet selection buttons.
	Manual selection of sample inlet.
	Manual selection of zero inlet.
	Manual selection of span inlet.
	Display of analyzer status.
	Indication of analyzer model and time.
	Display of analyzer cycle launching buttons.
	Manual start of calibration cycle.
	Manual start of zero reference cycle.
	Display of contextual help.
	Display of analyzer information panel.
	Access to the advanced analyzer functionalities.
	Access to homepage.

The information panel displays the software version, analyzer serial number, TCP/IP analyzer address, current date programmed in the analyzer, memorization period for data archiving, remaining days up to next maintenance, if YES or NO an USB function is active on the analyzer, the number of connected clients.



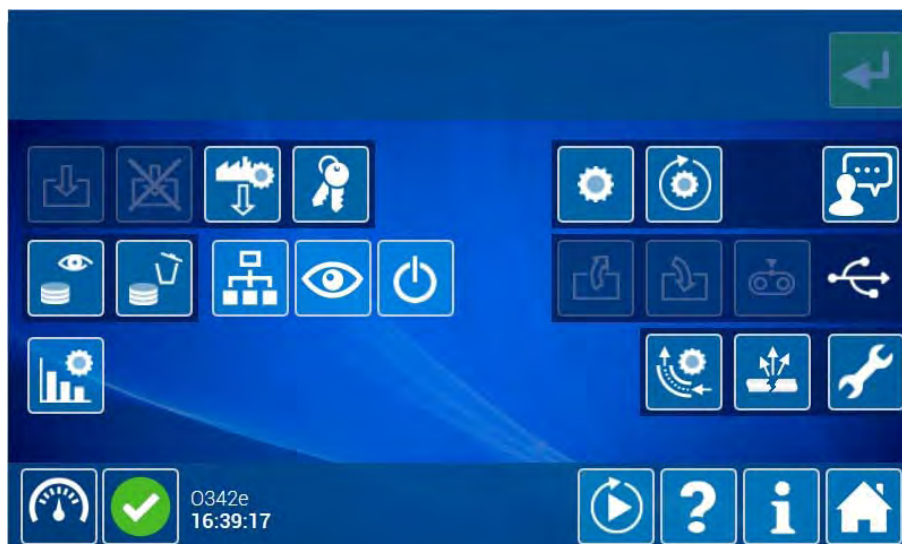
Icon and button definition specific to the information panel

	Means that the analyzer is equipped with a WIFI key for remote connection from a tablet or a smartphone. This icon goes with WIFI TCP/IP identifier and address required for connection.
	Allows to email analyzer status (mux signal values).
	Allows to email recorded events (history)
	Sound signal to locate the corresponding analyzer when not fitted with screen.
	Closes the information panel.

The advanced function access is selectively assigned to password bearers only. There are three possible authorization levels for passwords: user level, advanced level, expert level.

The user-accessible functions are white in blue background. They depend on the hierarchical level of the assigned password.

The access page to the advanced functions is the following:



The analyzer advanced functions are the followings:

	Undo last upgrade
	Restore factory settings. It is recommended to backup before executing this operation
	Passwords management
	General configuration (Address, Protocol, Measurement channels...)
	Automatic cycles configuration
	Language selection
	Find and display recorded averages
	Delete all recorded averages (Warning ! this operation can't be cancelled)
	List of connected clients
	Diagnostic functions (alarm, input / output, mux...)
	Set the analyzer in standby mode
	Advanced analyzer configuration
	Automatic linearization of flow

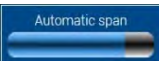
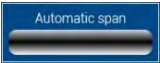
	Test for leaks
	Set the analyzer in maintenance mode
	Return to standard functionality home page
	Give the USB characteristics (free and total memory) when it is connected to the analyzer.
	Backup of configuration and software on USB when it is connected to the analyzer.
	Restore software and configuration from USB when it is connected to the analyzer
	Record instantaneous measurement on USB when it is connected to the analyzer

3.4. ANALYZER FUNCTION DESCRIPTION

3.4.1. STANDARD FUNCTIONS

3.4.1.1. Analyzer controls

The analyzer controls are the followings:

Controls	Function description
 Selects sample inlet	This function switches the analyzer to the sample gas inlet. The gas is continuously sampled through the dust inlet filter. This mode can be interrupted at any time by manually selecting another gas inlet (zero or span). The unit is selected in the « <i>Advanced analyzer configuration</i> » screen.
 Selects zero inlet	This function switches the analyzer to the external zero gas inlet. This operation allows stability checking and zero drift to determine the need to launch a zero reference cycle or program its repetition period. An activated carbon filter or zero air must be connected to this inlet.
 Selects span gas inlet	This function switches the analyzer to the span inlet. The measured span gas value, possibly increased by a programmed offset, is displayed on the screen. This operation allows checking the stability and the span drift to determine the necessity of launching an auto-span cycle or programming its repetition period. An external calibrated span must be connected to this inlet.
 Launches a calibration cycle	This function launches manually an automatic span cycle. The user activates  and views cycle progression in the scroll bar . At any time, it is possible to abort the current cycle by touching , or stop and validating the current cycle by touching . The current cycle stop results in « K » value modification. During an automatic calibration cycle, the analyzer adjusts its span factor K automatically to equal its reading value (minus the programmed offset) with span gas concentration. The span gas concentrations are programmable in the « <i>Advanced analyzer configuration</i> » screen. Adjustment of the cycle duration takes place in the « <i>Duration</i> » field of the « <i>Automatic cycle configuration</i> » screen. The cycle is finished when the scroll bar is completely grey, , it closes automatically.



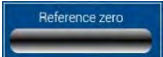
Launches a zero reference cycle

This function launches manually an automatic zero correction cycle to adjust the zero drift.

The user activates  and visualizes cycle progression in the scroll bar



At any time, it is possible to abort the current cycle by touching , or stop and validating the current cycle by touching .

The cycle is finished when the scroll bar is completely grey, , it closes automatically.

3.4.1.2. Measurement channel display

By default, the homepage displays the compound channel to be measured:



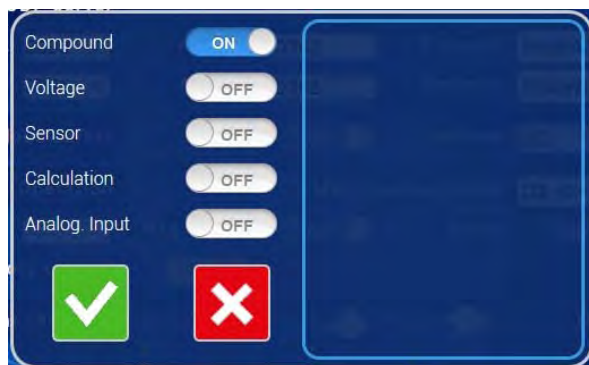
To display additional measurement channels, it is first necessary to select them in the « *Measurement channel* » section of the « *General configuration* » screen shown below (see outlined frame with arrow):



Touch button definitions specific to this section:

	Deletes the selected measurement channel
	Moves upwards the selected measurement channel
	Moves downwards the selected measurement channel
	Adds a measurement channel

To select additional channels, activate and open the selection pop-up. There are five channel groups: compound, voltage, sensor, calculation, analog input.



Then, to activate  to  in order to display the corresponding group elements, touch/click on the elements to be selected : they become sky-blue highlighted.

Then, activates  to validate this selection and close the pop-up. Do the same for the other channel groups: it is possible to select the whole available channels. The selected measurement channels take place in the numbered fields (see outlined frame with arrow):



When selection is finished, activates  to definitively validate the whole selection, then activates  to display the measurement channels in the home screen (home page):



Three measurement channels are displayed per page. To display the whole selected channels, the GUI (graphic user interface) adds as many pages as necessary. These additional pages are symbolized by the buttons  and  displayed at the screen top (see the outlined frame with arrow). The user clicks-on/touches these buttons to display pages successively.

3.4.1.3. Measurement channel curve display

The user views the measurement channel graphs by touching / clicking-on the next-to-last button in the browsing bar at screen top (see outlined frame with arrow). If no channel was previously selected, the following message is displayed:



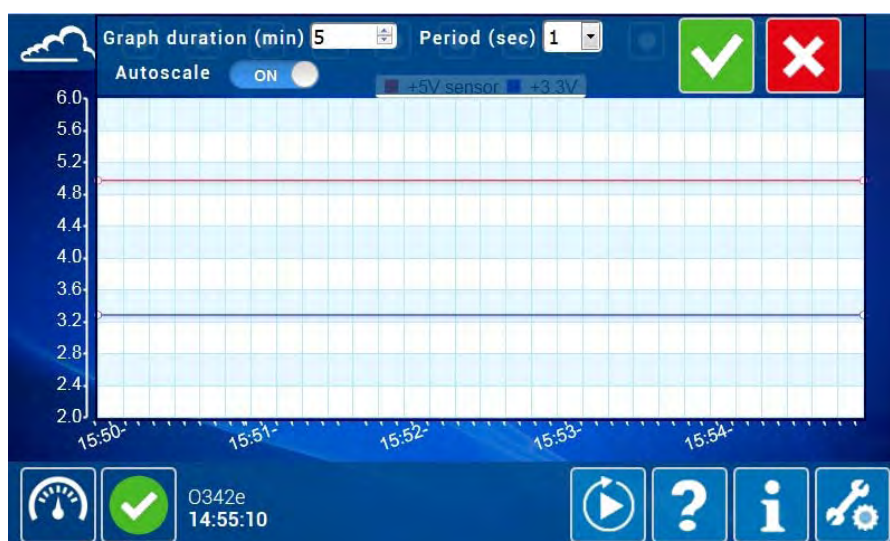
Touch / click-on the channel name to select: they change color and are displayed in green, as shown in the screen below:



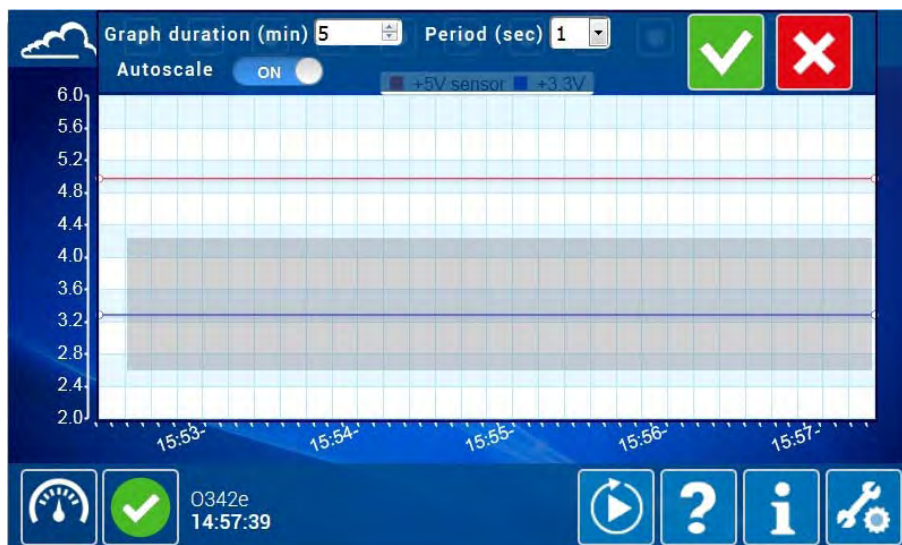
Go back to the next-to-last page to display the corresponding curves:



The Y-axis scale setting is automatic and adjusted in real time for current display optimization. Maximum six curves can be displayed simultaneously. By touching / clicking in the graph, the user activates the zoom function with the manual scale, as shown in the screen below:



The user selects the screen zone he wants to zoom:



Then, he activates  to validate: the selected zone is zoomed-in in the screen.



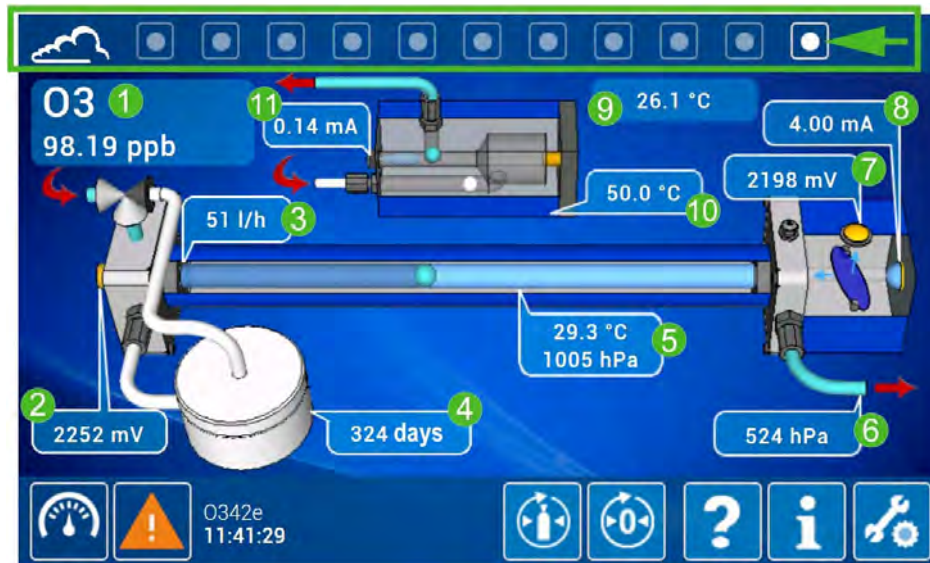
To exit zoom function, the user clicks in the screen two times, then he activates the action with .

This zoom function is available for X-axis and Y-axis.

Moreover, the user can deactivate the auto scale and he can set the desired Y-axis scale.

3.4.1.4. Analyzer synoptic diagram

The user views the synoptic diagram by clicking-on / touching the last button in the browsing bar at screen top (see outlined frame with arrow). This screen displays the entire fluid circuit and the significant operating parameter values.

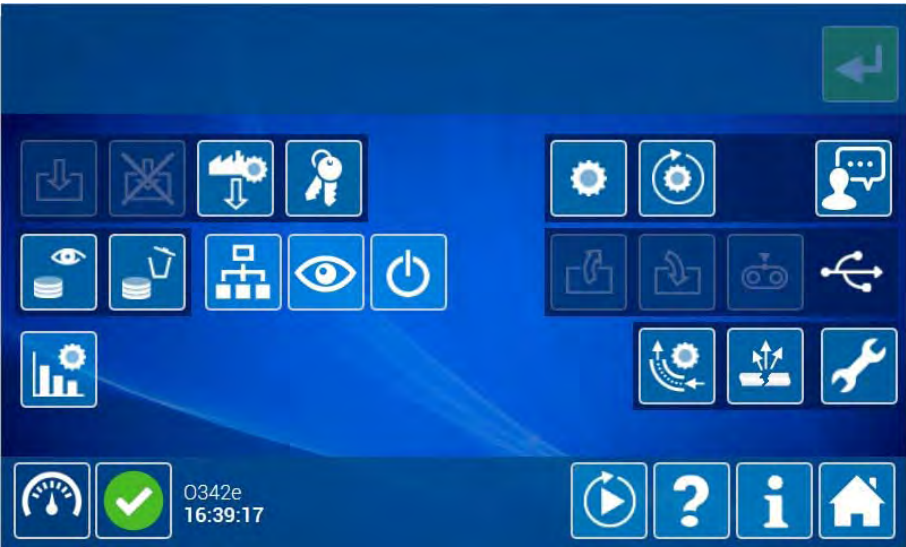


Screen legends are as follows:

- (1): instantaneous measurement of ozone concentration
- (2): UV measure signal
- (3): sample flow rate
- (4): ozone scrubber lifespan (in days),
- (5): absorption medium pressure (hPa) and temperature (°C)
- (6): pump pressure
- (7): UV reference signal
- (8): LED current
- (9): internal temperature of the analyzer
- (10): ozone generator (option) temperature
- (11): current of ozone generator lamp (option)

3.4.2. ADVANCED FUNCTIONS

The user accesses the advanced functions homepage by touching / clicking-on .



The advanced functions can only be accessed by authorized users with password. Three different password levels are available to authorize selective access: user level, advanced level, expert level. The whole advanced functions authorized to users with expert level password are described below.

3.4.2.1. Password management

The « expert » user only can activate this function by touching/double-clicking on  to display the password management pop-up.



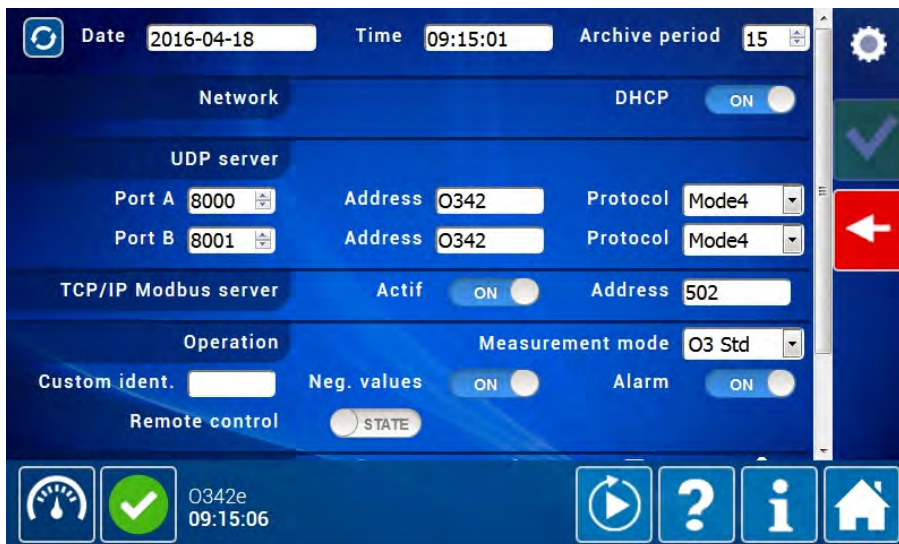
To modify passwords, the user fill-in the corresponding fields, then touch/click-on  to validate modifications and close the window.

Touch/click-on  to close the window without validating.

3.4.2.2. General configuration

This screen is accessed by touching/clicking-on .

This screen is used to configure the addresses and protocols necessary for communication and the measurement channels.



DATE and TIME fields: they indicate the current date and hour of the analyzer. They are modifiable with the « expert » level password only.

 : This icon is displayed when the analyzer is connected to a PC or a tablet. By clicking on it, the user updates the date and hour of the analyzer with the current date and hour of the PC or the tablet.

ARCHIVE PERIOD field: it indicates the analyzer archive period. It is modifiable with the « expert » level password only.

DHCP field: it is used to activate the automatic network configuration.

When ON, it allows to connect the analyzer to a TCP/IP network whose address is generated by a DHCP server.

When OFF, the ADDRESS, MASK and GATEWAY fields are displayed to allow the user to define the TCP / IP address, gateway and mask necessary for connection. This can be done the « expert » level password only.

WIFI/WPA field: it indicates the WPA encryption key of WiFi. It is modifiable with the « expert » level password only (8 to 63 characters, 0-9 and A-F authorized).

UDP SERVER section: it allows configuration of the two available UDP ports, Port A and Port B.

- Port number can be fixed within the range 1000 to 9999.
- Address is used only for Mode4 (four alphanumeric characters) and JBUS (number 0000 to 0255).

They are modifiable with the « expert » level password only

OPERATION section:

- NEG. VALUES field ON/OFF indicates if negative values are authorized or not.
- ALARM field ON/OFF enables or disables the alarm management.
- MEASUREMENT MODE field is displayed when an ozone generator fits the analyzer. It is used to select either the standard mode or the measurement mode with ozone generator.

MEASUREMENT CHANNEL section: it is used to configure the measurement channel display.



3.4.2.3. Automatic cycle configuration

This screen is accessed by touching/clicking-on .

This screen allows configuration of the four periodic cycles of the analyzer: Zero, Span, Zero-Ref., and Calibration. These cycles can be triggered in automatic or manual mode.



INLET field: allows selecting the channel used for the cycle (Zero, Span, Ref-Zero, Calibration).

PROGRAMMED INLET field: when ON, the analyzer uses the programmed inlet when a cycle is launched manually.

REMOTE CONTROL field: allows triggering cycle on a remote-control input (when optional ESTEL/SOREL board(s) is/are available...).

CYCLIC field: enables or disables cycle triggering in automatic mode.

DURATION field: allows fixing purge time.

PURGE DURATION field: it fixes purge time. Purge is applied before or after the cycle depending on the analyzer.

PERIOD field: allows fixing periodicity of automatic triggering.

START HOUR field: allows fixing start hour to trigger cycles automatically. If several cycles are in automatic mode, they will be triggered in the following order: Zero-Ref, Calibration, Zero, and Span.

AT STARTUP field (optional): allows to force a Zero-Ref cycle triggering at analyzer startup (when warming-up cycle is finished).

3.4.2.4. Language selection

The available languages are English and French.

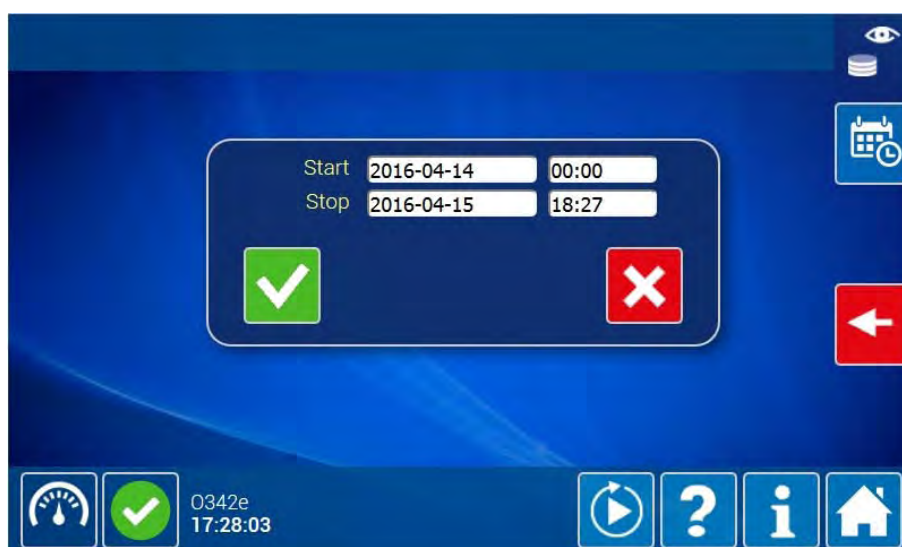
The user touches/double-clicks on  to display the language selection pop-up. The flag surrounded by a red frame indicates the language in use. The user touches / clicks-on the other flag to select the desired language and close the pop-up.

The message « Language selection » is displayed at screen top when English is selected, and « Choix de la langue » when French is selected.



3.4.2.5. Find and display recorded averages

This screen is accessed by touching/clicking-on . It allows to set the data period the user wants to watch, and to display these data.



The user touches/clicks-on  to validate the selection, close the pop-up and display the screen to visualize the recorded averages for the period. Touch/click-on  to close the pop-up without validating modifications.



Button definition specific to this screen



: exports the displayed data to a TXT file.



: opens the selection popup of dates and times period the user wants to view.

Icon meaning



: indicates an average stored with an alarm status. Alarm details are displayed by touching this icon.



: indicates an average stored with a warning status. Warning details are displayed by touching this icon.



: indicates an average stored with a "calibration" status. Calibration details are displayed by touching this icon (zero, span).

3.4.2.6. Delete all recorded averages

Touching/ double clicking-on this button  delete all the recorded averages.

WARNING : **this action is irreversible.**

3.4.2.7. List of connected clients

The user touches/double-clicks on  to display the pop-up giving the IP addresses and inactivity duration of the currently connected clients.

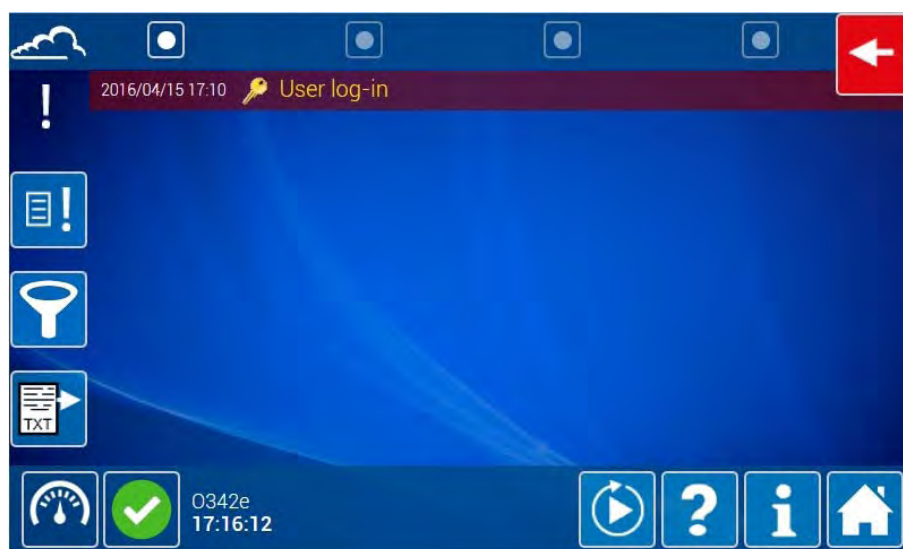


Click-on/touch  to close the pop-up.

3.4.2.8. Diagnostic functions

This screen is accessed by touching/clicking-on . It consists of four pages displayed by touching/clicking-on the white points in the screen top browsing bar.

The first page lists the current alarms and significant events on the analyzer and allows access to page 4 for alarm solving.



Button definitions specific to this screen



Displays or hides all the archived events.



Displays or hides the simple events. It acts as filter.



Exports the history of the recorded events in a TXT file.

The second page displays the analyzer signal values:

Power supplies			
+24V	24.4 V	+12V	12.0 V
+5V	5.0 V	+5V sensor	5.0 V
+4V	4034.8 mV	+3.3V	3.3 V
I +24V	0.3 mA		
Optical bench			
I sig.	4e+9 pts	I0 sig.	4e+9 pts
UV sig.	4e+9 pts	UV0 sig.	4e+9 pts
UV Ref.	2198.2 mV	UV Measure	2214.9 mV
I LED	10.5 mA	I LED status	ON

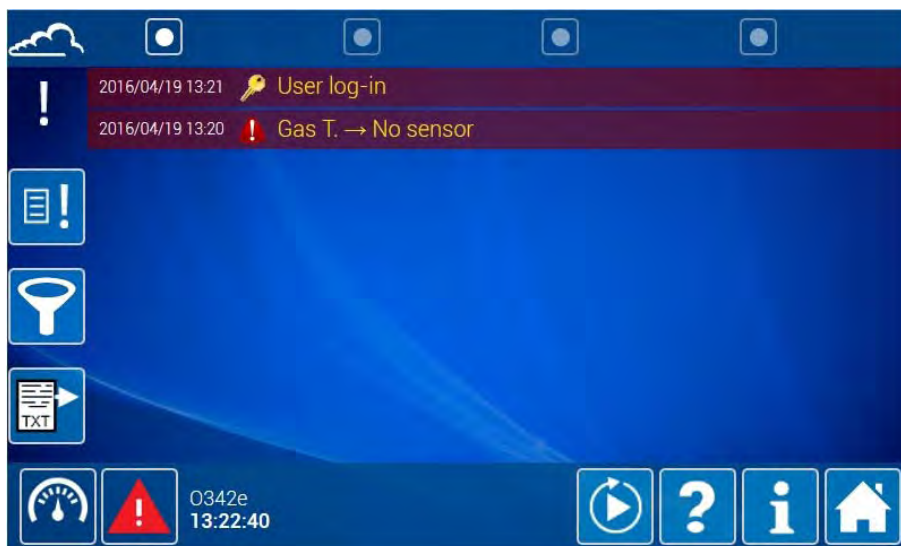
Sample			
Pump	ON	SV zero	OFF
SV span	OFF	Internal T.	27.2 °C
Pump Pr.	557 hPa	Flow	49.1 l/h

The third page displays the control status, the component list and status when detected on analyzer startup:

SV zero	OFF	SV span	OFF
+24V	ON	+5V sensor	ON
Pump	ON	Fan	ON
SV zero detect	ON	SV span detect	ON
Gene. detect	ON	Pump detect	OFF
Fan detect	ON	SV cycle detect	ON
PWM pump detect	ON	SV cycle	ON

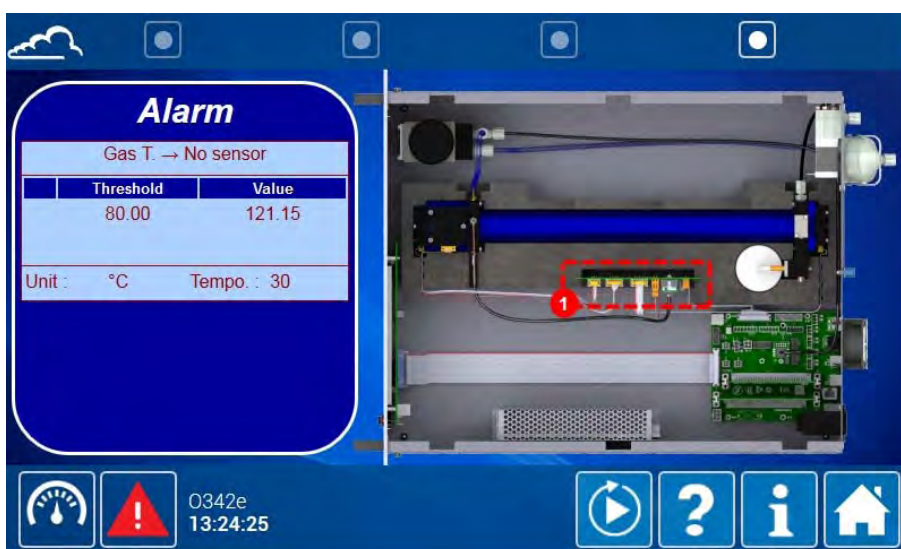
Alarm solving :

When an alarm triggers, the alarm icon  is displayed at the bottom left side of the screen. The user touches/clicks-on to access the diagnostic function first page displaying the current alarm and event list, as shown below: fields indicate triggering hour and fault nature. The user touches/clicks-on  to access the diagnostic function fourth page.

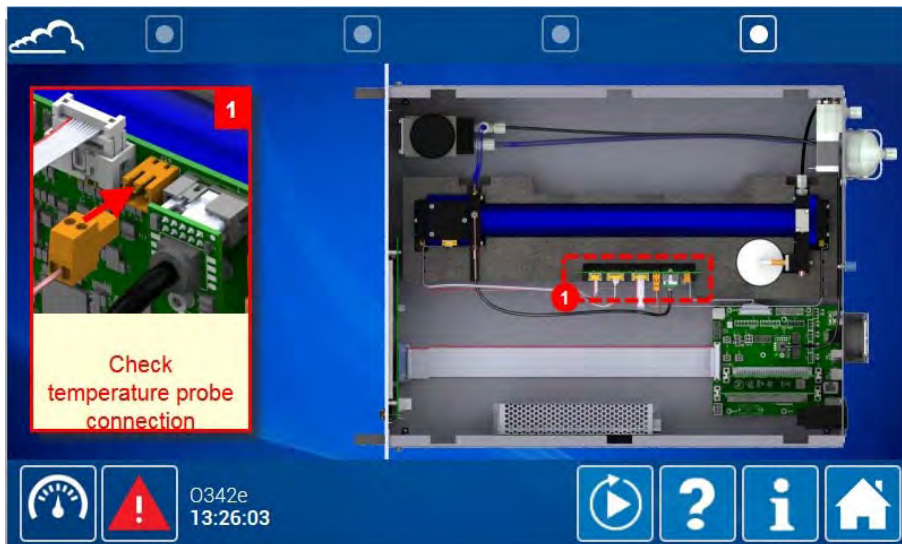


The fourth diagnostic function page displays the alarm threshold exceedance values, and shows the concerned faulty element in the internal analyzer view.

As example, the view below indicates a gas temperature alarm due to a sensor fault on the red-outlined board and focused with .



The user touches/clicks-on  to display the screen explaining what to do for alarm solving : « Check temperature probe connection » is written in the red frame of internal view left side, as shown below :



When the alarm is cleared, the analyzer returns to its normal status, and the icon  replaces the icon  in the bottom left side.

3.4.2.9. Set the analyzer in standby mode

To activate stand-by mode, the user double-clicks on/touches  which becomes .

Standby mode is used to stop the pump while all the other controls remain in operation. To reactivate the measurement mode, double-click / touch .

When the analyzer is switched to standby mode, the status icon (in bottom left side of screen) is modified to  indicating standby status. However, alarm or control indication is a priority.

Consequently, if the analyzer is switched to standby mode while in alarm or control is active, the alarm icon  or control icon  will remain displayed and the standby icon  will not be displayed.

Under these conditions, to verify if the measurement or standby status is active, the user checks the button  is red-outlined or not in the advanced function page.

3.4.2.10. Set the analyzer in maintenance mode

To activate maintenance mode, the user double-clicks on/touches  which becomes .

Maintenance mode activation is used to flag that the analyzer is in maintenance. When maintenance mode is activated, the recorded data are tagged with a status mark: "Maintenance" appears next to the concerned data when they are downloaded.

Double-click on/touch  to reactivate the measurement mode.

As the analyzer is switched to maintenance mode, the status icon in the left bottom side of the screen changes to  indicating maintenance status. However, alarm or control indication is a priority.

Consequently, if the analyzer is switched to maintenance mode while in alarm or control is active, the

alarm icon  or control icon  will remain displayed and the maintenance icon  will not be displayed.

Under these conditions, to verify if the measurement or maintenance status is active, the user checks the button  is red-outlined or not in the advanced function page.

3.4.2.11. Advanced analyzer configuration

This screen is accessed by touching/clicking-on . This screen allows configuration of the metrological parameters.



COMPOUND O3 section:

- « *Unit* » field: selects the compound measurement unit.
- « *Threshold 1* » and « *Threshold 2* » fields: fix a limit value to the measurement. If measurement value is out of this limit, an alarm triggers.
- « *Conversion Coef.* » field: displays the coefficient used for unit conversion (for ex. : ppb → µg/Nm³). This function is only applied if unit is µg/m³, mg/m³, g/m³.
- « *Offset* » and « *Span Coef.* » fields: these two values are used to calibrate the measurement with the linearization curve $Y = Ax + B$. A is the span coefficient, B is the offset. The span coefficient can be modified by the user or by a calibration cycle.
- « *Zero adjust* » field: allows setting manually the compound baseline (zero). As for the « *Span Coef.* », the analyzer automatically calculates this value when the user launches a Ref-Zero.
- « *Delta (%)* » field: when the span coefficient is modified by a calibration cycle, this value indicates the difference between the new coefficient and the previous coefficient. If this value is greater than 50%, a span alarm is triggered. It is necessary to re-validate manually the span coefficient to disable this alarm. When the analyzer is in alarm, it does not take account the new calculated calibration coefficient, but keeps the previous calibration coefficient.

« *Span gas value* » field: span gas cylinder concentration used to perform calibration check. This concentration is the reference value to be reached.

« *Calib. ON/OFF* » field: this field is used when the analyzer can measure many gases. It allows to perform a gas-by-gas calibration with no effect on the other gases. It invalidates or not the auto-calibration for the considered gas.

« *Response time* » field: shows response time value. Refer to Chapter 2 of this manual for more details.

« *Filter autonomy (d)* » field: shows the remaining day number before carrying out the next filter maintenance. This value is decreased daily, and an alarm triggers when it reaches 1.

« *Init. Counter OFF/ON* » field: reset filter autonomy to the programmed value after a filter change.

« *Dilution OFF/ON* » field: used to activate the dilution function and to set the dilution coefficient.

ADJUST LED section:

This section allows to configure the LED control according to the selected mode:

- « *Current mode* » selected: configure the setting in mA on the LED current in the « I LED consign » field.
- « *UV mode* » selected: configure the setting in mV on the UV signal in the « UV REF consign » field.

3.4.2.12. Automatic linearization of the flow

This screen is accessed by touching/clicking-on . It allows to linearize the flow value as function of the corresponding pump set point.

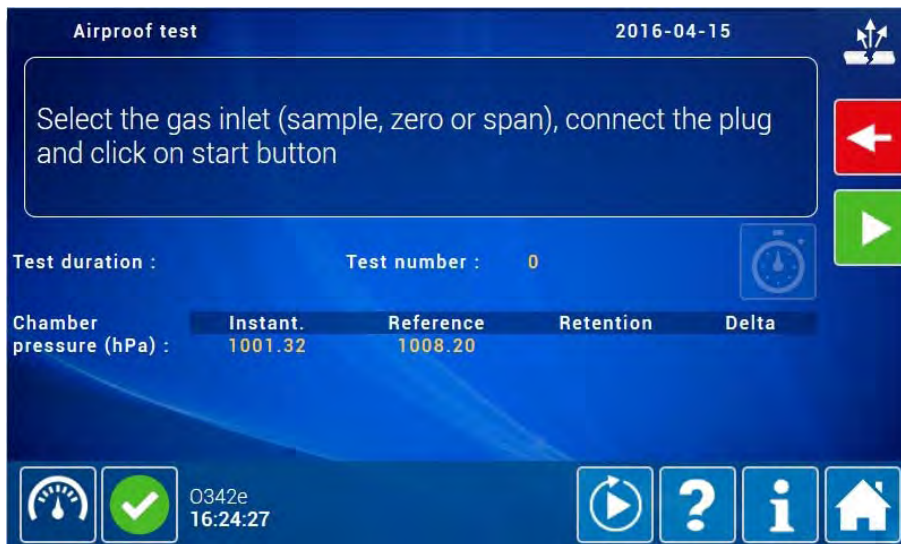
Before linearization launching, the user must connect a flowmeter (0-100 nl/h) to the suitable fluid inlet. Then, he clicks-on/touches the corresponding fluid inlet button , or , or , and he inputs in the « Nominal » field the read value on the flowmeter. Then, he launches linearization by touching/clicking on . Then, the analyzer will be in a maintenance mode for around 5 minutes.

REMARK: the in-factory pump setting is 70% by-default.



3.4.2.13. Airproof test

This screen is accessed by touching/clicking-on . It explains the procedure to follow for performing the airproof test of the analyzer fluid circuit. The user must follow the instructions detailed in the white-outlined frame.



When the user launches the airproof test, the analyzer switches to maintenance mode.

3.5. CALIBRATION

IMPORTANT NOTICE:

Use of the analyzer, as an equivalent method for EPA reporting, requires periodic multipoint calibration and subsequent zero/span checks as described below. All gases for calibration must be traceable to a reference photometer.

3.5.1. OVERVIEW OF CALIBRATION AND CONCEPTS

To ensure the accuracy of performed measurements using the O342e analyzer, the unit must be regularly checked, calibrated and adjusted, following the user quality assurance plan.

- **Zero and span check :**

This operation consists of comparing the analyzer response, for zero air and span point of the used range, to the used gas standards.

This check is used to measure the monitor drift in time without modifying the span coefficient (K span).

This check can be performed by using the internal zero air and one of the ozone concentrations settings (C1, C2, C3 and C4) that correspond to the internal ozone generator (optional), or by using zero air and an external O₃ concentration.

Periodicity: generally 24 hours in automatic cycle mode.

- **2-point calibration :**

This procedure is used for checking and correcting the monitor response to zero and span points located at about 0 % and 80 %, respectively of the full scale of the measurement range used.

Periodicity: monthly, or more frequently if the installation allows it.

- **Span (multipoint-point calibration) :**

This involves a complete checkup of the monitor linearity performance.

Periodicity: quarterly, or following out-of-tolerance calibration check results requiring an intervention on the analyzer.

3.5.1.1. Zero air generation

Zero air is purified, dried and free of any traces of ozone (< 1 ppb) air, connected to the analyzer zero air inlet (with optional zero solenoid valve, Span External Control option), or directly to the sample gas inlet, at atmospheric pressure. Zero air should also be free of any substances potentially absorbed at 255 nm (such as Benzene, m-xylene and toluene) and of any substances reacting with Ozone (such as NO and C₂H₄).

Zero air can be obtained using one of the following methods:

- A zero air generator with ozonizer, activated charcoal and molecular sieve (type JPAG/INSAT).
- A cylinder of reconstituted pure air.
- A set of filtering columns (see Figure 3–4) made of inert-to-O₃ material and whose efficiency is periodically checked by comparison to a zero air generator, at the time of a multipoint calibration (for zero and span checks and 2 points calibration only).

These filtering columns contain: 450 cc silicagel and 450 cc activated vegetable charcoal (grain 22-631-362)

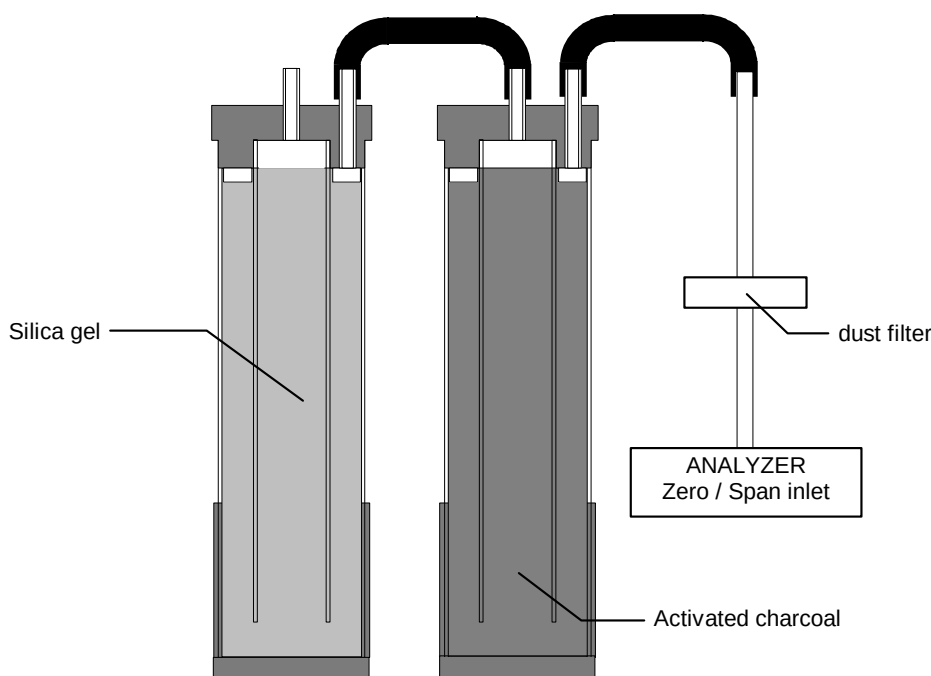


Figure 3–4 – Filtering columns

3.5.1.2. Ozone standard generation

Use one of the three methods below, depending on the application:

- A complete calibration system containing a photometer :

This apparatus is described in section 3.5.1.3.

It is possible to use a modified version of the O342e (O342e/C) and a source of pressurized dry clean air to comply with the requirements for an ozone calibration system. Note however that such a photometer must never be used to sample ambient air and must always be used with dry clean air.

- A reference ozone generator certified :
- An internal analyzer ozone generator (for span checks only).

3.5.1.3. Description of a complete calibration apparatus

A complete UV calibration system consists of an ozone generator, an output port or manifold, a photometer, an appropriate source of zero air, and other components as necessary. The configuration must provide a stable ozone concentration at the system output and allow the photometer to accurately assay the output concentration to the precision specified for the photometer (see 3.5.1.3.a). Figure 3–4 shows a commonly used configuration and serves to illustrate the calibration procedure which follows. Other configurations may require appropriate variations in the procedural steps. All connections between components in the calibration system downstream of the O₃ generator should be of glass, Teflon, or other relatively inert materials.

3.5.1.3.a. *UV photometer*

The photometer consists of a low-pressure mercury discharge lamp, (optional) collimation optics, an absorption cell, a detector, and signal-processing electronics. It must be capable of measuring the transmittance, I/I_0 , at a wavelength around 255 nm with sufficient precision such that the standard deviation of the concentration measurements does not exceed the greater of 0.005 ppm or 3% of the concentration. Because the low-pressure mercury lamp radiates at several wavelengths, the photometer must incorporate suitable means to assure that no O₃ is generated in the cell by the lamp, and that at least 99.5% of the radiation sensed by the detector is 255 nm radiation. (This can be readily achieved by careful selection of optical filter and detector response characteristics). The length of the light path through the absorption cell must be known with an accuracy of at least 99.5%. In addition, the cell and associated plumbing must be designed to minimize loss of O₃ from contact with cell walls and gas handling components.

3.5.1.3.b. Air flow controllers

Devices capable of regulating air flow as necessary to meet the output stability and photometer precision requirements.

3.5.1.3.c. Ozone generator

Device capable of generating stable levels of O₃ over the required concentration range.

3.5.1.3.d. Output manifold

The output manifold should be constructed of glass, Teflon, or other relatively inert material, and should be of a sufficient diameter to insure a negligible pressure drop at the photometer connection and other output ports. The system must have a vent designed to insure atmospheric pressure in the manifold and to prevent ambient air from entering the manifold.

3.5.1.3.e. Two-way valve

Manual or automatic valve, or other means to switch the photometer flow between zero air and the O₃ concentration.

3.5.1.3.f. Temperature indicator

Accurate to ±1 °C.

3.5.1.3.g. Barometer or pressure indicator

Accurate to ±2 torr.

3.5.1.3.h. Option 1

The various O₃ concentrations required in steps 11 of section 3.5.1.5 and 3.a of section 3.5.5.1 may be obtained by dilution of the O₃ concentration generated.

With this option, accurate flow measurements are required. The dynamic calibration system may be modified to allow for dilution air to be metered downstream of the O₃ generator. A mixing chamber between the O₃ generator and the output manifold is also required. The flow rate through the O₃ generator (F₀) and the dilution airflow rate (F_D) are measured with calibrated mass flowmeters. Each O₃ concentration generated by dilution is calculated by:

$$[O_3]'_{OUT} = [O_3]_{OUT} \left(\frac{F_0}{F_0 + F_D} \right)$$

Where:

[O₃]'_{OUT} = diluted O₃ concentration, ppm

F₀ = flow rate through the O₃ generator, liter/min

F_D = diluent air flow rate, liter/min

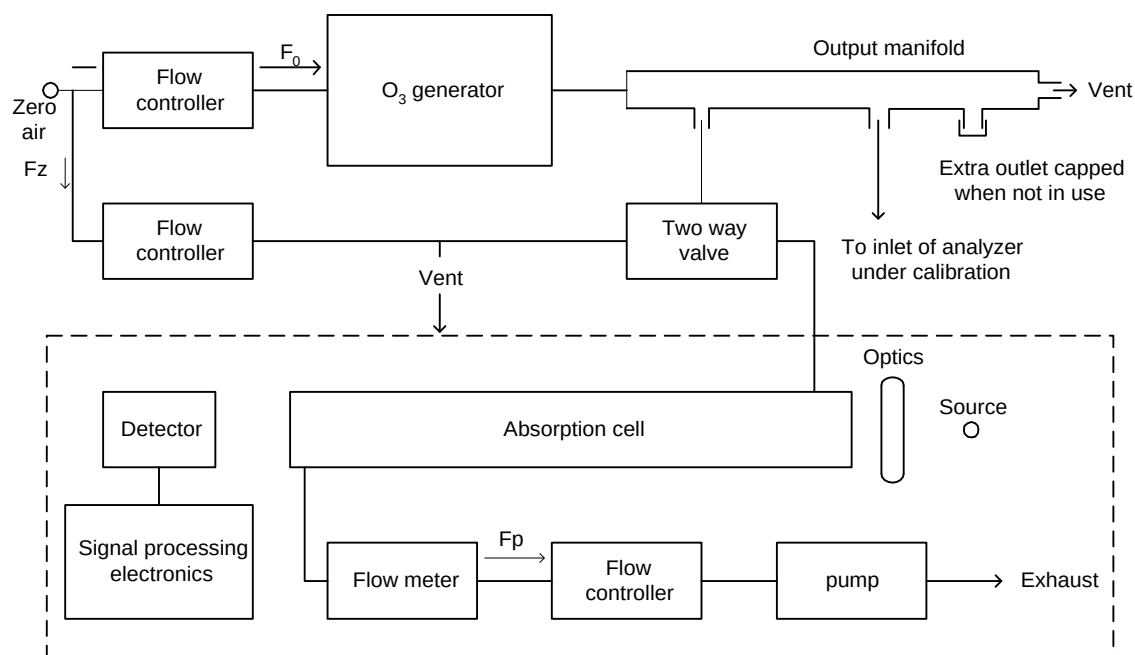


Figure 3-5 – Schematic diagram of a typical UV photometric calibration system (1)

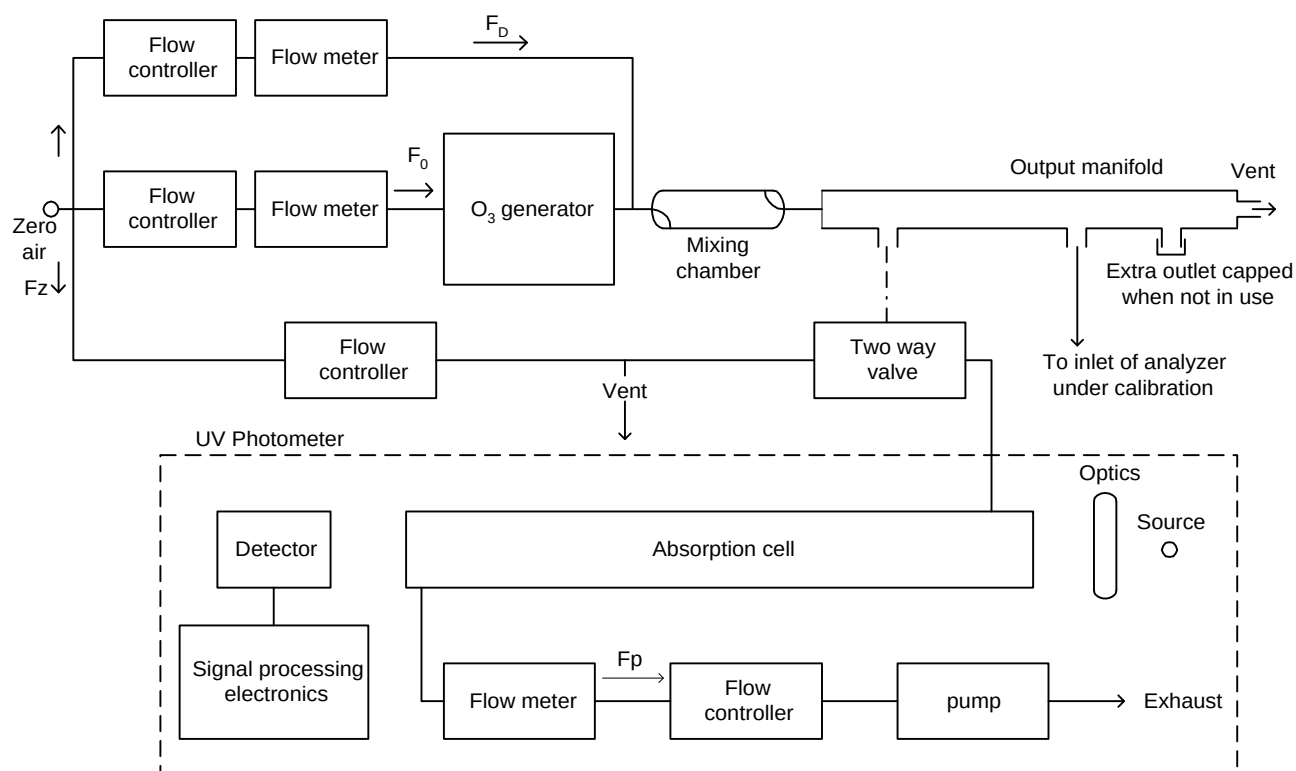


Figure 3-6 – Schematic diagram of a typical UV photometric calibration system (2)

3.5.1.4. Preparation and verification of calibration apparatus**3.5.1.4.a. General operation**

The calibration photometer must be dedicated exclusively to use as a calibration standard. It should always be used with clean, filtered calibration gases, and never used for ambient air sampling. Consideration should be given to locating the calibration photometer in a clean laboratory where it can be stationary, protected from physical shock, operated by a responsible analyst, and used as a common standard for all field calibrations via transfer standards.

3.5.1.4.b. Preparation

Proper operation of the photometer is of critical importance to the accuracy of this procedure. The following steps will help to verify proper operation. The steps are not necessarily required prior to each use of the photometer.

Upon initial operation of the photometer, these steps should be carried out frequently, with all quantitative results or indications recorded in a chronological record either in tabular form or plotted on a graphical chart.

As the performance and stability record of the photometer is established, the frequency of these steps may be reduced consistent with the documented stability of the photometer.

b.1) Instruction manual:

Carry out all set up and adjustment procedures or checks as described in the operation or instruction manual associated with the photometer.

b.2) System check:

Check the photometer system for integrity, leaks, cleanliness, proper flow rates, etc. Service or replace filters and zero air scrubbers or other consumable materials, as necessary.

b.3) Linearity:

Verify that the photometer manufacturer has adequately established that the linearity error of the photometer is less than 1.5%, or test the linearity by dilution as follows:

Generate and assess an O₃ concentration near the upper range limit of the system (0.5 or 1.0 ppm), then accurately dilute that concentration with zero air and evaluate it again. Repeat at several different dilution ratios. Compare the evaluation of the original concentration with the one of the diluted concentration divided by the dilution ratio, as follows:

$$E = \frac{A_1 - A_2 / R}{A_1} \times 100\%$$

Where:

E = linearity error, percent

A₁ = assay of the original concentration

A₂ = assay of the diluted concentration

R = dilution ratio = flow of original concentration divided by the total flow

The linearity error must be less than 5%. Since the accuracy of the measured flowrates will affect the linearity error as measured this way, the test is not necessarily conclusive.

b.4) Intercomparison

When possible, the photometer should be occasionally intercompared, either directly or via transfer standards, with calibration photometers used by other agencies or laboratories.

b.5) Ozone losses

Some portion of the O₃ may be lost upon contact with the photometer cell walls and gas handling components. The magnitude of this loss must be determined and used to correct the calculated O₃ concentration. This loss must not exceed 5%.

3.5.1.5. O₃ concentrations assessment using a photometer

To generate and evaluate the ozone concentrations using a photometer or a complete calibration apparatus, follow the procedure below.

- (1) Allow the photometer system to warm up and stabilize.
- (2) Verify that the flow rate through the photometer absorption cell, F allows the cell to be flushed in a reasonably short period of time (2 liter/min is a typical flow). The measurement precision is inversely related to the time required for flushing, since the photometer drift error increases with time.
- (3) Insure that the flow rate into the output manifold is at least 1 liter/min greater than the total flow rate required by the photometer and any other flow demand connected to the manifold.
- (4) Insure that the flow rate of zero air, FZ, is at least 1 liter/min greater than the flow rate required by the photometer.
- (5) With zero air flowing in the output manifold, actuate the two-way valve to allow the photometer to sample first the manifold zero air, then FZ. The two photometer readings must be equal ($I = I_0$).

NOTE : In some commercially available photometers, the operation of the two-way valve and various other operations of this section (3.5.1.4) may be carried out automatically by the photometer.

- (6) Adjust the O₃ generator to produce an O₃ concentration as needed.
- (7) Actuate the two-way valve to allow the photometer to sample zero air until the absorption cell is thoroughly flushed and records a stable measured value of I_0 .
- (8) Actuate the two-way valve to allow the photometer to sample the ozone concentration until the absorption cell is thoroughly flushed and records a stable measured value of I .
- (9) Record the temperature and pressure of the sample in the photometer absorption cell.
- (10) Calculate the O₃ concentration from the equation below. An average of several determinations will provide better precision.

$$[O_3]_{OUT} = \left(\frac{-1}{\alpha l} \ln \frac{I}{I_0} \right) \left(\frac{T}{273} \right) \left(\frac{760}{P} \right) \times \frac{10^6}{L}$$

Where:

- $[O_3]_{OUT}$ = O_3 concentration, ppm
- α = absorption coefficient of O_3 at 254 nm = 308 atm⁻¹ cm⁻¹ at 0 °C and 760 torr (O342M/C), or at 255 nm = 309 atm⁻¹.cm⁻¹ (O342e/C)/
- l = optical path length, cm
- T = sample temperature, K
- P = sample pressure, torr
- L = correction factor for O_3 losses from section 3.5.1.4.b.5 = (1-fraction O_3 lost).

NOTE : Some commercial photometers may automatically evaluate all or part of the equation above. It is the operator's responsibility to verify that all of the information required for this equation is obtained, either automatically by the photometer or manually. For automatic photometers which evaluate the first term of the equation above based on a linear approximation, a manual correction may be required, particularly at higher O_3 levels.

NOTE : Model O342M uses the exact formula (no linear approximation to replace the logarithm) for calculating $[O_3]_{OUT}$.

- (11) Obtain additional O_3 concentration standards as necessary by readjusting the ozone generator and following all the steps (6) to (10) described above or by dilution (see Option 1 in section 3.5.1.3.h)

3.5.1.6. Internal valve for zero and span switching

When performing a multipoint calibration the user should connect the zero and span gas sources to the sample gas inlet. After this calibration the zero and span sources should be connected to the corresponding inlets of the analyzer. The O342e should give identical responses whether the sources are connected to the sample or the zero and span inlets of the analyzer. If not, then internal valves should be serviced. Subsequently internal valves can be used for zero and span checks and two point calibrations.

3.5.2. CHECK OF ZERO AND SPAN POINT

3.5.2.1. Required Equipment

- zero air :
Use any of the methods described in section 3.5.1.1.
- Span point :
Use any of the methods described in section 3.5.1.2.

3.5.2.2. Procedure

- Zero check
 - With standard O342e, apply zero gas to sample gas inlet at atmospheric pressure.
 - For O342e with zero/span solenoid valve option or internal Ozone generator option, apply zero gas to zero inlet and use zero key to select the zero air inlet on analyzer.

Connect an activated charcoal filter and wait for measurement to stabilize. The measurement should be within ± 2 ppb of zero and then select the monitor zero air inlet using the  key (taking account the offset if it is programmed).

If this is not the case (measurement not comprised within ± 2 ppb), a zero adjustment is necessary.

Check that the ozone scrubber lifespan is greater than 30 days.

- Check of span point

Use one of the methods described in section 3.5.1.2 to generate ozone.

Apply ozone to the analyzer inlet at atmospheric pressure.

Wait for the reading to stabilize.

Select the gas inlet where the span gas is connected using the  key or the  key. The result will be compared to the concentration generated by the device used, taking account its accuracy as well as a possible programmed offset.

If the difference between the span gas and the reading is more than + 10 %, a span adjustment is necessary.

3.5.2.3. Use of automatic cycles

To program the cycles, touch  to access the « *Automatic cycle configuration* » screen. Refer to information detailed in paragraph 3.4.3.2.

- *Zero cycle:*

The « zero » air filter is permanently connected to the analyzer « zero air » inlet. The minimum recommended duration of the zero check is 600 seconds, as indicated in the "Duration" field of the "Cycles/Zero" column in the below screen.

- *Calibration cycle:*

The span point generator is permanently connected to the monitor zero/span gas inlet. The generated concentration must be below the full scale of the range used for the measurement. The recommended duration of the check is 600 seconds.

- *Calibration cycle:*

The span point generator is permanently connected to the analyzer zero/span gas inlet. The generated concentration must be below the full scale of the range used for the measurement. The recommended duration of the check is 600 seconds.



NOTE : The zero gas and calibration gas must both be supplied to the zero/span inlet one after the other at atmospheric pressure. This can be automated since the analyzer features two, zero and span, control contacts accessible on the analyzer rear panel.

3.5.3. TWO POINT CALIBRATION

3.5.3.1. Equipment required

- Zero reference:
Use one of the methods described under section 3.5.1.1
- Span point:
Use one of the first two methods described under section 3.5.1.2

NOTE : Materials in contact with zero or span gases must be made of Teflon, glass or other inert material.

3.5.3.3. Procedure

– Zero reference:

- If the « *Programmed inlet* » field is switch to ON in the « *Cycles/Zero-Ref* » column of the « *Automatic cycles configuration* » screen, connect the zero air generator to the programmed inlet.

If the « *Programmed inlet* » field is switch to OFF in the « *Cycles/Zero-Ref* » column of the « *Automatic cycles configuration* » screen, select the appropriate inlet between  (switch to sample inlet),  (switch to zero inlet),  (switch to span inlet).

- Touch  to carry out a zero-reference cycle (minimum duration: 600 seconds, as indicated in the “Duration” field of the « *Cycles/Zero-Ref* » column).

– Span point correction:

- If the « *Programmed inlet* » field is switch to ON in the « *Cycles/Calibration* » column of the « *Automatic cycles configuration* » screen, connect the gas generator to the programmed inlet.

If the « *Programmed inlet* » field is switch to OFF in the « *Cycles/Calibration* » column of the « *Automatic cycles configuration* » screen, select the appropriate inlet between  (switch to sample inlet),  (switch to zero inlet),  (switch to span inlet).

Program the O₃ concentration in the « *Span gas value* » field of the « *Compound: O₃* » section in the « *Advanced analyzer configuration* » screen, as shown below.



- Automatic correction:

Touch  key. The analyzer automatically modifies its calibration coefficient as a function of the programmed gas concentration. The recommended duration for self-calibration is 600 seconds, as indicated in the « *Duration* » field of the « *Cycles/Calibration* » column.

3.5.4. MULTIPOINT CALIBRATION

3.5.4.1. Equipment required

- Zero air

Use any of the following methods:

- A zero air generator with ozonizer, activated charcoal and molecular sieve (type JPAG/INSAT).
- A cylinder of reconstituted pure air.

- Span gas

Use one of the two methods below:

- A complete calibration system containing a photometer :

This apparatus is described in section 3.5.1.3.

It is possible to use a modified version of the O342e (O342e/C) or of the O342M, and a source of pressurized dry clean air to comply with the requirements for an ozone calibration system. Note however that such a photometer must never be used to sample ambient air and must always be used with dry clean air.

- A certified ozone transfer standard :

A transfer standard is certified by relating the output of the transfer standard to one or more ozone standards. The exact procedure varies depending on the nature and design of the transfer standard.

3.5.5. OVERVIEW

Gases will be applied at atmospheric pressure to the sample gas inlet of the analyzer.

Materials in contact with zero or span gases must be made of Teflon, glass or other inert material.

Calibration of the monitor requires precision gas generation of 7 points including zero (example: 0, 15, 30, 45, 60, 75 and 90% of the full scale of the range used), the dilution carrier gas must be the same as the one used for the zero measurement.

The multipoint calibration consists of 3 steps.

- (1) Install and check ozone generation apparatus.
- (2) Perform a two point calibration procedure (section 3.5.3.3).
- (3) Generate several ozone concentrations and plot a calibration curve to check linearity.

It is recommended to connect a strip chart recorder to the analog outputs of the analyzer for data gathering and processing. If the RS232-422 or RJ45 option is available, data can be collected by a computer.

3.5.5.1. Procedure

- (1-a) Install and check ozone generation apparatus according to the instruction manual of the ozone standard use for calibration.
- (1-b) Allow sufficient time for the O342e analyzer and the photometer or the transfer standard to warm-up and stabilize.
- (2-a) Allow the analyzer to sample zero air until a stable reading is obtained and adjust the analyzer zero control.
- (2-b) Generate an ozone concentration standard, $[O_3]_{OUT}$, of approximately 80% of the full scale of the range used. Allow the O342e to sample this ozone concentration standard until a stable reading is obtained. Note the obtained reading as $[O_3]_{READ}$.
- (2-c) Adjust the analyzer's span as described in section 3.5.3.2 following the formula :

$$K_{(new)} = K_{(former)} \times \frac{\text{span gas value}}{\text{read value (without offset)}}$$

- (3-a) Generate several ozone concentrations following the instruction manual of the ozone standard used for the calibration. For each ozone concentration record the corresponding analyzer response.
- (3-b) Plot the O342e responses versus the corresponding generated O_3 concentrations and draw the analyzer's calibration curve or calculate the appropriate response factor using least squares regression.
- (3-c) Fine adjust the K span factor if necessary according to the result of (3-b).

NOTE : The K span is usually set in factory. Should the results of the calibration lead to a deviation from this value of more than $\pm 5\%$, then extreme care has to be exercised in re-checking the whole process since such a result is likely to be inconsistent.

- Check the quality of zero air.
- Check certification of Ozone transfer (if applicable) and construction and set up of calibration apparatus.
- Check for leaks.
- Check for appropriate pressure at the used inlet of the analyzer.
- If everything is correct then the analyzer needs to be serviced (see chapter 4)

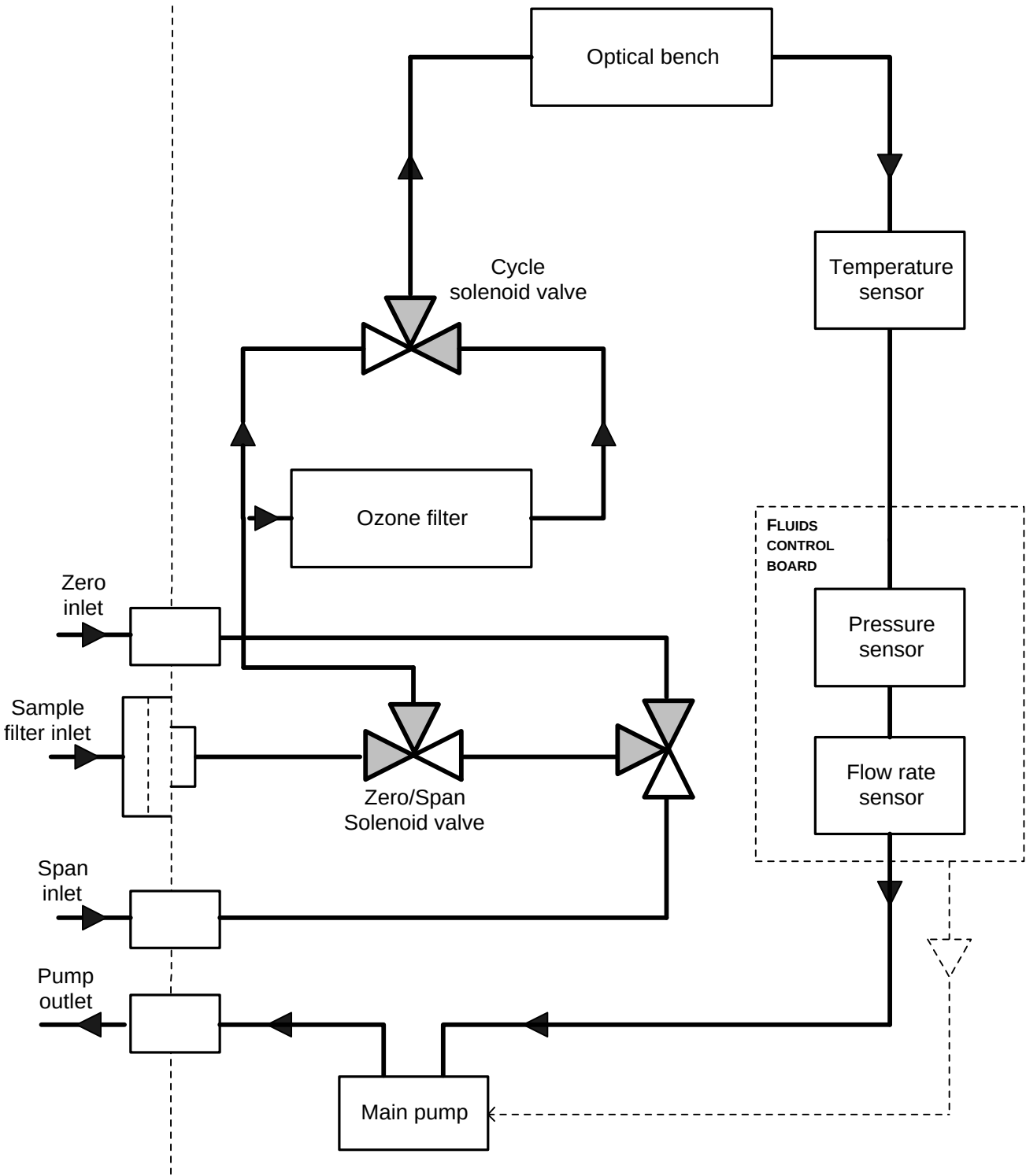


Figure 3-7 – Gas flow system with zero span solenoid valve

3.5.6. INTERNAL OZONE GENERATOR

Refer to Figure 3–8.

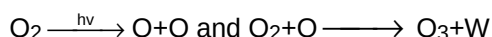
CAUTION : This feature must not be used for span adjustment in case of reference method analyzer (compliance) monitoring. The span point check can be performed using the internal Ozone generator.

3.5.6.1. General operating principle

The zero air, obtained by filtering the ambient air, is drawn through a chamber containing a low pressure mercury vapor lamp.

This lamp is operated by a current regulated power supply. The temperature of the unit is kept constant.

The 184.9 nm wavelength UV rays emitted by the lamp convert part of the oxygen contained in the zero air into Ozone, through the following reactions:



A fixed Ozone concentration is therefore distributed between the analyzer optical bench by means of the zero/span solenoid valve and the ozone outlet available on the rear part of the analyzer.

3.5.6.2. Description of the operating mode

Refer to 3.5.6.4.

3.5.6.3. Remarks on use of Ozone generator as calibration standard

This option should not be used when using analyzer for EPA compliance purposes.

In principle, the analyzer cannot calibrate itself. It can nevertheless serve as a system for making performance checks.

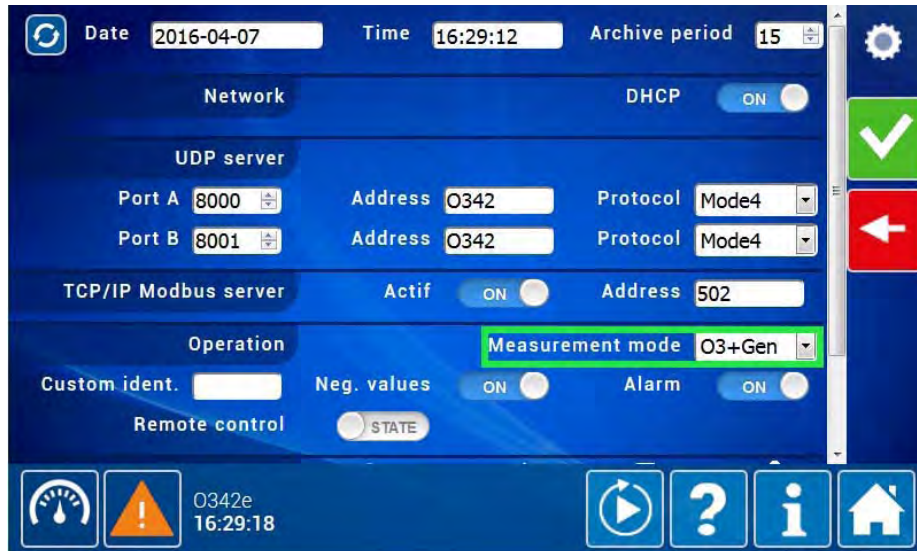
In any case, some restrictions must be indicated in order to guarantee the highest efficiency of the system:

- high system repeatability, differences not exceeding the highest of the values between $\pm 5\%$ and ± 5 ppb, requires operation of lamp under rated conditions, i.e. at a temperature of about 50°C and a regulated supply current of approximately 2 to 10 mA (depending on the UV lamp model). To do this, the required concentrations will preferably be selected within the range of 40 to 300 ppb.

3.5.6.4. Ozone generator configuration and starting-up

After having installed the ozone generator in the unit, proceed in the following order:

- From the home page, touch/ click on  to access the advanced functions, then on  to display the « General configuration » screen and configure the « Measurement mode » field to « O3+Gen », as shown below (see green square).



- Valid the action by touching/clicking on the button .
- Close the span inlet on the analyzer rear panel to avoid dilution of the generated ozone concentration.
- Wait until the ozone generator temperature reaches 50°C. Check this temperature in the « Synoptic » screen (10) : this screen is displayed in touching / clicking on , then on the last button  in the top right part of the screen, which is then activated to  (see the green arrow below).



- When the temperature of 50° C is reached, touch/click on  to return to the advanced functions, then on  to display the "Advanced analyzer configuration" screen. The "Ozone Generation", "Concentration 1" and "Concentration 2" fields appear at screen bottom. The "Concentration" fields allow to enter the desired concentration between 30 and 200 ppb.



- Enter the desired concentration 1 and validate by clicking/touching . Then press  to start the ozone generator calibration. During calibration, the analyzer switches into maintenance mode for about 25 minutes. Do the same with concentration 2 if necessary.

When this calibration is finished, the user must update the "Ozone Generation" field: to do this, he click/ touch  to exit the screen then click/touch  to return again to the "Advanced analyzer configuration" screen. The concentrations 1 and 2 are refreshed in the "Ozone Generation" field (see green frame). The user will select them when a new calibration is needed.

NOTE : The orange-written values are obtained from the previous ozone generator calibration and are reminded here.



- Switch on the generator by clicking/touching  to select the span inlet: Ozone is generated 2 minutes later.

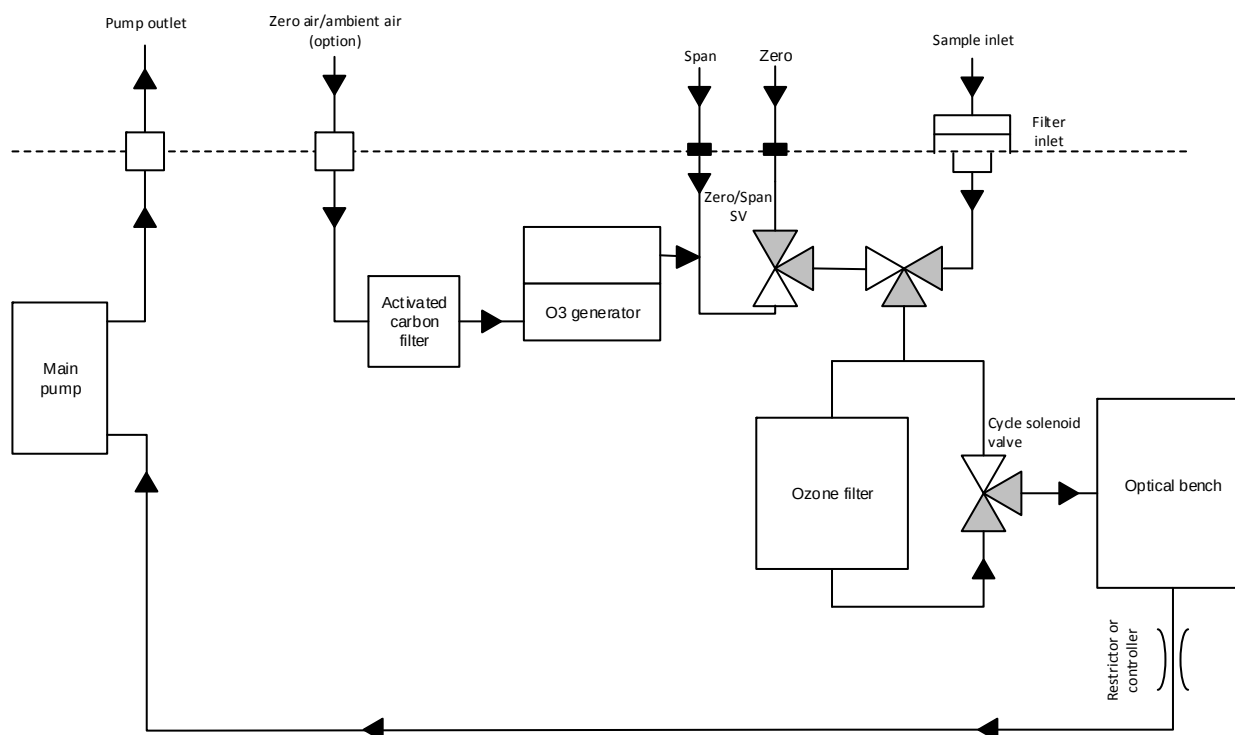


Figure 3-8 – Gas flow system with Ozone generator

CHAPTER 4

PREVENTIVE MAINTENANCE

4.1	SAFETY INSTRUCTIONS	4-3
4.2	MAINTENANCE CALENDAR	4-4
4.3	MAINTENANCE OPERATION SHEETS	4-5
4.4	KITS AND SPARE PARTS SET FOR O342E MAINTENANCE	4-16

Figure 4-1 – Replacing of sample inlet dust filter	4-5
Figure 4-2 – KNF head pump dismantling	4-9
Figure 4-3 – Diaphragm replacing	4-10
Figure 4-4 – Replacing of valves and O-rings	4-11
Figure 4-5 – Pump reassembly	4-12
Figure 4-6 – Cleaning of flow restrictor	4-13
Figure 4-7 – Replacing of Teflon solenoid valve	4-14
Figure 4-8 – Replacement of selective filter grids	4-15

Page intentionally left blank

4 PREVENTIVE MAINTENANCE

4.1 SAFETY INSTRUCTIONS

The user must follow these safety instructions at all times:

- Always turn off the power supply when performing analyzer maintenance.
- Personnel should be trained in the proper operation of this equipment before attempting to operate it.
- The manufacturer shall not be responsible for any adverse outcomes resulting from the following:
 - Use of the monitor by unqualified service personnel.
 - Use of the monitor under conditions other than those specified in this document.
 - Use of components or accessories not manufactured by ENVEA, failure to use recommended parts may reduce the safety features.
 - Use of this equipment in a manner not approved by ENVEA, as it can cause harm to the equipment or operating personnel.
 - Inappropriate maintenance of the analyzer.
- A periodic inspection is required.

4.2 MAINTENANCE CALENDAR

By design, the O342e requires very limited maintenance. However, the unit must be regularly serviced to ensure proper performance over time. The routine maintenance schedule shown below is an example, and this schedule can vary according to operating conditions.

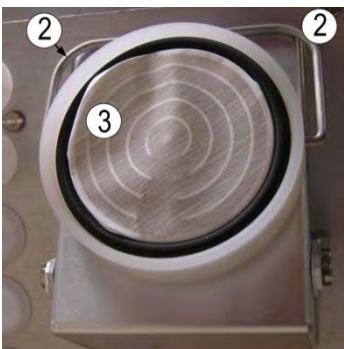
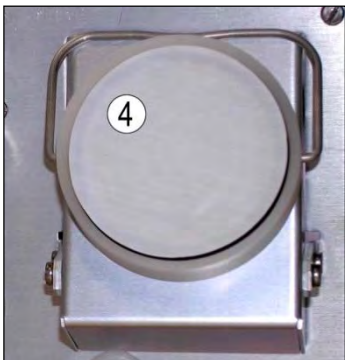
Nature of operations	Periodicity	Sheet N°
Replacing of sample inlet dust filter	2 to 4 weeks	4.3.1
Check of optical, electric and fluid parameters	1 month	4.3.2
Maintenance of KNF pump	1 year	4.3.3
Cleaning of flow restrictor	1 year	4.3.4
Replacing of Teflon solenoid valve	2 years	4.3.5
Replacing the selective filter grids	1 an	4.3.6

Annual check

The monitor must undergo thorough cleaning (absorption medium, restrictors, flow circuit, etc.) in the laboratory at least once a year. All metrological parameters and tightness of fittings should be checked at this time.

4.3 MAINTENANCE OPERATION SHEETS

MAINTENANCE SHEET

MONITOR Serial No.:		OPERATION SHEET: 4.3.1	
Scope: Replacing of sample inlet dust filter		PAGE: 1/1	Periodicity: 2 to 4 weeks according to dust amount
<u>Proceed in the following order:</u> – Turn off the analyzer power switch and disconnect the sample inlet. – Unlock the filter holder cover (1) by pulling the spring (2) upwards. – Remove the worn filter (3). – Place the new filter (4) on the filter holder without the separating blue sheet. – Make sure the filter correctly covers the filter-holder surface and that it is correctly placed flat on top of it. – Reposition the cover of the filter holder, and lock the cover of the filter holder by lowering the spring (2). – Connect the sample inlet and power on the analyzer.    (1) filter holder cover, (2) spring, (3) worn filter, (4) new filter Figure 4-1 – Replacing of sample inlet dust filter			Date
<u>Tools and parts required:</u> • Tweezers • Sample inlet filter (Teflon filter - porosity 5 µm - Ø 47 mm) - Ref.: F05-11-842			

MAINTENANCE SHEET

MONITOR Serial No.:		OPERATION SHEET: 4.3.2	
Scope: Check of optical, electric and fluid parameters		PAGE: 1/2	Periodicity : 1 month
– Flow rate check - Connect a flow meter on the sample inlet of the O342e.  - Make sure the flow rate value indicated on the flow meter is 55 ± 5 NI/h.			Date
<u>Tools required</u> Flow meter, 100 liters/hour			

MAINTENANCE SHEET

MONITOR Serial No.:		OPERATION SHEET : 4.3.2																																																																																	
Scope : Check of optical, electric and fluid parameters		PAGE : 2/2	Periodicity : 1 month																																																																																
- MUX signals check: - Switch the analyzer to zero air. - In the « Synoptic diagram of the analyzer » screen and the « Diagnostic functions » screen, write down the signal values indicated on the test report and compare them with the limits given below: <table border="1"> <thead> <tr> <th>Signals</th> <th>Min. values</th> <th>Typical values</th> <th>Max values</th> </tr> </thead> <tbody> <tr> <td>Reference zero filter</td> <td>0</td> <td>365</td> <td>N. A.</td> </tr> <tr> <td>UV Measure signal</td> <td>500 mV</td> <td>1200 mV</td> <td>2650 mV</td> </tr> <tr> <td>UV Reference signal</td> <td>500 mV</td> <td>1500 mV</td> <td>2650 mV</td> </tr> <tr> <td>LED current</td> <td>2 mA</td> <td>30 mA</td> <td>80 mA</td> </tr> <tr> <td>Internal temperature</td> <td>5 °C</td> <td>30 °C</td> <td>45 °C</td> </tr> <tr> <td>Absorption medium temperature</td> <td>+ 5 °C</td> <td>+ 25 °C</td> <td>+ 45 °C</td> </tr> <tr> <td>Flow rate</td> <td>30 NL/h</td> <td>55 NL/h</td> <td>90 NL/h</td> </tr> <tr> <td>Atmospheric pressure</td> <td>300 hPa</td> <td>1002 hPa</td> <td>1200 hPa</td> </tr> <tr> <td>Pump pressure</td> <td>200 hPa</td> <td>340 hPa</td> <td>600 hPa</td> </tr> <tr> <td>Absorption medium pressure</td> <td>300 hPa</td> <td>990 hPa</td> <td>1200 hPa</td> </tr> <tr> <td>+4 V supply</td> <td>3950 mV</td> <td>4000 mV</td> <td>4050 mV</td> </tr> <tr> <td>+3.3 V supply</td> <td>3.2 V</td> <td>3.3 V</td> <td>3.4 V</td> </tr> <tr> <td>+12 V supply</td> <td>11.8 V</td> <td>12 V</td> <td>12.2 V</td> </tr> <tr> <td>+24 V supply</td> <td>23,4 V</td> <td>24,0 V</td> <td>25.0 V</td> </tr> <tr> <td>+5 V supply</td> <td>4.8 V</td> <td>5 V</td> <td>5.2 V</td> </tr> <tr> <td>Sensor +5 V supply</td> <td>4.9 V</td> <td>5 V</td> <td>5.1 V</td> </tr> <tr> <td>+24V current</td> <td>- 0.1 A</td> <td>0,5 A à 3 A</td> <td>> 6 A</td> </tr> <tr> <td>GND</td> <td>-1 mV</td> <td>0 mV</td> <td>+1 mV</td> </tr> <tr> <td>GNDA</td> <td>-1 mV</td> <td>0 mV</td> <td>+1 mV</td> </tr> </tbody> </table>			Signals	Min. values	Typical values	Max values	Reference zero filter	0	365	N. A.	UV Measure signal	500 mV	1200 mV	2650 mV	UV Reference signal	500 mV	1500 mV	2650 mV	LED current	2 mA	30 mA	80 mA	Internal temperature	5 °C	30 °C	45 °C	Absorption medium temperature	+ 5 °C	+ 25 °C	+ 45 °C	Flow rate	30 NL/h	55 NL/h	90 NL/h	Atmospheric pressure	300 hPa	1002 hPa	1200 hPa	Pump pressure	200 hPa	340 hPa	600 hPa	Absorption medium pressure	300 hPa	990 hPa	1200 hPa	+4 V supply	3950 mV	4000 mV	4050 mV	+3.3 V supply	3.2 V	3.3 V	3.4 V	+12 V supply	11.8 V	12 V	12.2 V	+24 V supply	23,4 V	24,0 V	25.0 V	+5 V supply	4.8 V	5 V	5.2 V	Sensor +5 V supply	4.9 V	5 V	5.1 V	+24V current	- 0.1 A	0,5 A à 3 A	> 6 A	GND	-1 mV	0 mV	+1 mV	GNDA	-1 mV	0 mV	+1 mV	Date
			Signals	Min. values	Typical values	Max values																																																																													
Reference zero filter	0	365	N. A.																																																																																
UV Measure signal	500 mV	1200 mV	2650 mV																																																																																
UV Reference signal	500 mV	1500 mV	2650 mV																																																																																
LED current	2 mA	30 mA	80 mA																																																																																
Internal temperature	5 °C	30 °C	45 °C																																																																																
Absorption medium temperature	+ 5 °C	+ 25 °C	+ 45 °C																																																																																
Flow rate	30 NL/h	55 NL/h	90 NL/h																																																																																
Atmospheric pressure	300 hPa	1002 hPa	1200 hPa																																																																																
Pump pressure	200 hPa	340 hPa	600 hPa																																																																																
Absorption medium pressure	300 hPa	990 hPa	1200 hPa																																																																																
+4 V supply	3950 mV	4000 mV	4050 mV																																																																																
+3.3 V supply	3.2 V	3.3 V	3.4 V																																																																																
+12 V supply	11.8 V	12 V	12.2 V																																																																																
+24 V supply	23,4 V	24,0 V	25.0 V																																																																																
+5 V supply	4.8 V	5 V	5.2 V																																																																																
Sensor +5 V supply	4.9 V	5 V	5.1 V																																																																																
+24V current	- 0.1 A	0,5 A à 3 A	> 6 A																																																																																
GND	-1 mV	0 mV	+1 mV																																																																																
GNDA	-1 mV	0 mV	+1 mV																																																																																
Tools required Flow meter, 100 liters/hour																																																																																			

MAINTENANCE SHEET

MONITOR Serial No.:		OPERATION SHEET: 4.3.3	
Scope: Maintenance of KNF pump		PAGE: 1/5	Periodicity: 1 year
Before any intervention, unplug the power cable and make sure the pump is turned off. The diaphragm and the valves represent the only wearing parts of the pump. The maintenance operation consists of cleaning and/or replacing them. <u>Preliminary operation:</u> – Turn off the analyzer and unplug the power cable. – Unscrew the fixation screw of the cover located on the rear panel of the analyzer. Remove the protective cover. – Disconnect the pump from its electrical connection. Unscrew the fluid inlet and the fluid outlet of the pump. – Unscrew and remove the three screws to release the pump from its holder, and carry out the maintenance out of the analyzer.			Date
<u>Tools and parts required</u> • Two flat screwdrivers, 1 x 5,5 mm and 0,4 x 2,5 mm • Hexagon socket head wrench n°3 • Maintenance kit of KNF pump. Ref. : V02-0199 • Cross-tip screwdriver, 4,5 x 75 mm			

MAINTENANCE SHEET

MONITOR Serial No.:	OPERATION SHEET: 4.3.3	
Scope: Maintenance of KNF pump	PAGE: 2/4	Periodicity: 1 year
<u>Dismantling the head pump</u> – Draw a line with a pen (7) to mark the cover (8), the cylinder head (9), and the body of the pump (10). This will avoid wrong positioning of the parts when the pump is reassembled. Remove the cover (11) by delicately turning the blade of a screwdriver in the slots (12) in order to access the eccentric (13). – Unscrew the fixation screws (14) of the pump head to remove the cover (8) and the cylinder head (9) from the pump body (10) in order to release the diaphragm (15), the valves (16), and the O-rings. (7) pen mark, (8) cover, (9) cylinder head, (10) body pump, (11) eccentric cover, (12) slots, (13) eccentric, (14) fixation screws, (15) diaphragm, (16) valves. Figure 4-2 – KNF head pump dismantling		Date
<u>Tools and parts required</u> • Two flat screwdrivers, 1 x 5,5 mm and 0,4 x 2,5 mm • Hexagon socket head wrench, n°3 • Maintenance kit of KNF pump. Ref. : V02-0199 • Cross-tip screwdriver, 4,5 x 75 mm		

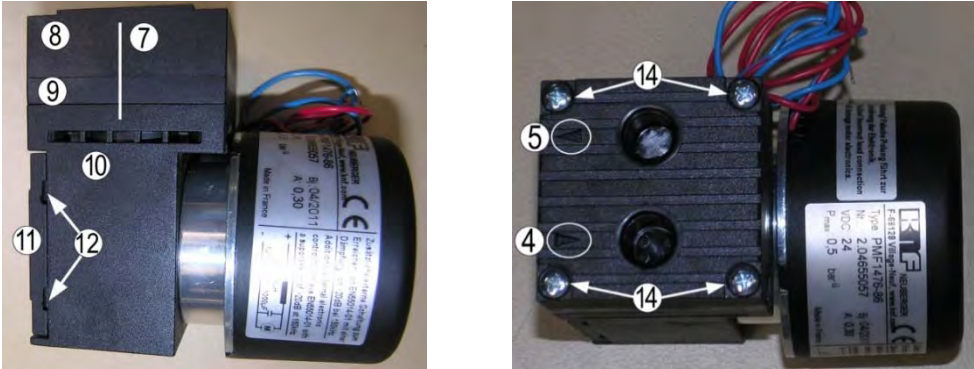
MAINTENANCE SHEET

MONITOR Serial No.:		OPERATION SHEET: 4.3.3	
Scope: Maintenance of KNF pump		PAGE: 3/5	Periodicity: 1 year
Diaphragm replacing - Turn the wheel of the eccentric (13) to position the diaphragm (15) to the upper position, so the diaphragm can be easily grasped by its circumference (17) and turn it counter clockwise to unscrew it. - The diaphragm is placed on a cupel (18). Remove the washers (19) located between the diaphragm and the cupel, and the washers (20) located between the cupel and the threading hole (21). Note carefully the number of washers at each location because it is critical to install the same number of washers at the same places during reassembly, - To change the diaphragm (15), put the washers (19) between the diaphragm and the cupel (18) back in place, and install the diaphragm on the cupel. Put the washers (20) on the external side of the cupel back in place so they are intercalated between the cupel and the threading hole (21) upon reassembly. - Completely rescrew the assembly (diaphragm + cupel + washers) in the threading hole (21).			Date
 (13) eccentric, (15) diaphragm, (17) the user grasps the diaphragm by its circumference, (18) cupel, (19) washers, (20) washers, (21) threading hole. Figure 4-3 – Diaphragm replacing			
Tools and parts required - Two flat screwdrivers, 1 x 5,5 mm and 0,4 x 2,5 mm - Hexagon socket head wrench, n°3 - Maintenance kit of KNF pump. Ref. : V02-0199 - Cross-tip screwdriver, 4,5 x 75 mm			

MAINTENANCE SHEET

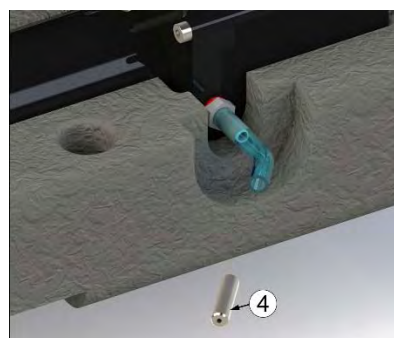
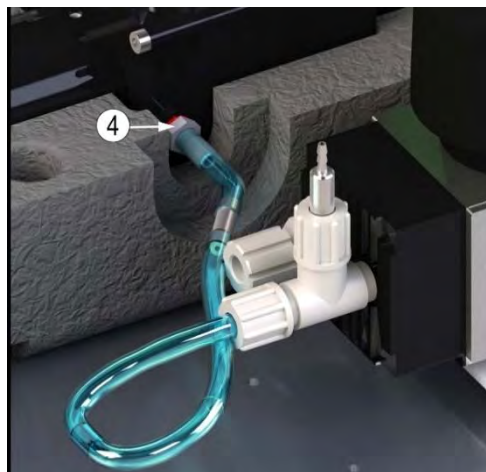
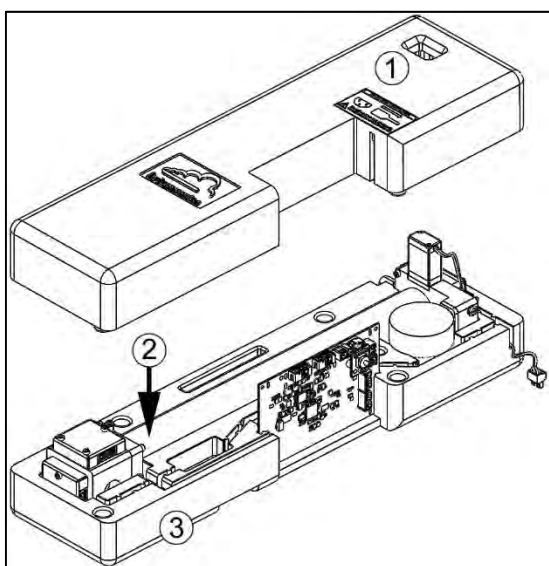
MONITOR Serial No.:		OPERATION SHEET: 4.3.3	
Scope: Maintenance of KNF pump		PAGE: 4/5	Periodicity: 1 year
<u>Replacing of valves (22) and O-rings (23):</u> The valves are located in the cylinder head. Replace the valves (22) and the O-rings (23), taking care to correctly fit together the O-rings in their housings. (22) valves, (23) O-rings Figure 4-4 – Replacing of valves and O-rings			Date
<u>Tools and parts required</u> • Two flat screwdrivers, 1 x 5,5 mm and 0,4 x 2,5 mm • Hexagon socket head wrench, n°3 • Maintenance kit of KNF pump. Ref. : V02-0199 • Cross-tip screwdriver, 4,5 x 75 mm			

MAINTENANCE SHEET

MONITOR Serial No.:		OPERATION SHEET: 4.3.3	
Scope: Maintenance of KNF pump		PAGE: 5/5	Periodicity: 1 year
<u>Pump reassembly</u> Reassembly is carried out in the reverse order from dismantling. – Reassemble the cover (8) and the cylinder head (9) on the pump body (10) according to the line drawn to mark the various parts (7). Put the four screws of the pump head (14) in place and then screw them clockwise. – Replace the eccentric cover (11). – Reassemble the pump in its holder inside the analyzer and replace the screws. Reconnect the electric connector, the fluid inlet line, and the fluid outlet line. – Replace the cover of the analyzer, replace the screw, and power on the analyzer.  (4) fluid inlet, (5) fluid outlet, (7) pen mark, (8) cover, (9) cylinder head, (10) body pump, (11) eccentric cover, (12) slots, (14) screws of the pump head. Figure 4-5 – Pump reassembly		Date	
<u>Tools and parts required</u> • Two flat screwdrivers, 1 x 5,5 mm and 0,4 x 2,5 mm • Hexagon socket head wrench, n°3 • Maintenance kit of KNF pump. Ref. : V02-0199 • Cross-tip screwdriver, 4,5 x 75 mm			

MAINTENANCE SHEET

MONITOR Serial No.:	OPERATION SHEET: 4.3.4	
Scope: Cleaning of flow restrictor	PAGE: 1/1	Periodicity : 1 year
<u>Proceed in the following order:</u> – Turn off the analyzer and unplug the power cable. – Unscrew the fixation screw of the cover located on the rear panel of the analyzer. Remove the protective cover. – Remove the protective foam cover (1) of the absorption medium (2), and release it from its housing (3). – Mark the flow restrictor (4), then unscrew to remove it from the fluid circuit. – Clean the flow restrictor by immersing in alcohol, then dry it with dusted and oil-free compressed air. – Reassemble the flow restrictor. – Replace the absorption medium (2) in its housing, and replace the protective foam cover (1) above. – Reconnect the power cable and power on the analyzer.		Dates



(1) foam cover, (2) absorption medium, (3) absorption medium housing, (4) restrictor

Figure 4-6 – Cleaning of flow restrictor

Tools required :

- Alcohol
- Dusted and oil-free compressed air
- Wrench, 8 mm diameter

MONITOR Serial No.:		OPERATION SHEET: 4.3.5	
Scope: Replacing of Teflon solenoid valve		PAGE: 1/1	Periodicity : 2 years
– Turn off the analyzer and unplug the power cable. – Unscrew the fixation screw of the cover located on the rear panel of the analyzer. Remove the protective cover. – Remove the protective foam cover (1) of the absorption medium (2), and lay it flat in its housing (3). – Unplug the electrical connector from J11 of the interconnection board. Unscrew the fixing screws (4) of the Teflon solenoid valve and remove it. – Change the Teflon solenoid valve, rescrew the fixing screws (4), and plug the electric connector to J11 of the measurement board. – Replace the protective foam cover (1) above the absorption medium (2). – Reconnect the power cable and power on the analyzer.			Dates
(1) foam cover, (2) absorption medium, (3) absorption medium housing, (4) screw of the Teflon solenoid valve Figure 4-7 – Replacing of Teflon solenoid valve			
Tools and parts required : • Screwdriver, 5 x 100 mm • Teflon solenoid valve – Ref. : D01-0749			

MAINTENANCE SHEET

MONITOR Serial No.:	OPERATION SHEET: 4.3.6	
Scope: Replacement of the selective filter grids	PAGE: 1/1	Periodicity : 1 an
Proceed in the order: – Turn off the analyzer and unplug the power cable. – Unscrew the cover fixing screws on the analyzer rear panel. Remove the analyzer cover. – Remove the foam protective cover (1) of the absorption medium (2), and leave it flat in its housing (3). – Carefully lift up the measurement module from its housing (4). – Unscrew the screws (5), remove the fixing plate (6) and the two seals (7). – Remove the selective filter (8), and open it carefully. – Replace the grids in the following order: 5 flat grids (9) – 1 honeycomb (10) – 5 flat grids (11). – Close the selective filter, and assemble it again on the measurement module: replace the fixing plate (6), the two seals (7) and screw the screws (5). – Replace the protective foam (1) above the absorption medium (2). – Connect the power cord and switch on the analyzer. (1) foam cover, (2) absorption medium, (3) absorption medium housing, (4) detail of the absorption medium, (5) screw, (6) fixing plate, (7) seals, (8) selective filter, (9) and (11) flat grids, (10) honeycombs. Figure 4-8 – Replacement of selective filter grids		Date
Parts required : • Set of 10 treated grids - Ref. : P01-1731		

4.4 KITS AND SPARE PARTS SET FOR O342E MAINTENANCE

Maintenance kit for 1 year - Ref.: O342e -K

Line Nb	Designation	Reference	Qty
1	Teflon filter (porosity 5 µm - Ø 47 mm)	F05-11-842	25
2	O-ring o int. :6.0 cord 2.0 viton	G06-006_0-2_2-V	2
3	Set of 10 treated grids	P01-1731	1
4	Diaphragm Kit for pump V02-0199/V02-0236/V02-0259	V04-0005	1

CHAPTER 5.

CORRECTIVE MAINTENANCE

Figure 5–1 – Interconnection board diagram	5–13
Figure 5–2 – Automaton board diagram (above view)	5–15
Figure 5–3 – Photodiode board diagram (above view)	5–16
Figure 5–4 – LED UV board diagram (above view)	5–17
Figure 5–5 – Measurement board diagram	5–19
Figure 5–6 – Supply board diagram	5–20
Figure 5–7 – ARM20 board	5–21

Table 5–1 – List of faults and corrective actions	5–4
Table 5–2 – Interconnection board configuration	5–12
Table 5–3 – Automaton board configuration	5–14
Table 5–4 – Photodiode board configuration	5–16
Table 5–5 – LED UV board configuration	5–17
Table 5–6 – Measurement board configuration	5–18
Table 5–7 – Supply board configuration	5–20
Table 5–8 – ARM20 board description	5–21

Page intentionally left blank

5. CORRECTIVE MAINTENANCE

Analyzer corrective maintenance must be carried out only by qualified people using the information provided in this document.

The monitor automatically and continuously self-tests its main components. It indicates any malfunction detected: the alarm icon  is displayed and blinks on the home page and in the menu bar of all software screens, next to the date and analyzer name.



Then, it is just necessary to touch the alarm icon to directly access the « *DIAGNOSTIC FUNCTIONS* » screen and check the malfunction origin.

Table 5–1 summarizes the main faults indicated by the unit with their corresponding possible corrective actions.

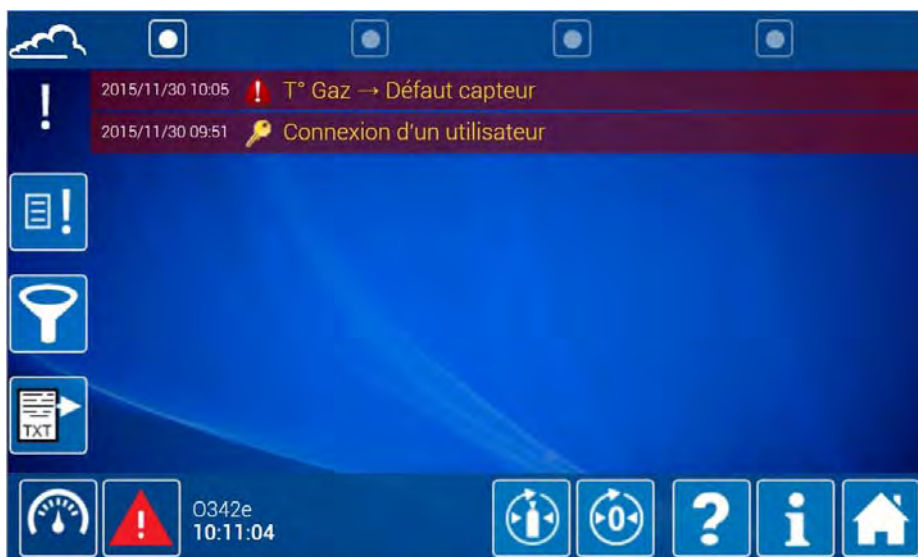


Table 5–1 – List of faults and corrective actions

Alarm message Min./Max. values	Possible cause(s)	Possible actions(s)
Overrange Max. value: 4095 points	– Wirings between measurement board and photodiode board are damaged. – Measurement board is out of service.	– Identify the wrong wire and change it. – Change measurement board.
Ref. zero filter Min. value: 0 Normal value: 365 days Max. Value: N.A.	– Zero reference filter is clogged, or it has reached its lifetime	– Change zero reference filter.
Chamber temp. Min. value: + 5°C Average value: + 25°C Max. value: + 45°C	– If $45^{\circ}\text{C} < T^{\circ}\text{C} < 80^{\circ}\text{C}$: the ambient room temperature, where the analyzer is installed, may be non-compliant. – If $80^{\circ}\text{C} \leq T \leq 100^{\circ}\text{C}$: temperature sensor of absorption medium is not connected to measurement board.	– Compare absorption medium temperature with internal analyzer temperature: • If $\Delta T = \pm 3^{\circ}\text{C}$: maintain room temperature between 5°C and 45°C. • If $\Delta T > \pm 3^{\circ}\text{C}$: replace temperature sensor of absorption medium connected to J16 of measurement board. – Connect temperature sensor of absorption medium to J16 of measurement board.
Internal temperature Min. value: +5°C Normal value: +30°C Max. value: + 45°C	– If $45^{\circ}\text{C} < T^{\circ}\text{C} < 80^{\circ}\text{C}$: ambient room temperature where analyzer is installed, may be non-compliant. – If $80^{\circ}\text{C} \leq T \leq 100^{\circ}\text{C}$: internal temperature probe is not connected to interconnection board.	– Compare absorption medium temperature with internal analyzer temperature: • If $\Delta T = \pm 3^{\circ}\text{C}$: maintain room temperature between 5°C and 45°C. • If $\Delta T > \pm 3^{\circ}\text{C}$: replace internal temperature sensor connected to J29 of interconnection board. – Connect internal temperature sensor to J29 of interconnection board.

Alarm message Min./Max. values	Possible cause(s)	Possible actions(s)
Chamber pressure Min. value: 300 hPa Normal value: 990 hPa Max. value: 1200 hPa	– Pressure sensor is not connected. – Sampling tube is dirty or too worn. – Dust filter is clogged or dirty. – Solenoid valve block is clogged or dirty. – There is a leak in fluid circuit between solenoid valve block and measurement module inlet. – Pressure sensor of absorption medium is faulty.	– Connect pressure sensor to J5 of measurement board. – Check sample tube status: clean or replace if necessary. – Check dust filter status: replace if necessary. – Check solenoid valve block status: clean if necessary, and test for leaks. – Check tightness of fluid circuit between solenoid valve block and measurement module inlet, and remedy if necessary. – Replace pressure sensor of absorption medium connected to J5 of measurement board.
Pump pressure Min. value: 200 hPa Normal value: 310 hPa Max. value: 600 hPa	– If $0 \text{ hPa} \leq P < 200 \text{ hPa}$: • Pump pressure sensor is not connected to interconnection board. • Pump pressure sensor is faulty. – If $600 \text{ hPa} < P \leq P_{\text{atm}}$: • Fluid pipe connected to pump pressure sensor is disconnected. • Restrictor is clogged. • Pump aspiration pipe is disconnected.	• Connect pump pressure sensor to J9 of interconnection board. • Replace pump pressure sensor connected to J9 of interconnection board. • Connect fluid tube to pump pressure sensor connected to J9 of interconnection board, and test tightness of fluid circuit between pump and pump pressure sensor. • Clean restrictor (see maintenance sheet in chapter 4). • Reconnect aspiration pipe of pump, and test tightness of fluid circuit between pump and measurement module restrictor.

Alarm message Min./Max. values	Possible cause(s)	Possible actions(s)
Atmo. pressure Min. value: 300 hPa Normal value: 1002 hPa Max. value: 1200 hPa	– If $0 \text{ hPa} \leq P < 300 \text{ hPa}$: • Pressure sensor is not connected to interconnection board. • Pressure sensor is faulty.	• Connect pressure sensor to J5 of interconnection board. • Replace pressure sensor connected to J5 of interconnection board.
Gas flow fault Min. value: 30 NL/h Normal value: 55 NL/h Max. value: 90 NL/h	– Pump pressure sensor is disconnected or out of service. – Pressure sensor of absorption medium is disconnected or out of service. – Restrictor is dirty or clogged.	– Reconnect or replace pump pressure sensor. – Reconnect or replace pressure sensor of absorption medium. – Check restrictor status. Clean if necessary (see maintenance sheet in Chapter 4).
I. LED fault Min. value: 2 mA Normal value: 3 mA Max. value: 90 mA	– If $UV_{\text{Measurement}} \approx UV_{\text{Reference}} \leq 100 \text{ mV}$ • Electrical connection between the LED board and measurement board is fault. • LED is faulty. • LED board is faulty. – If $I_{\text{LED}} > 90 \text{ mA}$ • LED has reached its lifetime. – If $I_{\text{LED}} < 2 \text{ mA}$ • Ambient room temperature, where analyzer is installed, is very low.	• Check the electrical connection between the LED board and the measurement board. Reconnect the flat wire between the LED board and the measurement board, if necessary. • Replace LED. • Replace LED board. • Replace LED. • Maintain ambient room temperature between $+ 5^{\circ}\text{C}$ and $+ 45^{\circ}\text{C}$.

Alarm message Min./Max. values	Possible cause(s)	Possible actions(s)
Measure signal fault Min. value: 550 mV Normal value: 1000 mV Max. value: 3000 mV	– If $100 \text{ mV} \leq UV_{\text{Measurement}} < 550 \text{ mV}$ • Electrical connection between measurement photodiode board and measurement board is faulty. • Measurement photodiode is defective. • Measurement photodiode board is defective. – If $UV_{\text{Measurement}} < 100 \text{ mV}$ • Electrical connection between LED board and measurement board is faulty. • Electrical connections between reference photodiode and measurement photodiode boards, and measurement board are faulty.	• Check electrical connection between photodiode measurement board and measurement board. Reconnect the flat wire between photodiode measurement board and measurement board, if necessary. • Replace measurement photodiode. • Replace measurement photodiode board. • Check electrical connection between LED board and measurement board. If necessary, reconnect the flat wire between LED board and measurement board. • Check electrical connections between reference photodiode and measurement photodiode boards and measurement board. If necessary, reconnect flat wires between reference photodiode and measurement photodiode boards, and with measurement board.
Reference signal fault Min. value: 550 mV Normal value: 1500 mV Max. value: 3000 mV	– If $100 \text{ mV} \leq UV_{\text{Reference}} < 550 \text{ mV}$ • Electrical connection between reference photodiode board and measurement board is faulty. • Reference photodiode is faulty. • Reference photodiode board is faulty.	• Check electrical connection between reference photodiode board and measurement board. Reconnect the flat wire between reference photodiode board and measurement board, if necessary. • Replace reference photodiode. • Replace reference photodiode board.

Alarm message Min./Max. values	Possible cause(s)	Possible actions(s)
	– If $UV_{reference} < 100 \text{ mV}$, • Electrical connection between LED board and measurement board is faulty. • Electrical connections between reference photodiode and measurement photodiode boards, and measurement board are faulty.	• Check electrical connection between LED board and Measurement board. Reconnect the flat wire between LED board and measurement board, if necessary. • Check electrical connections between reference photodiode and measurement photodiode boards, and measurement board. Reconnect the flat wires between reference photodiode and measurement photodiode boards, and with measurement board, if necessary.
GND fault Min. value: - 1 mV Normal value: 0 mV Max. value: + 1 mV	– Measurement board is out of service.	– Change measurement board.
GNDA fault Min. value: - 1 mV Normal value: 0 mV Max. value: + 1 mV	– Measurement board is out of service.	– Change measurement board.
+ 3.3 V fault Min. value: 3,2 V Normal value: 3,3 V Max. value: 3,4 V	– Power supply board is out of service. – Measurement board is out of service. – Controller board is out of service.	– Change power supply board. – Change measurement board. – Change controller board.
+ 4 volts fault Min. value: 3950 mV Normal value: 4000 mV Max. value: 4050 mV	– Measurement board is out of service.	– Change measurement board.

Alarm message Min./Max. values	Possible cause(s)	Possible actions(s)
+ 5 V fault Min. value: 4,8 V Normal value: 5 V Max. value: 5,2 V	– Power supply board is out of service.	– Change power supply board.
+ 5 V sensor fault Min. value: 4,9 V Normal value: 5 V Max. value: 5,1 V	– At least one temperature sensor is out of service. – At least one pressure sensor is out of service. – Controller board is out of service.	– Identify the faulty temperature sensor, and change it. – Identify the faulty pressure sensor, and change it. – Change controller board.
+ 12 V fault Min. value: 11,8 V Normal value: 12 V Max. value: 12,2 V	– Power supply board is out of service.	– Change power supply board.
+ 24 V fault Min. value: 23,4 V Normal value: 24,0 V Max. value: 25,0 V	– Power supply board is out of service. – Mains block is out of service.	– Change power supply board. – Change mains block.
+24V I. fault Min. value : - 0,1 A Normal value : from 0,5 A to 3 A Max. value : > 6 A	– If I > 6 A : • Power supply board is out of service. • Controller board is out of service. • +24 V power supply is out of service.	• Change power supply board. • Change controller board. • Identify the out-of operation element (pump, fan, solenoid valve).

Alarm message Min./Max. values	Possible cause(s)	Possible actions(s)
Automate card fault LD12 LED indicator is OFF instead of ON.	– Communication between controller board and interconnection board does not work. – Controller board is out of service (HUB USB). – Power supply board is out of service.	– Check connection from controller board to interconnection board. – Change controller board. – Change power supply board.
Measure O₃ card fault LD3 LED indicator is OFF instead of ON.	– Controller board is out of service (HUB UCB). – Connection between LED board and measurement board is faulty. – O₃ measurement board is out of service.	– Change controller board. – Check connection between LED board and measurement board, and change connection cable if necessary. – Change measurement board.

Page intentionally left blank

Table 5-2 – Interconnection board configuration

Jumpers	Connections
J1	Controller board link
J2	Power supply board link
J3	Link between interconnection board and ARM20 board
J4	Input of 24 V supply block connection
J5	Pressure sensor board link
J6	SV link
J7	Pump supply link
J8	SV link
J9	Pressure sensor board link
J10	Fan supply link
J11	SV link
J12	Peltier supply link
J13	Cycle 1 SV link
J14	Peltier temperature regulation
J15	24 V external supply in rear panel
J16	Cycle 2 SV link
J17	Ozone generator block supply link
J18	External converter temperature link
J19	Copy of Zero/Span SV
J20	PWM pump supply link
J21	Analog supply inlet link
J22	PWM 24V pump motor supply
J23	TCP/IP Ethernet link
J24	RJ45 without LED link
J25	Not used
J26	O3 generator board link

Test Point	Signal
PT1	0 V (GND)

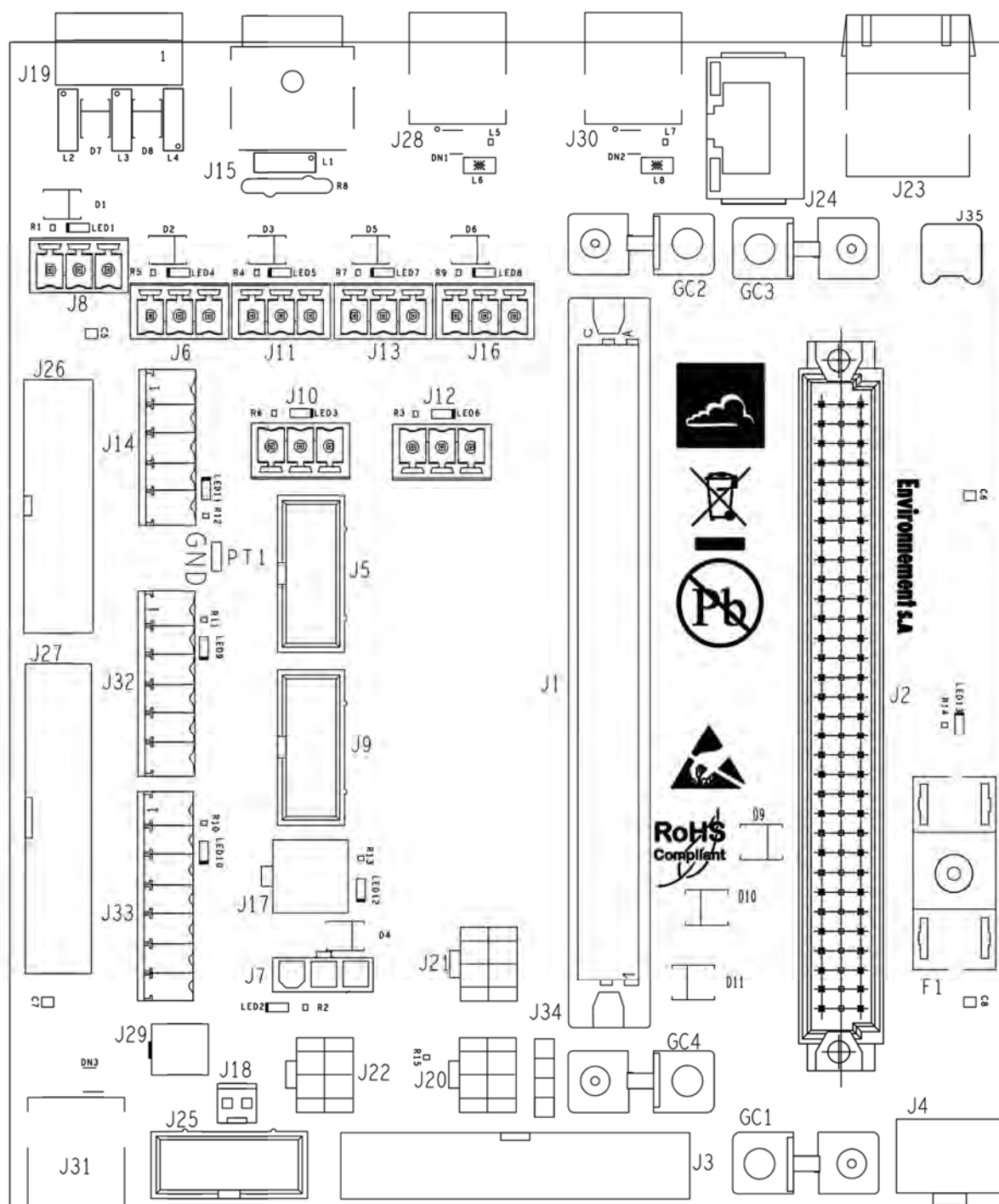


Figure 5-1 – Interconnection board diagram

Table 5–3 – Automaton board configuration

Jumpers	Connections
J1	Main connector

Test Points	Signals
PT1	0 V (GND)
PT2	0 V (GND)
PT3	+24 V Supply
PT4	+12 V Supply
PT5	+5 V Supply
PT6	+3.3 V Supply

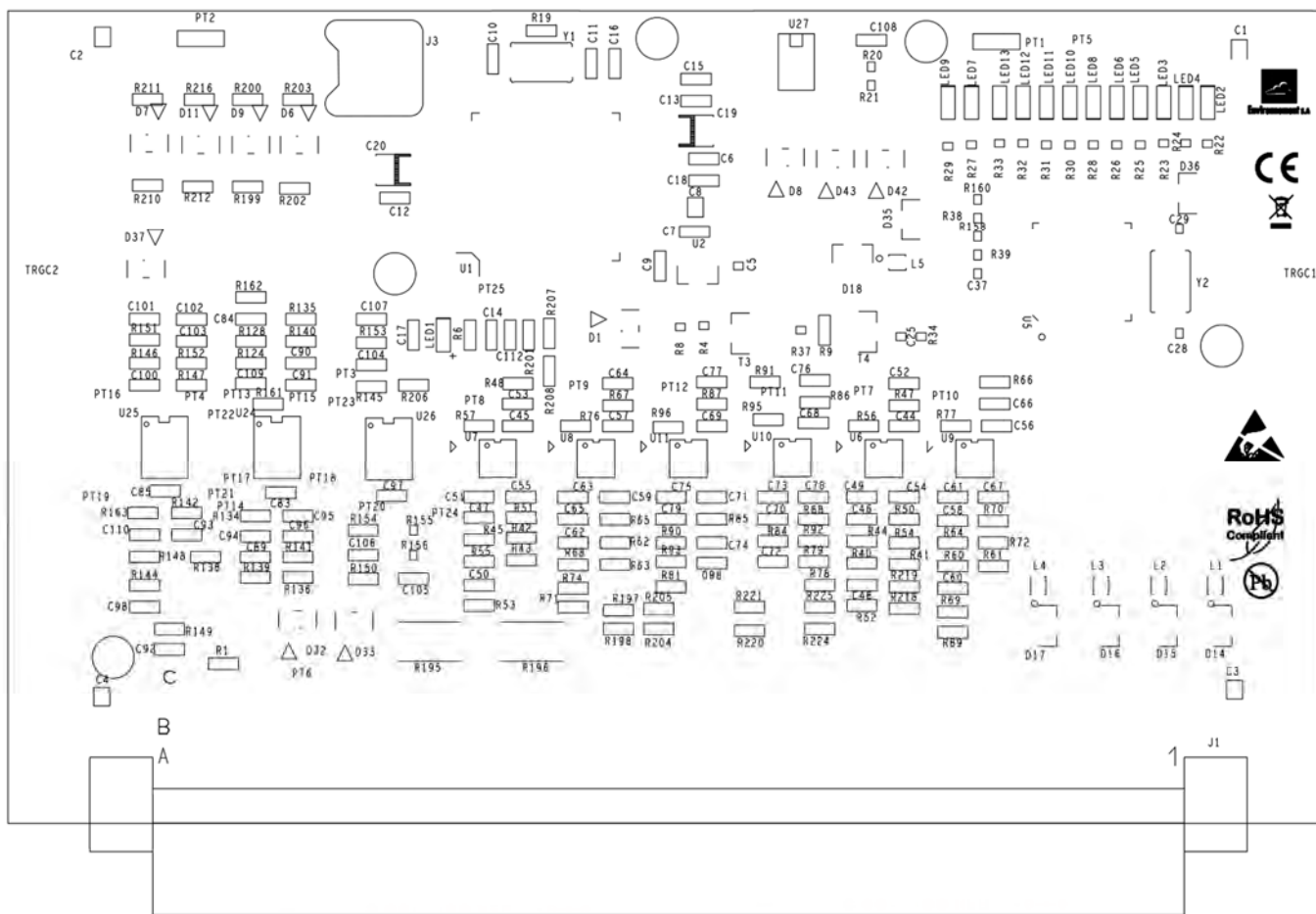


Figure 5-2 – Automaton board diagram (above view)

Table 5-4 – Photodiode board configuration

Jumpers	CONNECTION
J1	Main connector

Test Point	Signal
PT1	0 V (GND)
PT2	Photodiode Signal

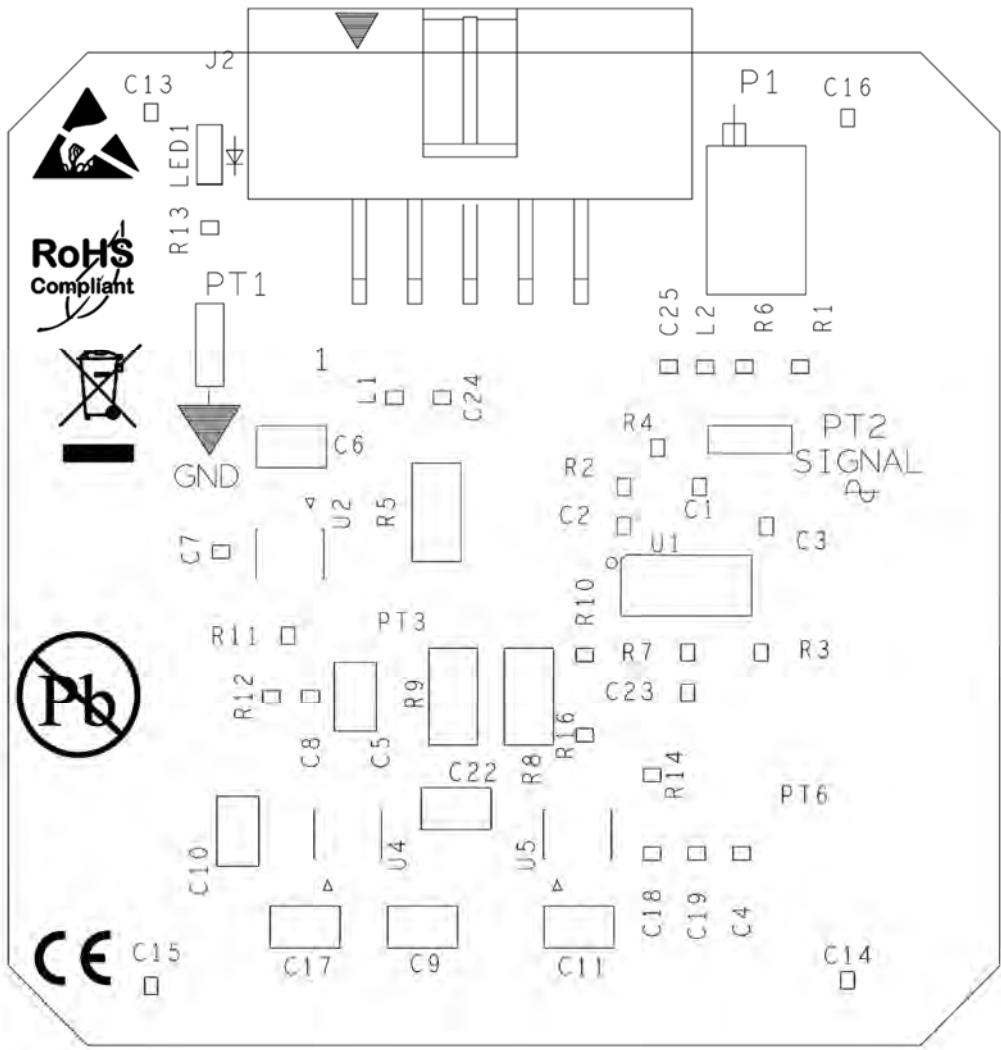


Figure 5-3 – Photodiode board diagram (above view)

Table 5-5 – LED UV board configuration

Jumpers	Connection
J1	Main connector

Test Points	Signals
PT1	0 V (GND)
PT2	LED signal

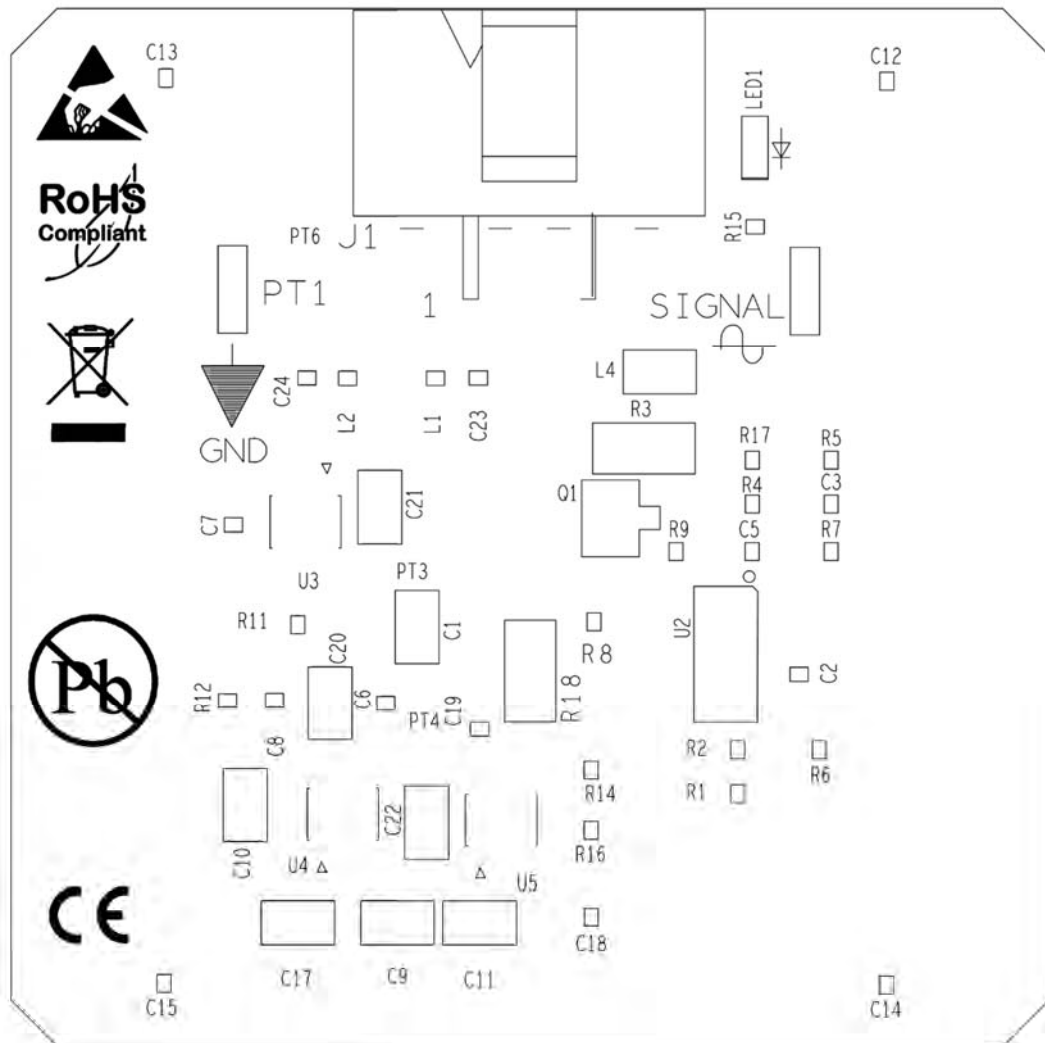


Figure 5-4 – LED UV board diagram (above view)

Table 5-6 – Measurement board configuration

Jumpers	Connections
J1	Interconnection board link
J2	USB-PC link
J3	Option link
J5	Pressure sensor connector
J7	Temperature AUX
J8	Heating AUX
J16	PT1000 absorption medium temperature link
J41	Connector of the reference photodiode board
J42	Connector of the measurement photodiode board
J43	Connector of UV LED board

Test Points	Signals
PT1	0 V (GND)
PT2	0 V (GND)
PT3	Supply +5 V
PT4	Supply +3.3 V

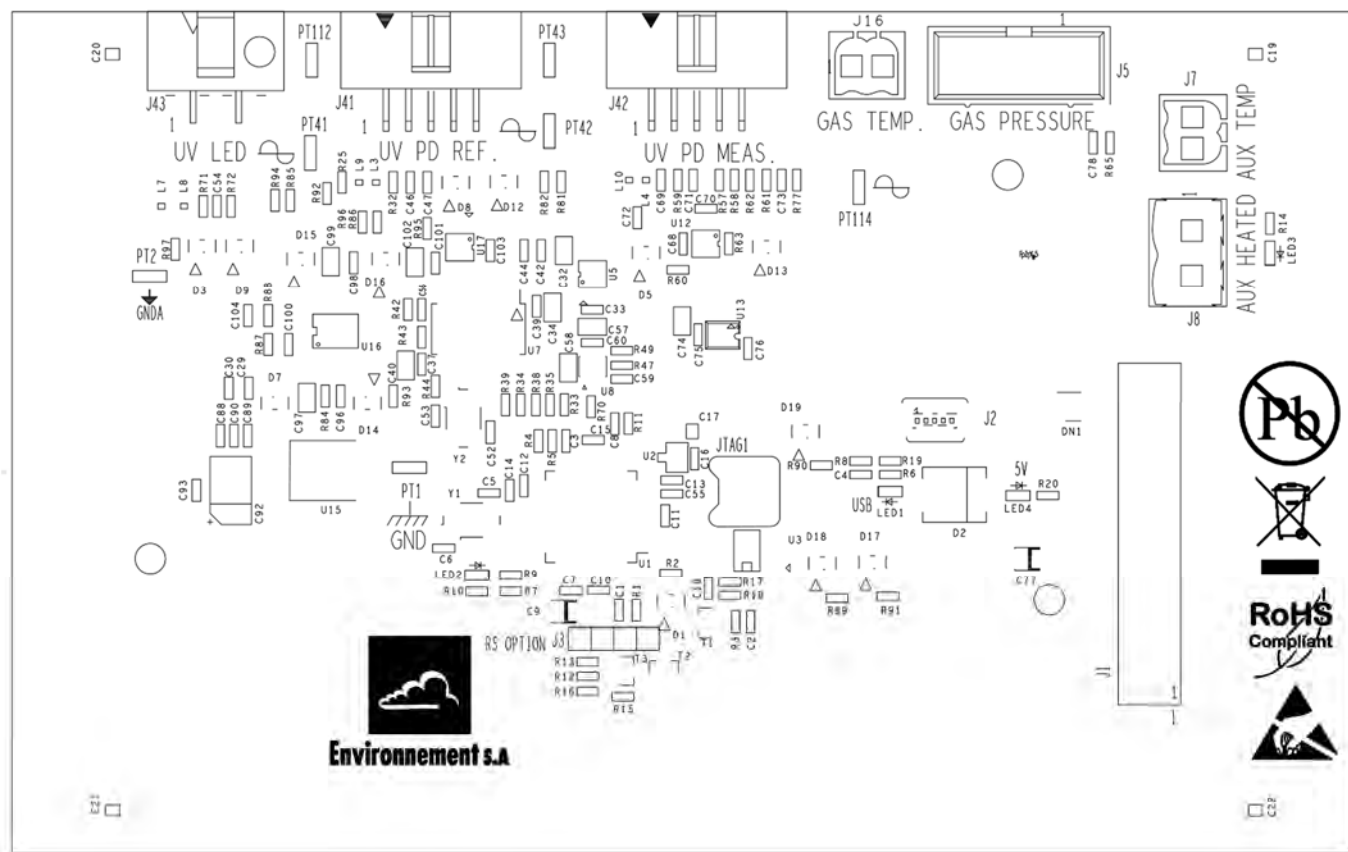


Figure 5-5 – Measurement board diagram

Jumpers	Connection
J1	Main connector

Test Point	Signal
PT1	0 V (GND)

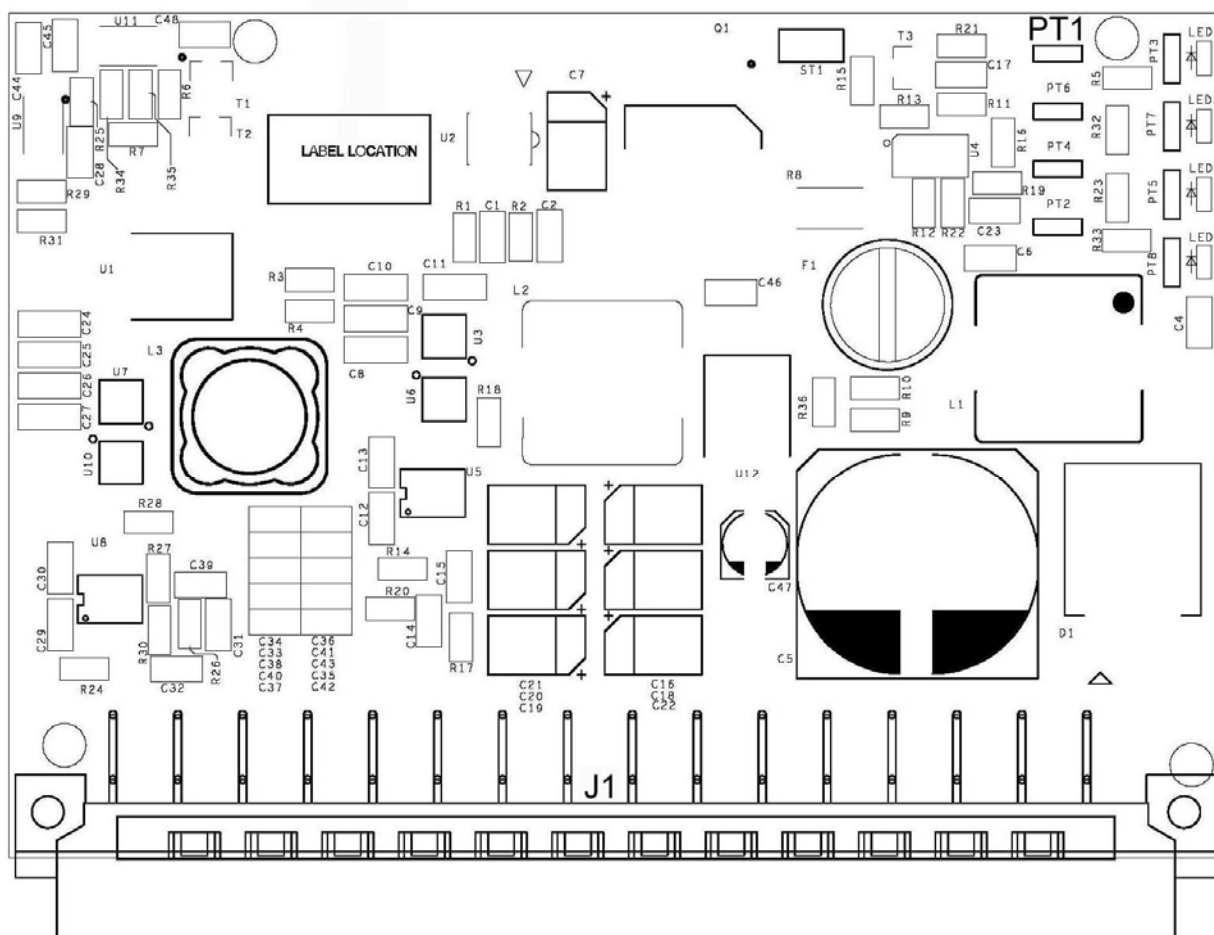


Figure 5–6 – Supply board diagram

Table 5-8 – ARM20 board description

Identification	Functions
J1	LCD screen
J2	Interconnection board link
J5	Touchscreen
J9	Push button link

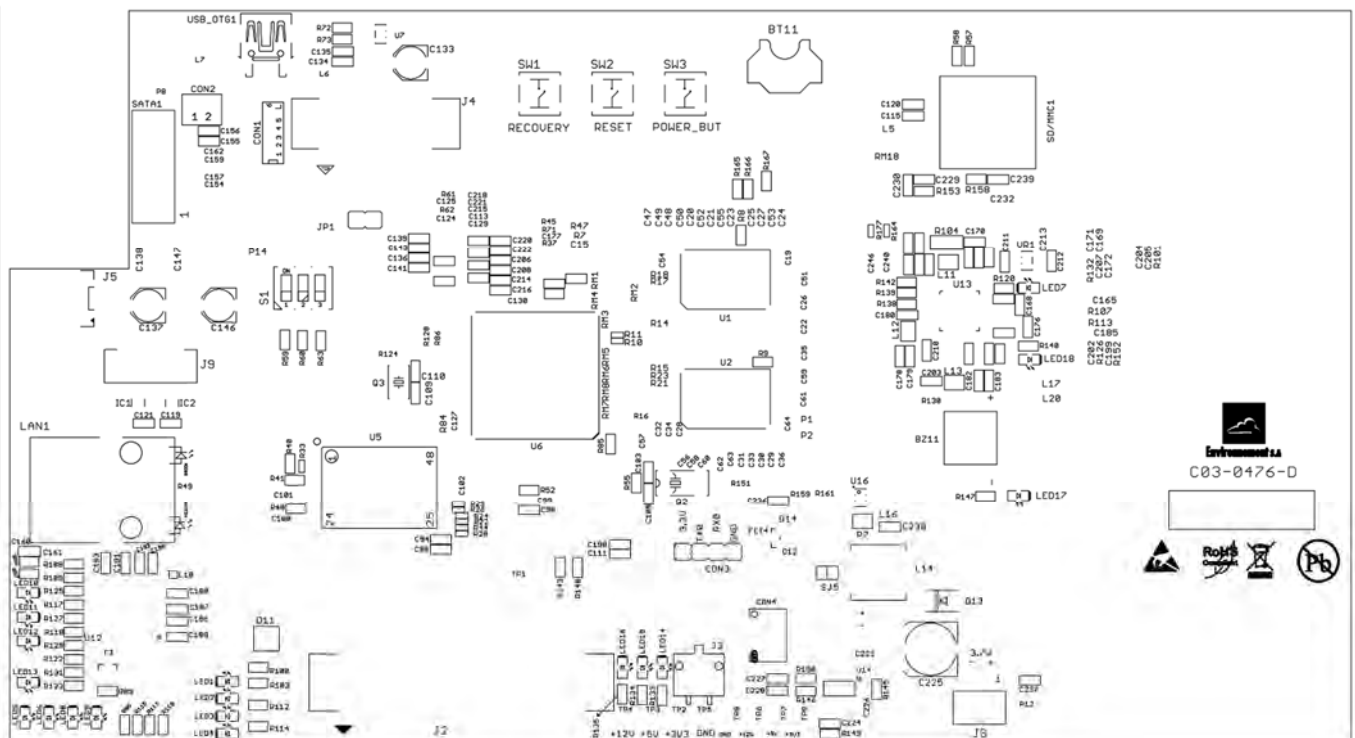


Figure 5-7 – ARM20 board

Page intentionally left blank